# Managing Expectations with Exchange Rate Policy*

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**Abstract**

We study a small open economy in which the exchange rate aggregates private information about macro fundamentals, yet imperfectly because of non-fundamental capital flows. Foreign exchange interventions influence economic conditions via traditional portfolio balance and novel informational effects. Publicly announced interventions reveal the central bank's knowledge about macro fundamentals, while secret interventions alter the informational content of the exchange rate. When the private sector overreacts to public information, the optimal policy is to intervene secretly, tolerating some non-fundamental capital flows to mitigate mis-learning from the exchange rate. Our results speak to practical questions about exchange rate policy design.

**Keywords**: exchange rates, informational frictions, foreign exchange interventions, central bank communication.

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Despite a historical consensus favoring freely floating exchange rates, central banks of several emerging and advanced economies routinely intervene in foreign exchange (FX) markets by buying and selling foreign currencies (Adler et al., 2021). A remarkable feature of these interventions is the wide variation in central banks' communication practices. While some central banks transparently announce the timing and size of interventions, about two-thirds either do not disclose these activities or reveal them only after a delay, de facto conducting *secret* interventions (Patel and Cavallino, 2019).

These practices point to underexplored channels of transmission. FX interventions may influence the economy not only via conventional portfolio-balance effects, but also through their impact on expectations—whether by revealing information or by shaping how market participants interpret price signals such as exchange rates. Yet, leading macro models largely abstract from these informational dimensions, and thus from the design of interventions when agents learn from prices or announcements.

To address these questions, this paper develops a micro-founded small open economy model in which agents learn from both exchange rates and FX interventions. The backbone of our model is a standard dynamic small open economy featuring tradable and non-tradable goods, and segmented international asset markets as in Gabaix and Maggiori (2015), Fanelli and Straub (2021), and Itskhoki and Mukhin (2022). In this class of models, financial flows contribute to determining the exchange rate. To introduce learning, we depart from the full information rational expectation assumption along two key dimensions. First, we relax the assumption of full information. Instead, agents observe private signals about future productivity—the economy's fundamental. The exchange rate aggregates this dispersed information and serves as a public signal about productivity. However, this signal is noisy, as the presence of noise-trading shocks blurs the relationship between exchange rates and productivity, similar to Bacchetta and Wincoop (2006).[1] Second, we relax the rational expectations assumption. Agents overreact to their private signals, as in Bordalo et al. (2018), while mistakenly thinking that both they and others are rational. As a result, in equilibrium they overreact to the information contained in the exchange rate. Expectations overreaction is well documented in the literature (Bordalo et al., 2019, 2020, 2024; Afrouzi et al., 2023; Bianchi et al., 2025), and we provide novel evidence that speaks to our specific mechanism. Using survey data from professional forecasters in a sample of small open

[1]The notion that exchange rates partly reflect expectations about a country's future productivity or macroeconomic conditions is consistent with the analysis of Engel and West (2005) and the evidence in Chahrour et al. (2024) and Stavrakeva and Tang (2023).

economies, we show that GDP growth forecasts are excessively optimistic when the exchange rate has recently appreciated.

The paper provides three main contributions. First, we formalize the informational role of the exchange rate, an aspect often emphasized by policymakers and commentators yet largely overlooked in international macroeconomics. In our model, agents use the exchange rate as a public signal to form their expectations of future productivity, influencing their external borrowing, consumption, and capital investment. This channel is distinct from traditional mechanisms of exchange rate transmission, such as expenditure switching, and its strength hinges on the *informational content of the exchange rate*—the amount of independent information that the exchange rate contains about future productivity. Because the informational content of the exchange rate is endogenous, it also depends on the conduct and communication of FX interventions.[2]

Second, we show that FX interventions have radically different macroeconomic effects depending on whether they are revealed to the public or conducted secretly. We consider a central bank that observes aggregate financial flows, either perfectly or imperfectly, consistent with the supervisory role many monetary authorities hold over financial institutions. We thus assume that the central bank buys and sells foreign bonds according to a publicly known rule that responds to both noise-trading and exchange rate fluctuations. This rule is general, as the central bank, by responding to financial flows and the exchange rate, implicitly responds to both productivity and noise-trading shocks. The central bank can choose the strength of the reaction to either shock as well as whether to communicate the size of the intervention. If the size of the intervention is publicly announced, it becomes an additional public signal about the beliefs of the central bank. Consequently, the intervention assumes a "signaling role," which expands the information set of agents, akin to the closed-economy models of Melosi (2017) and Tang (2015).[3] In contrast, if the intervention is not publicly announced, *i.e.*, it is conducted secretly, the central bank alters the informational content of the exchange rate through the intervention rule. By deciding how much to lean against noise-trading shocks the central bank effectively controls the accuracy of the exchange rate as a signal about productivity. Secret interventions, in this sense, represent a tool to "manage expectations" about the state of the economy.

[2]Despite this endogeneity, our model admits a unique equilibrium because it abstracts from complementarities studied elsewhere in the literature, e.g., in Amador and Weill (2010).

[3]Our notion of signaling differs from contexts in which FX interventions signal future changes in monetary policy (Mussa, 1981), or new target levels for the exchange rate (Bhattacharya and Weller, 1997; Vitale, 1999).

Third, the suboptimal use of public information by the private sector can justify the optimal choice of conducting secret FX interventions, a common but not fully understood policy practice. The underlying friction stems from expectations' overreaction to public news—a well-documented fact that policymakers often cite as a source of concern. To understand this normative result, it is useful to highlight that our small open economy features two sources of inefficiency relative to the first-best allocation with frictionless financial markets and full-information rational expectations. First, asset market segmentation gives rise to expected excess currency returns that result in suboptimal external borrowing, as in Fanelli and Straub (2021) and Itskhoki and Mukhin (2022). Second, dispersed information leads agents to underreact to productivity shocks, while behavioral biases cause them to overreact to new information, and together these frictions drive a wedge between expectations of future productivity and the full-information rational value. These expectations misalignments result in suboptimal consumption and investment for a given level of external borrowing.[4]

Publicly announced interventions cannot manipulate the informational content of the exchange rate and therefore their optimal design is aimed exclusively at reducing suboptimal external borrowing. To do so, the central bank offsets, as best as it can, inefficient financial flows due to noise-trading shocks by appropriately buying or selling foreign currency. While this policy corrects, partially or entirely, inefficient external borrowing, it nevertheless cannot mitigate beliefs overreaction, implying that the ensuing consumption and investment allocations are generally distorted.

In contrast, *secret* interventions can exploit a novel trade-off between inefficient external borrowing and expectations' overreaction by altering the informativeness of the exchange rate. Intuitively, when the central bank does not announce the size of its interventions, a policy that dampens the effects of noise-trading shocks makes the exchange rate more informative about productivity and causes agents to place more weight on a public signal to which they overreact. This mechanism entails a trade-off: mitigating this overreaction requires intentionally reducing the signal content of the exchange rate at the cost of allowing some inefficient, noise-driven exchange rate fluctuations.

We show that when expectations' overreaction is sufficiently strong, the central bank optimally chooses to exploit this trade-off through secret interventions. Under

[4]As in Gabaix (2020) and most of the behavioral literature, we assume that agents operate under non-rational beliefs but experience utility as rational agents, and that the central bank is aware of this feature.

these circumstances, agents effectively place an excessive weight on the exchange rate in forming expectations; thus, lowering its informational content *reduces*, rather than increases, the volatility of expectational mistakes. In such scenarios, it is indeed optimal to limit the informativeness of the exchange rate. To achieve this, the central bank allows some degree of non-fundamental variation in the exchange rate arising from noise trading, even though it results in suboptimal external borrowing.

Our model reconciles two seemingly conflicting aspects of central banking practices. On the one hand, surveys of central bankers point to a consensus that FX interventions work primarily by affecting market expectations through a signaling channel. On the other hand, most central banks conduct secret interventions in the FX market. Together, these observations compose a *secrecy puzzle* (Sarno and Taylor, 2001) as the signaling channel is generally thought to work through transparent policy announcements. Our model rationalizes these puzzling observations. Publicly announced interventions indeed have signaling effects. Yet, in environments in which market participants overreact, it can be desirable to conceal interventions, and thus their signaling role, to control the amount of information available to the market. The model thus offers a new interpretation of the widespread practice of "systematic managed floating," whereby central banks regularly respond to changes in market pressure, with a portion reflected in the exchange rate and the rest absorbed through changes in FX reserves (Frankel, 2019).

**Relation to the literature** This paper contributes to several strands of the literature in international macroeconomics and information economics.

First, a growing body of work in open-economy macroeconomics has highlighted motives for FX interventions, including alleviating intermediaries' financial constraints (Chang and Velasco, 2017), counteracting the effects of shocks originating in FX markets (Cavallino, 2019; Itskhoki and Mukhin, 2022; Basu et al., 2023), implementing a desired exchange rate policy when the interest rate is at the zero lower bound (Amador et al., 2019), mitigating the distributional effects of exchange rate fluctuations due to consumption externalities (Fanelli and Straub, 2021), or addressing the effects of production externalities (Ottonello et al., 2024). This paper contributes to this literature by studying the informational effects of FX interventions. Information and communication aspects are important in real-world FX policies, as indicated directly by central bankers in surveys and suggested by the observed opacity surrounding many interventions (Patel and Cavallino, 2019).[5] Our contribution is twofold: first, we describe a novel informational channel of FX policies, which applies irrespective of the rationale behind the intervention; second, we show how optimal exchange rate policy can leverage this informational channel to address inefficiencies stemming from frictions in belief formation.[6] This perspective reframes FX intervention as a tool for belief management—not just flow stabilization.

The paper contributes also to the literature on the social value of public information. With exogenous information, more precise public communication can be detrimental to welfare in the presence of a socially harmful desire to coordinate (Morris and Shin, 2002) or when economic fluctuations are driven primarily by shocks or other distortions that make the full-information equilibrium inefficient (Angeletos and Pavan, 2007). These conditions may arise, for example, in the presence of markup shocks (Angeletos et al., 2016), distortionary taxes (Fujiwara and Waki, 2020), nominal rigidities (Fujiwara and Waki, 2022), and even with supply shocks if they are inefficiently shared across countries (Candian, 2021). When information is endogenous as in learning from prices (Grossman, 1976), more public information can be harmful because of strategic complementarities that make agents overweight public signals (Amador and Weill, 2010), or correlated expectation errors that are common knowledge (Hassan and Mertens, 2017). Our paper formalizes a novel channel through which public information can reduce welfare, namely, through overreaction to endogenous public signals. We show that this channel is conceptually distinct from others in the aforementioned literature.

Finally, we contribute to the literature on the effects of macroeconomic policy and their communication by highlighting how opaque balance-sheet policies can limit price informativeness. Bond and Goldstein (2015) investigate how uncertain future government intervention affecting a firm's cash flows impacts price informativeness. Gaballo and Galli (2022) studies the informational channel of central bank's asset purchases, but in a closed economy setting where interventions are always publicly observed. Iovino and Sergeyev (2021) study the effects of central bank's balance-sheet policies in a model where people form expectations through iterative level-$k$ thinking. Angeletos and Sastry (2020) delve into the separate but complementary question of whether policy

[5] Recent empirical evidence on FX interventions includes Kearns and Rigobon (2005), Kuersteiner et al. (2018), Fratzscher et al. (2019), Menkhoff et al. (2021), Adler et al. (2021), and Ferreira et al. (2024).

[6] Earlier theoretical research has employed partial equilibrium models to explore other aspects of FX interventions such as central bank transparency and credibility about their exchange rate target (Stein, 1989; Bhattacharya and Weller, 1997; Vitale, 1999, 2003).

communications should anchor expectations about a policy instrument or the targeted outcome, and how it depends on a departure from rational expectations. Brunnermeier et al. (2021) study the incentives of market participants to acquire information when there is uncertainty about government interventions in financial markets via asset purchases. As in our environment, their government faces a trade-off between reducing (inefficient) price volatility and enhancing price efficiency. Differently from our model, this trade-off arises because the intervention itself introduces noise, rather than because the private sector processes the information contained in prices incorrectly.[7]

# 1 Belief formation in an open economy model

We consider a two-period small open economy model with a tradable sector and non-tradable sector, and extend it to incorporate two features of interest. First, market segmentation gives rise to a finite elasticity of demand for foreign bonds and, therefore, a scope for foreign exchange interventions. Second, the economy is affected by two aggregate shocks that are imperfectly observed by agents in the economy: productivity shocks and "noise-trading" shocks to the demand for foreign bonds.

## 1.1 Model setup

The small open economy is populated by private agents and a central bank. Private agents, that include households, firms, and financiers, are located on a continuum of atomistic islands, $i \in [0,1]$, as in Lucas (1972). Information is common within islands but heterogeneous across islands. In particular, on each island, households and financiers receive the same private noisy signal on next-period productivity of the small open economy. Agents observe local output and prices as well as the exchange rate, which serves as a noisy public signal about next-period productivity. Time is discrete and indexed by $t = [0,1]$. Foreign variables are denoted with a star symbol.

### 1.1.1 Households and goods markets

The utility function of the representative household of island $i$ is:

$$E_0^i \left\{ \frac{{C_0^i}^{1-\sigma}}{1-\sigma} + \beta \frac{{C_1^i}^{1-\sigma}}{1-\sigma} \right\}, \tag{1}$$

[7] See also Kimbrough (1983, 1984), Chahrour (2014), Paciello and Wiederholt (2014), Angeletos et al. (2020), Kohlhas (2020a), Dávila and Walther (2023), and Morelli and Moretti (2023).

where $C^i$ denotes consumption and $E_0^i$ is the subjective expectation operator conditional on the information set at $t = 0$.

Households have an initial endowment of capital, $K_0^i = K_0 > 0$ and must invest final goods to accumulate capital for the next period, $K_1^i$. The capital stock fully depreciates between periods and is used in the production of tradable goods:

$$Y_{T,0}^{i,H} = K_0^{i\,\alpha}, \qquad Y_{T,1}^{i,H} = A_1 K_1^{i\,\alpha}. \tag{2}$$

Above, $A_1$ represents stochastic period-1 productivity that is common across islands. In each period, the household also receives an endowment of the non-tradable good: $Y_{N,0} = (1+\alpha\beta\gamma)Y_{N,1}$.[8] Final goods that are used for consumption and period-1 capital are composites of tradable and non-tradable goods:

$$C_0^i + K_1^i = \left[(1-\gamma)^{\frac{1}{\theta}} Y_N^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} Y_T^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \quad C_1^i = \left[(1-\gamma)^{\frac{1}{\theta}} Y_N^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} Y_T^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}. \tag{3}$$

The parameter $\theta$ denotes the elasticity of substitution between tradable and non-tradable goods in the production of final goods while $\gamma$ governs the share of tradable goods in final goods. $Y_{T,t}^i$ represents domestic absorption of the tradable good, which is the sum of production and imports from the rest of the world $Y_{T,t}^i = Y_{T,t}^{i,H} + Y_{T,t}^{i,F}$. We assume that each island trades with the rest of the world but not with other islands to avoid information revelation by inter-island interactions, which would preclude any marginal informational role of public signals such as exchange rate or interventions. The household's budget constraints are:

$$\begin{aligned} P_0^i C_0^i + P_0^i K_1^i + \frac{B_1^i}{R_0} &= P_{N,0}^i Y_{N,0} + \mathcal{S}_0 Y_{T,0}^{i,H} + T_0^i, \\ P_1^i C_1^i &= B_1^i + P_{N,1}^i Y_{N,1} + \mathcal{S}_1 Y_{T,1}^{i,H} + T_1^i. \end{aligned} \tag{4}$$

where $P_t^i$ is the island-$i$ price of the composite good, and $P_{N,t}^i$ is the island-$i$ price of the non-tradable good. $\mathcal{S}_t$ is the nominal exchange rate, which is common across islands. We assume that the foreign-currency price of tradable goods is constant and equal to 1, *i.e.*, $P_{T,t}^\star = P_T^\star = 1$. The date-0 budget constraint assumes no initial debt and states that the household's income from the sale of tradable and non-tradable goods as well as from government nominal transfers, $T_0^i$, can be used to buy consumption goods,

[8] The endowment in period 0 relative to period 1 implies a steady state with $B_1^\star = 0, Q_0 = Q_1 = 1$ and $C_0 = C_1$.

invest in physical capital, or save in a domestic nominal bond, $B_1^i$, whose interest rate is $R_0$. At date-1 all the income of the household is instead used for consumption.

Implicit in equation (4) is the assumption that the household cannot hold foreign bonds (*e.g.*, Gabaix and Maggiori, 2015, Fanelli and Straub, 2021, Itskhoki and Mukhin, 2021). Market segmentation captures the idea that it is difficult for many households in emerging markets to access international financial instruments without financial intermediation, especially when borrowing in foreign currency.

Using market clearing for final goods and non-tradable goods, island $i$ households' budget constraints in equilibrium simplify to:

$$\frac{B_1^i}{R_0} = \mathcal{S}_0(Y_{T,0}^{H,i} - Y_{T,0}^i) + T_0^i; \qquad -B_1^i = \mathcal{S}_1(Y_{T,1}^{H,i} - Y_{T,1}^i) + T_1^i. \tag{5}$$

where households of each islands leave no debt at the end of period 1.

### 1.1.2 Financial market

Financiers from every island trade home and foreign bonds in the small open economy-wide financial sector. Along with financiers, the government and a set of noise traders also operate in the financial sector, as we describe next.

**Financiers** We assume that there are frictions in the financial sector which result in a downward-sloping demand for currency from financiers. In particular, we follow the formulation of Fanelli and Straub (2021). In each island a continuum of risk-neutral financiers owned by the island household trade home and foreign bonds subject to position limits and heterogeneous participation costs, as in Alvarez et al. (2009).[9]

In Appendix A.1, we derive the profit maximization problems of island-$i$ financiers, and show that it results in the following demand for the foreign currency bond:

$$\frac{D_1^{i\star}}{R_0^\star} = \frac{1}{\hat{\Gamma}} E_0^i \left( R_0^\star - R_0 \frac{\mathcal{S}_0}{\mathcal{S}_1} \right), \tag{6}$$

where the zero-capital portfolio island-$i$ financiers and their carry trade profits are,

[9] The assumption that financial intermediaries are owned by the household ensures that profits and losses from carry trade activity do not represent a net benefit or cost to the small open economy. Fanelli and Straub (2021) already studied the implications for FX interventions of such "leakages." Our novel focus is instead on the informational role of exchange rates and FX interventions.

respectively:

$$\frac{D_1^i}{R_0} + \mathcal{S}_0 \frac{D_1^{i\star}}{R_0{}^\star} = 0, \qquad\qquad \pi_1^{i,D\star} \equiv D_1^{i\star} + \frac{D_1^i}{\mathcal{S}_1} = \cdots = \tilde{R}_1^\star \frac{D_1^{i\star}}{R_0{}^\star}.$$

Aggregating across islands, the overall demand of financiers for foreign bonds is:

$$\frac{\int D_1^{i\star}\,\mathrm{d}i}{R_0^\star} = \frac{1}{\hat{\Gamma}} \bar{E}_0 \left( R_0^\star - R_0 \frac{\mathcal{S}_0}{\mathcal{S}_1} \right). \tag{7}$$

where $\bar{E}_0(X_t)$ is the average expectation of $X_t$ across islands, *i.e.*, $\bar{E}_0 X_t = \int E_0^i X_t \,\mathrm{d}i$.

Financiers' demand for foreign bonds has a finite (semi-)elasticity to the expected excess return, implying that changes in the net supply of foreign bonds, *e.g.*, induced by noise trading or FX interventions, affect the equilibrium exchange rate. The critical parameter in equation (7) is the inverse demand elasticity $\hat{\Gamma}$, which governs the strength of frictions in the international financial market. If $\hat{\Gamma}$ is large, *e.g.*, due to tight position limits, intermediation is impeded. In the extreme case where $\hat{\Gamma} \to \infty$ intermediation is absent, and the country is in financial autarky. By contrast, when $\hat{\Gamma} \to 0$, financiers are not subject to position limits and expected excess currency returns are driven to zero. Henceforth, we assume $\hat{\Gamma} \in (0, \infty)$.

**Noise traders** Noise traders exogenously demand foreign currency $\frac{N_1^\star}{R_0{}^\star}$. Here $\frac{N_1^\star}{R_0^\star} > 0$ means that noise traders short home-currency bonds to buy foreign-currency bonds. They also hold a zero-capital portfolio in home and foreign bonds denoted $(N_1, N_1^\star)$, which implies:

$$\frac{N_1}{R_0} + \mathcal{S}_0 \frac{N_1^\star}{R_0^\star} = 0. \tag{8}$$

**Central bank/Government** The economy-wide central bank holds a $(F_1, F_1^\star)$ bond portfolio. The value of the portfolio is $\frac{F_1^i}{R_0} + \mathcal{S}_0 \frac{F_1^\star}{R_0^\star}$. We assume that the government finances its operations with transfers:

$$\begin{aligned} \frac{F_1}{R_0} + \mathcal{S}_0 \frac{F_1^\star}{R_0^\star} &= -\int T_0^i \,\mathrm{d}i, \\ 0 = F_1 + \mathcal{S}_1 F_1^\star + \tau \mathcal{S}_1 \left( \int \pi_1^{i,D\star} \,\mathrm{d}i + \pi_1^{N\star} \right) &- \int T_1^i \,\mathrm{d}i, \end{aligned} \tag{9}$$

where $\pi_1^{i,D\star}$ is the income from financial transactions of financiers on island $i$, defined above, and $\pi_1^{N\star}$ is the income from financial transactions of noise traders. In addition, the central bank ensures stable aggregate price levels, *i.e.*, $P_0 = P_1 = 1$. This implies a real exchange rate of the small open economy $\mathcal{Q}_t = \frac{\mathcal{S}_t P_t}{P^\star_{T,t}} = \mathcal{S}_t$ in every period.

**Financial market clearing** Since home-currency bond are in zero net supply, market clearing implies

$$\int B_1^i \, \mathrm{d}i + N_1 + \int D_1^i \, \mathrm{d}i + F_1 = 0. \tag{10}$$

Combining the market clearing condition with households and government budget constraints, and the portfolios and income from financial transactions of financiers and noise traders, we obtain the aggregate position of the financiers, in foreign currency:

$$\frac{\int D_1^{i\star} \, \mathrm{d}i}{R_0^\star} = \int \left( Y_{T,0}^{i,H} - Y_{T,0}^i \right) \, \mathrm{d}i - \frac{F_1^\star + N_1^\star}{R_0^\star} \tag{11}$$

That is, the aggregate position of financiers equals the portion of households' bond demand—originating from the trade imbalance—that is not met by the foreign currency supplied by the government and noise traders.

## 1.2 Exchange rate determination

We solve the model using a log-linear approximation around a steady state with $A_1 = 1, N^\star = F^\star = 0$ and $B^i = D^i = 0$, $\forall i$. Lower-case variables denote log-deviation from steady state of corresponding upper-case variables. In Appendix A.2 we report the island-level equilibrium, and in Appendix A.3 we derive the equilibrium aggregate real exchange rate as a function of shocks and expectations thereof:

$$q_0 = \frac{\Gamma \omega_1}{\Gamma \tilde{\theta} \omega_1 + \omega_3} (n_1^\star + f_1^\star) - \frac{\omega_2}{\Gamma \tilde{\theta} \omega_1 + \omega_3} \bar{E}_0 a_1 \tag{12}$$

where $\bar{E}_0$ is the average expectation across all islands, $\Gamma \equiv \hat{\Gamma} \beta^2 (\alpha\beta)^{\frac{\alpha}{1-\alpha}}$, and $\omega_1 > 0, \omega_2 > 0, \omega_3 > 0$, and $\tilde{\theta} > 0$ are convolutions of parameters independent of $\Gamma$ defined in Appendix A.3.

Equation (12) says that, *ceteris paribus*, the exchange rate appreciates following a fall in foreign bond demand (by noise traders or by the central bank) or an improvement in expectations of productivity. Lower foreign bond demand requires financiers to absorb more foreign bonds, commanding an expected profit, in proportion to $\Gamma$. As a result, the exchange rate appreciates today, so that its expected depreciation earns financiers an expected return on their long foreign-bond position.[10]

Higher expected future productivity in the tradable sector implies a reduction in the future relative price of tradables, *i.e.* a future exchange rate appreciation. By uncovered interest parity (UIP), the exchange rate today appreciates. The equilibrium appreciation, however, is attenuated relative to UIP due to the effects of higher expected future productivity on financial flows: households increase imports and borrowing today, prompting financiers to absorb more domestic bonds. As a result, domestic bonds must offer a positive excess return, exerting downward pressure on the equilibrium exchange rate.

### 1.3 Information structure

We now consider how expectations are formed and introduce two realistic deviations from Full Information Rational Expectations (FIRE): dispersed information and distorted beliefs.

**Dispersed information** Households and financiers in each island $i \in [0,1]$ have a common prior about next-period productivity $a_1 \sim \mathcal{N}(0, \beta_a^{-1})$. Moreover, at $t = 0$ these agents observe local prices and quantities, in addition to a local signal about the future realization of productivity $a_1$:

$$v^i = a_1 + \epsilon^i, \qquad \epsilon^i \sim \mathcal{N}(0, \beta_v^{-1}), \tag{13}$$

with $\int_i \epsilon^i \, \mathrm{d}i = 0$. We can think of this noise either as the product of dispersed noisy information (Lucas, 1972) or as a metaphorical representation of rational inattention (Sims, 2003).

Regarding the observability of aggregates, we assume that while agents in each island cannot observe aggregate prices and quantities, they share the same currency and can therefore observe the aggregate exchange rate $q_0$. The exchange rate is an important endogenous signal because it carries information about the average expectation of the common future productivity shock $a_1$ as per eq. (12). Yet, it is a noisy signal because individual agents cannot directly observe the amount of aggregate noise trading $n_1^\star$.

[10] For completeness, the exchange rate appreciation also lowers the price of tradable goods, leading to higher imports and borrowing. The ensuing households' demand for more domestic bonds alters the position of financiers and attenuates the equilibrium appreciation of the exchange rate.

These assumptions capture the idea that economic agents, for various reasons, do not perfectly know the state of the economy but use easily accessible information, such as exchange rates, to improve their inference.

In addition, agents may also be able to observe a second endogenous public signal namely the quantity of FX interventions, $f_1^\star$, depending on the communication policy adopted by the central bank. Irrespective of such communication policy, the public information available to every island $i$ can be summarized by a sufficient statistic

$$z|a_1 \sim \mathcal{N}(\mu_z, \beta_z^{-1}),$$

whose form will be detailed in each intervention scenario analyzed below.

**Distorted beliefs** We posit that agents do not form beliefs rationally but instead overreact to news. Expectation overreaction has been widely documented in survey data (Bordalo et al., 2020), analyst forecasts (Bordalo et al., 2019, 2024), market reaction to economic news (Bianchi et al., 2025), and laboratory experiments (Afrouzi et al., 2023; Ba et al., 2024), while Candian and De Leo (2023) demonstrate that such bias helps explain key features of exchange rate dynamics.[11] In Section 2.3, we present new evidence on the relationship between macro forecast errors and exchange rate movements, which further supports the notion of overreaction in our context. Specifically, we make the following two assumptions.

*E1: Diagnosticity with respect to exogenous signals.* Agent $i$'s posterior belief upon observing a private signal $v^i$ is given by:

$$E[a_1|v^i] = \mathbb{E}[a_1|v^i] + \delta\left(\mathbb{E}[a_1|v^i] - \mathbb{E}[a_1]\right), \tag{14}$$

where $E$ denotes the subjective belief, $\mathbb{E}$ denotes the rational expectation operator and $\mathbb{E}[a_1]$ is the common prior mean (equal to zero here). While $\delta \neq 0$ encodes a general departure from rationality, we will focus on the case $\delta > 0$, which implies overreaction to private signals (relative to rational expectations). This case corresponds to the Diagnostic Expectations (DE) model of Gennaioli and Shleifer (2010) and Bordalo et al. (2018), in which agents overweight outcomes that are easily recalled or deemed 'representative'—in this case news—a bias rooted in selective memory retrieval (Kahneman

[11] Benhima and Bolliger (2022), using forecast data from *Consensus Economics* across multiple countries find substantial heterogeneity in the degree of under- and overreaction across countries.

and Tversky, 1972).[12] This model relies on psychological foundations and it has been widely adopted in the macroeconomic literature (Bordalo et al., 2021; Bianchi et al., 2024a,b; Maxted, 2024; L'Huillier et al., 2024; Krishnamurthy and Li, 2025).

An important novel feature of our setup is that it embeds belief biases in a model with endogenous signals such as exchange rates. This requires specifying what agents know about the belief distortions of others, which we now turn to.

*E2: Misperception about others.* We assume that agents are unaware not only of their own diagnosticity bias, but also of the bias of other agents in the economy. In other words, they perceive themselves and every other agent as rational, *i.e.*, as forming expectations under Bayes' rule with $\delta = 0$. We introduce the "$\stackrel{p}{=}$" notation to denote objects as *perceived* by agents in this economy. We thus assume:

$$E[a_1|\mathcal{I}^j] \stackrel{p}{=} \mathbb{E}[a_1|\mathcal{I}^j], \tag{15}$$

where $\mathcal{I}^j$ is agent's $j$ information set. This means that every agent in the economy perceives agent $j$ as forming expectations about productivity rationally. The misperception leads agents to interpret endogenous public signals as reflecting rational expectations rather than actual beliefs.

Combining the information of the private signal $v^i$ with the prior and and the public signal $z$, we can write individual agents' posterior beliefs as:

$$E_0^i a_1 \equiv E[a_1|v^i, z] = \frac{(1+\delta)\beta_v v^i + \beta_z z}{\beta_v + \beta_z + \beta_a} \tag{16}$$

Above, $\beta_v + \beta_z + \beta_a$ is the overall posterior accuracy. The term $\frac{\beta_v}{\beta_v+\beta_z+\beta_a}$ is the rational weight on the private signal $v^i$, which is multiplied by $(1+\delta)$ because of assumption E1 and zero common prior mean. Finally, $\frac{\beta_z}{\beta_v+\beta_z+\beta_a}$ is the weight on the public signal, $z$. As we explain in the next section, because of assumption E2, agents misinterpret the informational content of $z$.

We conclude by noting that our beliefs specification coincides with rational expectations in the special case of $\delta = 0$. In this case, agents are neither diagnostic with

[12] In our setting, agents are diagnostic with respect to the news contained in their private signals $v^i$: the probability assessment of $a_1$ after observing $v^i$ is $h^\delta(a_1|v^i) \propto h(a_1|v^i)\left[h(a_1|v^i)/h(a_1|v^i=0)\right]^\delta$, where $h(a_1|v^i)$ is the true Bayesian posterior distribution of $a_1$ after observing the signal $v^i$, and $h(a_1|v^i=0)$ is the posterior if the signal realization were to equal the prior mean, which in this case is zero. The DE framework, combined with normality of shocks, implies that the subjective posterior variance equals the rational one, while the subjective posterior mean equals eq. (14).

respect to their private signals nor do they misperceive others' beliefs.

# 2 Learning and mislearning from the exchange rate

A central feature of our model is that agents learn about the economy's fundamentals from exchange rate fluctuations. In turn, learning affects their consumption and investment decisions, which feed back into the exchange rate via foreign exchange market clearing as described in Section 1.2. To examine this general equilibrium feedback loop in the cleanest way and to show that this mechanism exists independently of exchange rate policies, in this section we abstract from interventions by setting $f_1^\star = 0$. We refer to this economy as the *laissez-faire* economy and use a superscript $l$ where useful.

## 2.1 Perceived and actual exchange rate process

In the laissez-faire economy the only source of public information is the exchange rate. Because agents fail to realize that others overreact to their private information, they perceive the exchange rate and any other public signals to follow a different process than the equilibrium one.

Accordingly, we begin with a guess for the *perceived* exchange rate process:

$$q_0 \overset{p}{=} \lambda_a^l a_1 + \lambda_n^l n_1^\star. \tag{17}$$

Given this perceived process, individuals form posterior beliefs in (16) using the signal:

$$z^l | a_1 \equiv \left. \frac{q_0}{\lambda_a^l} \right| a_1 \overset{p}{\sim} \mathcal{N}(a_1, ({}^{\lambda_n^l}/_{\lambda_a^l})^2 \beta_n^{-1}) \tag{18}$$

where $\overset{p}{\sim}$ introduces the distribution that agents perceive the signal to follow. In this formulation, $z^l$ represents a signal that is perceived to be centered around future productivity $a_1$ with a error variance of $\beta_z^{-1} \equiv ({}^{\lambda_n^l}/_{\lambda_a^l})^2 \beta_n^{-1}$.

Next, because agents are unaware of others' belief distortion (E2), they perceive the market clearing condition (12) to be

$$q_0 \overset{p}{=} \frac{\Gamma \omega_1}{\Gamma \tilde{\theta} \omega_1 + \omega_3} n_1^\star - \frac{\omega_2}{\Gamma \tilde{\theta} \omega_1 + \omega_3} \bar{\mathbb{E}}[a_1 | v^i, z^l], \tag{19}$$

where $\bar{\mathbb{E}}[a_1|v^i, z^l] \equiv \int^i \mathbb{E}^i[a_1|v^i, z^l] \, \mathrm{d}i = \frac{\beta_v a_1 + \beta_z z^l}{\beta_v + \beta_z + \beta_a}$ denotes the average of rational (instead of actual) expectations. Matching coefficients in equations (17) and (19) gives the solution $(\lambda_a, \lambda_n)$ to the *perceived* exchange rate process. One can now use these equilibrium beliefs to solve for the *actual* exchange rate process. This can be done by plugging actual average posterior beliefs, $\int_0^1 E^i[a_1|v^i, z^l]di = \frac{(1+\delta)\beta_v a_1 + \beta_z z^l}{\beta_v+\beta_z+\beta_a}$, into the actual market clearing condition (12). We collect these results in the next proposition.

**Proposition 1.** *Under the assumed information structure, there exists a unique symmetric linear equilibrium. In equilibrium, the exchange rate is perceived to follow:*

$$q_0 \overset{p}{=} \underbrace{-\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v+\beta_z}{\beta_a+\beta_v+\beta_z}}_{\lambda_a^l} a_1 + \underbrace{\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v+\beta_z}{\beta_v}}_{\lambda_n^l} n_1^\star, \tag{20}$$

*with $\beta_z \equiv \Lambda_l^2\beta_n$ and a unique $\Lambda_l^2 \equiv \left(\frac{\lambda_a^l}{\lambda_n^l}\right)^2$ implicitly defined by:*

$$\Lambda_l^2 = \left(\frac{\omega_2}{\Gamma\omega_1}\right)^2 \frac{\beta_v^2}{(\beta_a+\beta_v+\Lambda_l^2\beta_n)^2}. \tag{21}$$

*Instead, the actual exchange rate process is given by:*

$$q_0 = (1+\delta)\lambda_a^l a_1 + \lambda_n^l n_1^\star. \tag{22}$$

*Proof.* See Online Appendix D. □

The ratio $\Lambda_l = \lambda_a/\lambda_n$ is central to our analysis and governs the perceived endogenous informativeness of the exchange rate about the aggregate productivity: when it is large, the exchange rate is perceived to be very informative about productivity.

The unique equilibrium value of $\Lambda_l$ is pinned down by equation (21), which states that the informativeness of the exchange rate has to be consistent with how much weight agents place on their private information. Models with endogenous price signals are often prone to multiple equilibria due to the complementarities in information weighting that learning from prices may introduce. These complementarities typically arise in the presence of endogenous public and private signals, such as in Amador and Weill (2010), Angeletos et al. (2016), Kohlhas (2020b). In our setting these complementarities do not arise because the precision of private signals is not affected by observed aggregate prices such as exchange rate.[13] This is a deliberate modeling choice which allows us to

[13]In our model, when every other agents put less weight on their private signal, the exchange rate

isolate the welfare effects of public information due to new ingredients, such as beliefs overreactions, rather than to well-known strategic complementarities. A convenient implication of abstracting from complementarities is that we do not have to select a specific equilibrium in our welfare analysis because the equilibrium is unique.

**The role of endogenous overreaction** A comparison of the perceived and actual exchange rate processes highlights the implications of misperception about others' beliefs. From eq. (22), consider the actual, rather than perceived, properties of public signal that agents use to form their beliefs

$$z^l|a_1 \equiv \frac{q_0}{\lambda_a^l}\bigg|a_1 \sim \mathcal{N}((1+\delta)a_1, (\lambda_n^l/\lambda_a^l)^2\beta_n^{-1}) \tag{23}$$

where $\sim$ introduces the true distribution of the signal. If $\delta = 0$, beliefs are rational and, therefore, agents correctly interpret the public signal, *i.e.*, eq. (18) coincides with eq. (23). However, $\delta > 0$ leads to a misinterpretation of $z^l$.

First, agents fail to internalize that they are using a biased public signal. As agents overreact to their own *private* signal, the average belief loads more on the average private signal (equal to $a_1$) by a factor of $1+\delta$. Since agents are unaware of others' overreaction, a gap emerges between perceived and actual average beliefs, and therefore between the perceived and actual exchange rate process.[14] As a result, for any given movement in the exchange rate, agents mistakenly attribute a larger-than-rational fraction of that movement to fluctuations in underlying productivity.[15] The end result is that agents endogenously overreact to the information about productivity contained in the exchange rate.[16]

Second, the perceived accuracy of the public signal equals the actual accuracy: while agents misperceive the signal's mean, they correctly perceive its accuracy—a consequence of the particular structure of the diagnosticity bias (Bordalo et al., 2020).[17]

becomes relatively less informative and, as a result, a given agent wants to put more weight on his private signal, *i.e.* weighting decisions are strategic substitutes.

[14]The gap between perceived and actual average beliefs is $\bar{E}[a_1|v^i, z^l] - \bar{\mathbb{E}}[a_1|v^i, z^l] = {}^{\delta\beta_v a_1}/_{(\beta_v+\beta_z+\beta_a)}$.

[15]An implication is that agents systematically underestimate the response of others, similar to Angeletos and Sastry (2020).

[16]While we find misperception about others conceptually appealing, Online Appendix B shows that similar results obtain if diagnosticity applies to both private and public signals without higher-order belief bias. Moreover, we show that if agents use private signals optimally but fail to realize other agents learn from prices, as in Bastianello and Fontanier (2024), they also overreact to endogenous signals.

[17]A belief bias on the perceived accuracy of private signals, *e.g.* overconfidence, would affect also the

To sum up, since agents are unaware that the other agents in the economy overreact to their private information, they underestimate the covariance between exchange rate and productivity, and thus overreact to its informational content.

## 2.2 Informational role of the exchange rate

By aggregating information about productivity, the exchange rate in our model plays a novel important role in agents' expectations formation and decisions. The strength of this informational role depends on the weight agents place on the exchange rate as a signal: $\frac{\Lambda^2\beta_n}{\beta_a+\beta_v+\Lambda^2\beta_n}$. As the expression highlights, this weight hinges on how informative the equilibrium exchange rate is relative to private signals and to the common prior.

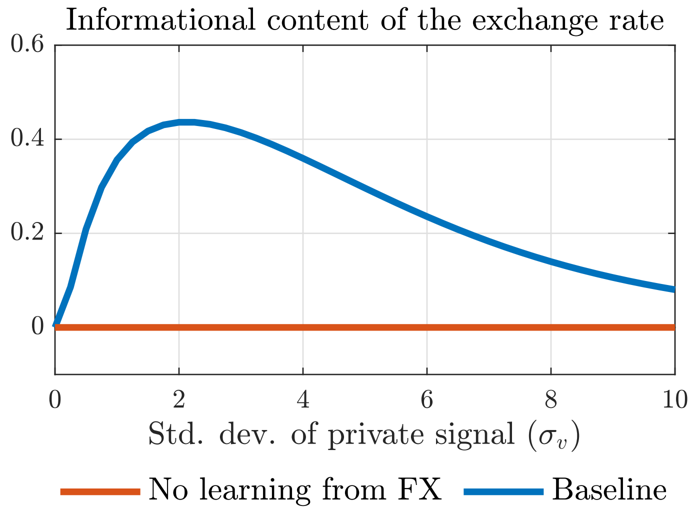
Figure 1 depicts the informational content of the exchange rate for different degrees of dispersion of private information under an illustrative calibration of the model with $\delta = 0$. When private information in infinitely precise ($\sigma_v^2 \equiv \beta_v^{-1} = 0$), the exchange rate does not carry any additional information, and agents therefore disregard it. At the other extreme, when private information is infinitely noisy ($\sigma_v^2 = \infty$), agents place no weight on their private signals. Thus, the exchange rate does not aggregate private information and also receives a zero weight in agents' posterior beliefs. For intermediate values of dispersion in private signals, the informational content is positive as the exchange rate meaningfully aggregates private information about future productivity.

By directly influencing agents' expectations, the exchange rate affects macroeconomic outcomes through a novel channel. To see this, examine the effects of an increase in noise-trading demand for home-currency bonds when agents learn from the exchange rate (Figure 2b, blue line). In each island, agents observe an exchange rate appreciation, but do not know whether it stems from an expected improvement in future productivity ($a_1 \uparrow$) or from higher noise-trading demand for home currency ($n_1^\star \downarrow$). They confound the effect of the noise-trading shock on the exchange rate with the effect of higher future productivity, thereby revising their beliefs about future productivity upward. Higher expected productivity leads households to increase their consumption and external borrowing, relative to the case without learning from the exchange rate (red line), which further appreciates the exchange rate.[18] The effect is analogous to an exogenous productivity news shock, but is due to the endogenous response of the

perceived accuracy of the endogenous public signal, as in Broer and Kohlhas (2022).

[18]The rational confusion between noise trading and expected productivity improvements amplifies the equilibrium effects of noise-trading shocks on the exchange rate, as originally highlighted in Bacchetta and Wincoop (2006), and on macro aggregates, as we emphasize using our fully-specified macro model.

Figure 1: Informational content of exchange rate under dispersed information



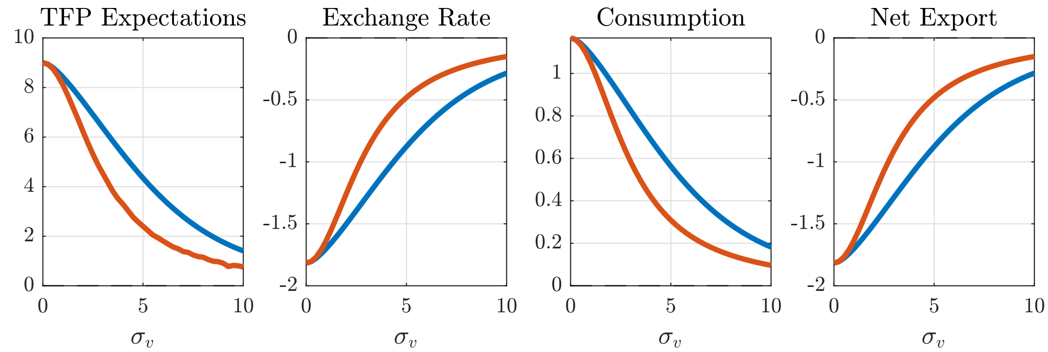

*Notes:* This figure reports the equilibrium value of the informational content of the exchange rate $\frac{\beta_z}{\beta_a+\beta_v+\beta_z}$ for different levels of the noise in private signal, $\sigma_v$, under laissez faire, and without belief distortions ($\delta = 0$). The rest of parameters are set according to Table G.1 (see Online Appendix).

exchange rate to higher home-currency demand from noise traders. This channel is in place every time agents use the exchange rate as a public signal, including in response to an increase in future productivity (see Figure 2a).

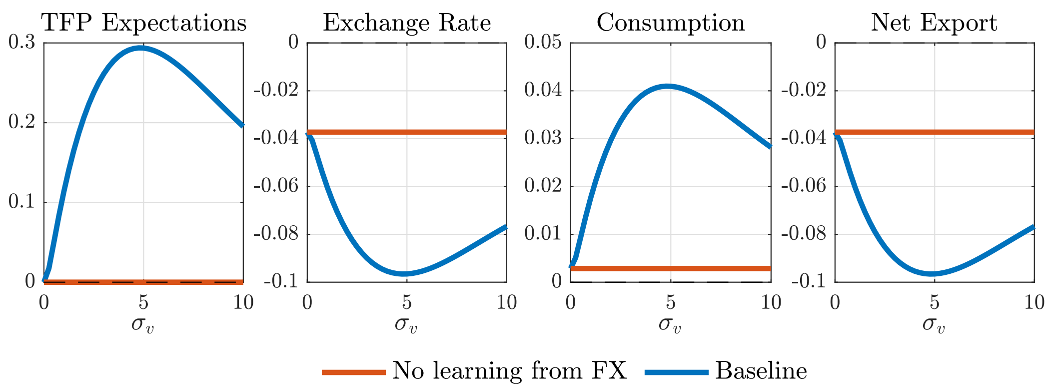
We make two related observations. First, the informational channel of the exchange rate just described does not rely on expectations' overreaction, as it operates even under rational expectations (indeed, $\delta = 0$ in Figure 2). Yet, expectations overreaction changes the quantitative role of the informational channel. When $\delta > 0$, the exchange rate is a *biased* public signal, resulting in amplified responses to productivity shocks (as depicted in Figure G.1 in the Online Appendix, where $\delta = 0.5$). This feature will turn out to be a key source of inefficiency that the policymaker considers when optimally choosing FX policies. Second, learning from the exchange rate blurs the relationship between productivity *expectations* and their subsequent *realizations*: because noise trading affects the exchange rate by altering the balance sheet of financiers, it also influences agents' expectations of productivity. This aligns with Chahrour et al.'s (2024) evidence that a substantial portion of the exchange rate variation can be attributed to both correctly anticipated changes in productivity and expectational "noise," which influences expectations of productivity but not the actual realization.

Figure 2: Equilibrium responses to shocks under dispersed information

(a) Response to one-standard deviation productivity shock ($a_1 \uparrow$)




(b) Response to one-standard deviation noise-trading shock ($n_1^\star \downarrow$)




*Notes:* This figure reports the equilibrium response of model variables to productivity and noise-trading shocks for different levels of the noise in private signal, $\sigma_v$, under laissez faire, and without belief distortions ($\delta = 0$). The rest of parameters are set according to Table G.1 (see Online Appendix).

## 2.3 Macro expectations and exchange rates: survey evidence

This section documents that the properties of expectations implied by the information structure in the model align with those of survey forecasts of a panel of small open economies. Forecasts are from *Consensus Economics*, which is a monthly survey of professional economic forecasters. From a sample of 20 small-open economies, we consider individual forecasts about next-year real GDP growth from 1998 to 2019 depending on data availability. We focus on countries/periods in which the exchange rate regimes is either "freely floating" or "managed floating around a moving band," according to the classification of Ilzetzki et al.'s (2019). Online Appendix F describes data sources and treatment in details.

Our model predicts that an exchange rate appreciation signals an improvement in expected productivity growth. In the absence of direct survey forecasts for productivity across countries, we use GDP growth forecasts as a proxy. This choice is supported by the fact that, under a broad range of parameterizations in our model, expected productivity growth and expected GDP growth are positively related.[19]

First, we investigate the relationship between GDP growth forecasts and exchange rate changes by estimating the following regressions

$$F^i_{m,t}[\Delta y_{j,t+1,t}] = \alpha_{i,j} + \beta_j \Delta s_{j,m,m-12} + \varepsilon^i_{j,m,t}, \tag{24}$$

where $\Delta y_{j,t+1,t}$ is the annual GDP growth rate in country $j$ from year $t$ to year $t+1$, $F^i_{m,t}[\Delta y_{j,t+1,t}]$ is the forecast provided in month $m$ of year $t$ by forecaster $i$ and $\Delta s_{j,m,m-12}$ is the 12-month percentage change of the country $j$ exchange rate (home currency price of one U.S. dollar). For each country, we include the forecaster fixed effect $\alpha_{i,j}$. The calendar-year structure of the survey implies that forecasts in every month within a year refer to the same horizon. To limit repetitions of forecasts about the same horizon, we consider only the first 6 months of each year. Figure 3a shows that, for a large majority of countries in our sample, exchange rate appreciations are associated with an increase in expected real GDP growth, which is consistent with the model's equilibrium exchange rate equation (12).

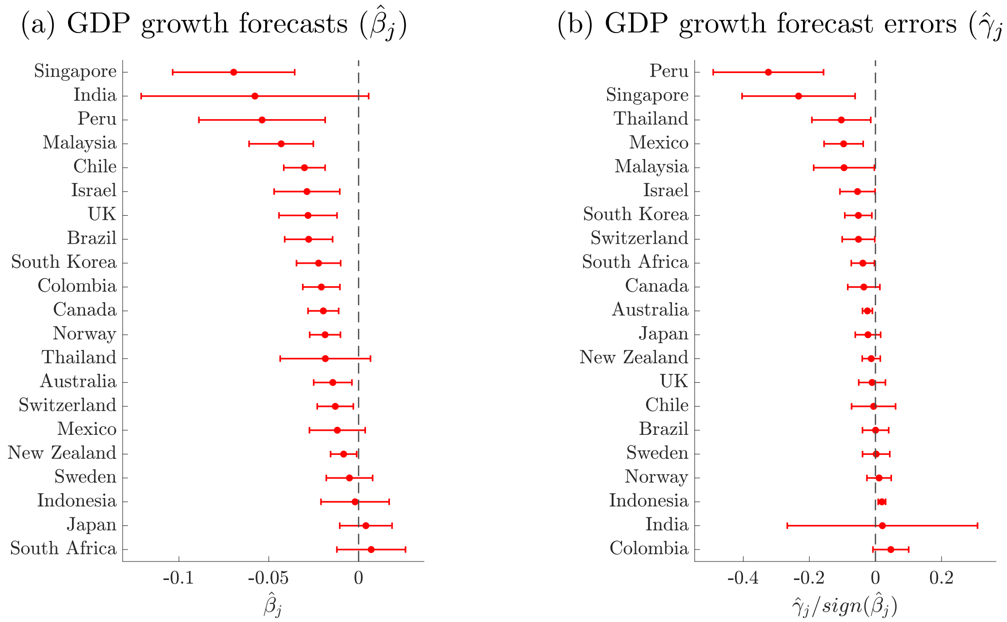
Second, we test whether macroeconomic expectations are excessively elevated when the exchange rate has recently appreciated. To do so, for each country, we implement the following regression of GDP growth forecast errors on exchange rate changes:

$$\Delta y_{j,t+1,t} - F^i_{m,t}[\Delta y_{j,t+1,t}] = \alpha_{i,j} + \gamma_j \Delta s_{j,m,m-12} + \varepsilon^i_{j,m,t}, \tag{25}$$

Figure 3b reports the estimated $\gamma_j$, normalized by the sign of the coefficient $\beta_j$ in regression (24). A coefficient $\gamma_j < 0$ implies that macro expectations are overly optimistic when the exchange rate has appreciated over the past year, which would be consistent with a mechanism of expectations' overreaction. The Figure shows that the coefficient $\gamma_j$ in regression (25) is negative and significant at the 10% level for about half of the countries in the sample, including, e.g., South Korea, Thailand, New Zealand, Mexico, and Israel. This evidence supports our choice of $\delta > 0$.

[19] The endogenous component of GDP in our model is production of tradable goods: $Y^H_{T,1} = A_1 K_1^\alpha$. Therefore, an improvement in expected productivity maps into an improvement in expected GDP, provided that wealth effects on capital accumulation are not too strong.

Figure 3: Real GDP growth forecasts and exchange rate changes




*Notes:* Panel (a) reports the estimated $\hat{\beta}_j$ from regression (24) for each country $j$ (the expected yearly real GDP growth rate, in %, in the following year associated with a 1% 12-month appreciation in the local currency/USD exchange rate). Panel (b) reports the estimated $\hat{\gamma}_j$ from regression (25) for each country $j$ normalized by the sign of the estimated $\hat{\beta}_j$ from regression (24), i.e. $\hat{\gamma}_j/\,\text{sign}(\hat{\beta}_j)$. Bars display the 90% confidence interval and Driscoll and Kraay (1998) standard errors. Data is monthly from 1998-2019, with observation varying depending on the country (See Online Appendix F).

**Alternative specifications** Online Appendix F contains various robustness tests. First, instead of using forecasts for the first 6 months of every calendar year, we use all months. Second, we extend the sample of countries to also include "crawling band" exchange rate regimes. Third, we conduct the baseline analysis with GDP growth data from the *IMF International Financial Statistics*, instead of first-release data from the annual *IMF World Economic Outlook*. The results, reported in Figure F.1 in the Online Appendix, are similar across these different specifications. Fourth, a possible concern with GDP growth forecasts is their "calendar year" structure, where in every month of a given year the forecasts horizon is fixed to the following year. While this structure is common to all macro variables in *Consensus Economics*, interest rates represent an exception. Figure F.2 in the Online Appendix presents the analysis using forecasts of the 12-month-ahead short-term interest rate. The results are qualitatively similar to the baseline findings.

### 2.3.1 Examples of informational role of FX in policy and media

To conclude this section, we provide three recent examples in which the informational role of the exchange rate is explicitly discussed by policy makers or commentators. During a recent meeting of African central bank governors, IMF African Department Director Abebe Aemro Selassie highlights that "The exchange rate in most developing countries is the most visible and important price in the economy and so helps to anchor expectations, facilitate planning, as well as investment, and consumption decisions." (Selassie, 2023). This observation aligns with survey evidence that the nominal exchange rate receives substantial attention by firms in emerging economies (Delgado et al., 2024). Recently, after the Euro depreciated following a increase in the ECB policy rate, *Financial Times* Markets Editor Katie Martin writes that "The euro's latest wobble also forms yet another big signal that investors think Europe's luck has run out." (Martin, 2023). In a recent report to the Bank of International Settlements, economists at the Central Bank of Korea verbally describe how they think depreciation pressure on the Korean Won transmit through financial markets, and highlight that "The depreciation of the won in turn sends negative signals about the Korean economy and can make banks' FX funding more difficult again." (Ryoo et al., 2013).

While anecdotal, these examples reveal that the informational role of the exchange rate we formalized is often taken into consideration in the media and policy circles.[20]

# 3 Foreign exchange interventions

We now introduce the possibility for the central bank to intervene in the foreign exchange market by purchasing foreign-currency bonds $f_1^\star$. We assume the central bank is able to observe aggregate financial flows, such as $n_1^\star$, and the exchange rate, $q_0$. Indeed many monetary authorities serve supervisory and regulatory roles over local financial institutions, providing them with information about aggregate patterns of

[20] Gholampour and van Wincoop (2019) use Twitter data to compute a measure of investors' private information about the fundamentals driving the Euro-Dollar exchange rate, and use it in a structural estimation of the dispersed information model of Bacchetta and Wincoop (2006). One of their main findings is that Twitter data imply a sizable degree of dispersion in private information.

bond holdings.[21] The central bank thus intervenes according to:

$$f_1^\star = \kappa_q q_0 + \kappa_w n_1^\star. \tag{26}$$

We assume throughout that the intervention form $(\kappa_q, \kappa_w)$ is known to agents while the actual quantity of foreign-currency bonds $f_1^\star$ purchased by the central bank may or may not be known, as we detail below.

Note that central bank's rule (26) is not restrictive, as it implicitly responds to all shocks in the economy. Indeed, by reacting to financial flows and the exchange rate, the central bank is effectively responding to the private sector's average expectations $\bar{E}_0[a_1]$. In fact, one can use eq. (12) to express eq. (26) as follows:

$$f_1^\star = -\frac{\omega_2 \kappa_q}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\bar{E}_0 a_1 + \frac{\Gamma\omega_1\kappa_q + \Gamma\tilde{\theta}\omega_1\kappa_w + \omega_3\kappa_w}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q} n_1^\star. \tag{27}$$

We also emphasize that our positive and normative results, outlined below, do not depend on the assumption that the central bank perfectly observes noise trading. Online Appendix E analyzes a setting in which the central bank observes only a noisy signal about $n_1^\star$. Under this more general information structure, all the qualitative conclusions of the baseline model remain unchanged.

In our model, FX interventions are allocative for two reasons. As in standard portfolio-balance models, interventions can affect the exchange rate, because they alter the balance-sheet position of financiers. The novelty of our model is that interventions may alter the information available to agents.

We consider two types of FX intervention communication policy: publicly announced and secretly conducted FX interventions. We define them below.

**Public FX intervention.** The central bank communicates the size of the intervention to the public, and thus $f_1^\star$ becomes common knowledge.

**Secret FX intervention.** The central bank does not reveal the size of the intervention to the public, and thus $f_1^\star$ is unobserved.

[21] In many emerging economies, in addition, monetary authorities mandate that they report details of their FX and bond transactions, such as counterparties and amounts, for regulatory purposes.

## 3.1 Public foreign exchange interventions

When the central bank communicates the volume of foreign bond purchase $f_1^\star$, the intervention becomes a public signal about average expectations, as evident from eq. (27). Agents can thus access two public signals, the exchange rate and the FX intervention, which contain *independent* information about the same two components, $\bar{E}_0[a_1]$ and $n_1^\star$. As a result, agents are able to perfectly back out average expectations, $\bar{E}_0[a_1]$. The sufficient statistic for all the public information about $\bar{E}_0[a_1]$ under public FX interventions can be formally defined as $z^p$, which follows:

$$z^p|a_1 \sim \mathcal{N}((1+\delta)a_1, \beta_z^{-1}), \qquad \text{with} \qquad \beta_z = \infty \tag{28}$$

If agents are rational ($\delta = 0$), they correctly understand the expectations mapping and therefore can perfectly back out productivity from average expectations. If $\delta > 0$, average beliefs exhibit overreaction ($\bar{E}_0[a_1] = (1+\delta)a_1$) and agent misinterpret the public signal by $\delta a_1$. Either way, under public communication, the intervention has a *signaling* effect that expands agents' information set by revealing average expectations.

**Proposition 2.** *(Public FX Interventions) Suppose the central bank adopts FX intervention rule (26) with $\kappa_w \neq 0$, where $f_1^\star$ is directly observed. Then, the exchange rate follows*

$$q_0 = \lambda_a^p(1+\delta)a_1 + \lambda_n^p n_1^\star,$$

*where:*

$$\lambda_a^p = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}; \qquad \lambda_n^p = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}(1+\kappa_w).$$

*The combined informational content of FX interventions and the exchange rate, summarized by $z^p$, perfectly reveals the average expectations $\bar{E}_0[a_1]$, and central bank's reaction function coefficients, $\kappa_q$ and $\kappa_w$, do not affect the accuracy of public information.*

*Proof.* See Online Appendix D. ☐

By announcing the quantity of bonds purchased, the central bank implicitly discloses its information set, which, together with the observed exchange rate, fully reveals average expectations. Notably, private agents can perfectly infer the central bank's knowledge of noise trading flows as long as $\kappa_w \neq 0$—whether such knowledge is precise, as in the baseline case, or imprecise, as in the extension in Online Appendix E.[22] In other words, the actual values of policy coefficients in eq. (26) influence equilibrium exchange rates and overall allocation only through a portfolio-balance channel. A key exception is exchange rate smoothing—the special case when $\kappa_w = 0$ in eq. (26) and thus $f_1^\star = \kappa_q q_0$. In this case, $f_1^\star$ perfectly comoves with the other public signal, $q_0$, and thus it does not convey additional, independent information. Analogously, an alternative policy of purely exogenous FX interventions, $f_1^\star = \varepsilon_f$ with $\varepsilon_f \sim \mathcal{N}(0, \sigma_f^2)$, would also not convey any information.

The result that public interventions fully reveal average expectations follows from the central bank perfectly observing and responding to noise trading. This simplifies the signal extraction problem and clarifies the exposition. In Online Appendix E, where the central bank instead reacts to a noisy signal of noise trading, public interventions no longer fully reveal average expectations but still enhance agents' forecasts. Generally, any public intervention based on information available to the central bank—but not to private agents—effectively transmits that information to the public, thereby expanding its information set. Our results do not hinge on full information revelation, but rather on the fact that public interventions increase private agents' information.

## 3.2 Secret foreign exchange interventions

Now consider the case in which the central bank does *not* reveal the size of FX interventions.[23] Substituting the central bank's reaction function (26) in the exchange rate equation (12), one obtains

$$q_0 = \left(1 - \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\kappa_q\right)^{-1} \left[\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}(1+\kappa_w)n_1^\star - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\bar{E}_0 a_1\right] \tag{29}$$

Equation (29) already suggests that, when the exchange rate is the only source of public information, secret FX interventions can alter the informational content of the exchange rate by changing the equilibrium relationship between the exchange rate and the two shocks. We formally prove this result in the following proposition.

[22] In the baseline, this follows from rearranging equation (26) as $(f_1^\star - \kappa_q q_0)/\kappa_w = n_1^\star$. A similar expression obtains when the central bank observes noise trading imperfectly (see Online Appendix E).

[23] Inferring the size of interventions from publicly available data is hard. Central banks publish data about international reserves only infrequently, and factors such as valuation effects and investment income flows make international reserves a poor proxy of FX interventions (Adler et al., 2021). We also note that central banks maintain secrecy around interventions by imposing a confidentiality requirement on counterparties.

**Proposition 3.** *(Secret FX Interventions) Suppose the central bank adopts FX intervention rule (26), where $f_1^\star$ is not observed. Then, the exchange rate follows:*

$$q_0 = \lambda_a^s (1+\delta) a_1 + \lambda_n^s n_1^\star, \tag{30}$$

*where*

$$\lambda_a^s = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\left(\frac{\beta_v+\beta_z}{\beta_v+\beta_z+\beta_a}\right); \quad \lambda_n^s = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\left(\frac{\beta_v+\beta_z}{\beta_v}\right)(1+\kappa_w);$$

*and $\beta_z \equiv \Lambda_s^2 \beta_n$, and a unique $\Lambda_s \equiv (\lambda_a^s/\lambda_n^s)$ implicitly defined by*

$$\Lambda_s^3 + \frac{\beta_v + \beta_a}{\beta_n}\Lambda_s + \frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{\beta_n} = 0 \tag{31}$$

*Under secret FX interventions, the relative informational content of the exchange rate and the overall posterior accuracy depend on the intervention rule via eqs. (30)-(31). A rule that strengthens the correlation between the exchange rate and productivity or weakens its correlation with noise-trading increases the exchange rate informativeness.*

*Proof.* See Online Appendix D. □

Proposition 3 frames our key positive result. Like in the public intervention case, FX interventions affect the equilibrium relationship between exchange rate and underlying shocks. Differently from the public communication case, however, FX interventions are unobserved and thus alter the informational content of the exchange rate.

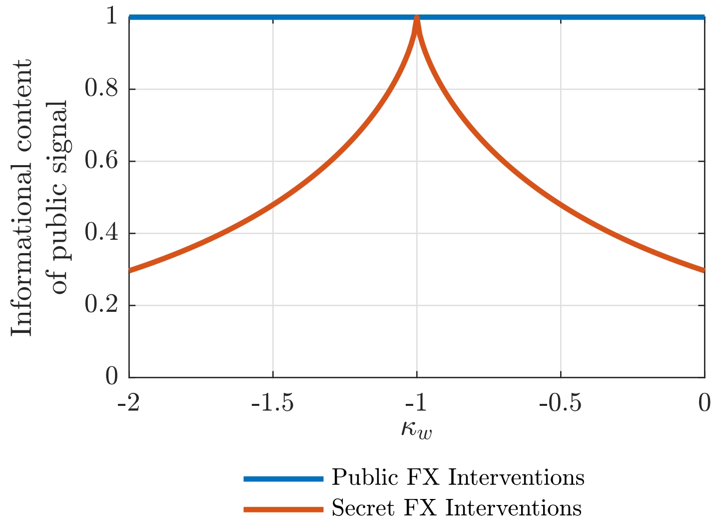
Formally, now the public signal sufficient statistics, defined as $z^s$, is only comprised of exchange rate information, and follows:

$$z^s | a_1 \equiv \left.\frac{q_0}{\lambda_a^s}\right| a_1 \sim \mathcal{N}((1+\delta)a_1, \beta_z^{-1}) \tag{32}$$

where $\lambda_a^s$, $\lambda_n^s$ and $\beta_z = \Lambda_s^2 \beta_n$ follow from equations (30)-(31). Differently from the public communication case, the precision of this public signal depends on FX intervention policy via (30)-(31).

When the exchange rate is more closely tied to future productivity rather than to noise trading, it becomes a more reliable signal of future productivity. By appropriately choosing the FX intervention policy coefficients—known to market participants—the central bank can influence the equilibrium relationship between the exchange rate and productivity, thereby shaping its informational content. More specifically, aside from

Figure 4: FX intervention and the informational context of the public signal




*Notes:* The figure reports values of the informational content of the public signal ($\frac{\beta_z}{\beta_a+\beta_v+\beta_z}$) for different values of the central bank's reaction function coefficient $\kappa_w$ under both public and secret intervention policy under rational expectations (that is, $\delta = 0$). The rest of the parameters are set according to Table G.1 (see Online Appendix).

the special case of an exchange rate peg ($\kappa_q = -\infty$), where the exchange rate becomes entirely uninformative, the informational content of the exchange rate depends solely on the extent to which the central bank responds to noise trading, as governed by $\kappa_w$ in equation (26). This is because any finite choice of $\kappa_q$, while affecting the equilibrium allocation through portfolio balance channels, does not influence the relative contribution of productivity versus noise trading to exchange rate fluctuations—captured by the ratio $\lambda_a^s/\lambda_n^s$ in equation (30). This ratio is the key determinant of the informational content of the exchange rate.

Figure 4 displays the equilibrium informational content of the exchange rate for different coefficients $\kappa_w$ under an illustrative calibration of the model. It is useful to highlight a number of interesting cases. First, secret intervention policy can make the exchange rate perfectly informative of average expectations if it completely offsets the noise trading variation in the exchange rate ($\kappa_w = -1$). Second, secret interventions can lead to an equilibrium in which the exchange rate is uninformative, by effectively eliminating the effect of productivity shocks on the exchange rate. This can be achieved either by setting $|\kappa_w| = \infty$, or by an *exchange rate peg* as discussed above. Outside of these two limit cases, there is a spectrum of cases in which the exchange rate is an *imperfect* signal of future productivity, achieved with appropriate choice of reaction function coefficients.[24] Note that *freely floating* (*i.e.*, laissez-faire) is a particular case of imperfectly informative exchange rate with $\kappa_q = \kappa_w = 0$. Under free floating, exchange rate fluctuations reflect both productivity and noise-trading, and thus their informational content is limited by the relative amount of noise-trading in exchange rate fluctuations.

In summary, we emphasize that secret FX interventions uniquely enable a central bank to "manage" the informativeness of the exchange rate. Next, we examine whether the central bank may find it preferable to intervene publicly or secretly.

# 4 Welfare and optimal policies

We now turn to the normative implications of FX interventions when agents learn from the exchange rate. Section 4.1 introduces the welfare function and describes the sources of inefficiencies relative to the frictionless allocation. Section 4.2 outlines the policy problem faced by central banks in their choice of FX intervention size and communication, and highlights the trade-offs involved in implementing optimal policies.

## 4.1 Welfare and sources of inefficiency

To characterize optimal policy we need a welfare criterion. Let the welfare of the small open economy be defined as the household's expected utility (1) under the objective (*i.e.*, rational) expectations.[25,26] We evaluate welfare by taking a quadratic approximation of the welfare function around the first-best allocation, that is the allocation that would arise under full-information rational expectations ($\beta_v = \infty$ and $\delta = 0$) absent any international asset market segmentation ($\Gamma = 0$).

[24] Hassan et al. (2022) explore how exchange rate policy influences the riskiness of that country's currency, by altering the stochastic properties of the exchange rate, and derive the implications for optimal exchange rate policy. We also emphasize that FX policy affects the macroeconomic allocation by altering the stochastic properties of the exchange rate, yet through a distinct, complementary channel: the informativeness of the exchange rate.

[25] Island-level expected utility (1) coincides with aggregate expected utility since islands are ex-ante identical.

[26] This choice follows Gabaix (2020) and most of the behavioral literature, which thinks of behavioral agents as using heuristics but experiencing utility from consumption like rational agents.

**Welfare function** *Welfare in deviation from the first best, $\widetilde{\mathbb{W}}$, can be written as:*

$$\widetilde{\mathbb{W}} = -\left[\Omega_1 \operatorname{var}(q_0^i - \bar{q}_0) + \Omega_2 \operatorname{var}(E_0^i a_1 - a_1)\right] \tag{33}$$

*where $\Omega_1 > 0$ and $\Omega_2 \geq 0$ are convolutions of parameters independent of policy, reported in Online Appendix C.*

Equation (33) reveals that there are two margins that are distorted in our small open economy. First, intermediation frictions induce inefficient fluctuations in external borrowing (as, *e.g.*, in Itskhoki and Mukhin, 2022), reflected in deviations of the island-level relative price of tradable goods from its first best, $q_0^i - \bar{q}_0$. These deviations distort domestic absorption. Second, welfare losses also stem from misalignment in expectations of future productivity, that is forecast errors, $E_0^i a_1 - a_1$, that are due to the departure from the FIRE assumption. Misaligned expectations lead to a suboptimal allocation of the final good into consumption and investment, conditional on a given level of external borrowing.[27] The extent of this misallocation depends on wealth and substitution effects, primarily shaped by the intertemporal elasticity of substitution, $^1/_\sigma$, and the price elasticity of tradables, $\theta$.

Interestingly, wealth and substitution effects cancel each other out when $\sigma = \theta = 1$ (Cole and Obstfeld, 1991; Corsetti et al., 2008). Under the Cole-Obstfeld calibration, for given domestic absorption, productivity expectations do not alter the consumption/investment allocation, and thus drop out of the welfare function (*i.e.*, $\Omega_2 = 0$). Under this knife-edge case, public or secret FX intervention policy can achieve the first-best allocation by appropriate intervention in the foreign exchange market that eliminate the external borrowing gap, $q_0^i - \bar{q}_0$. Outside of this knife-edge case, inefficient fluctuations in external borrowing and forecast errors have independent welfare effects, giving rise to a non-trivial trade off.

[27] The presence of capital in our model implies that expectations are a distinct concern for the policymaker because they determine the split of domestic spending between consumption and investment for given borrowing (determined by the exchange rate). In an alternative model without capital accumulation but with endogenous non-tradable goods, misaligned expectations would imply a suboptimal choice of the sum of tradable and non-tradable goods.

## 4.2 Optimal foreign exchange interventions

The optimal FX intervention policy is described by the values of $(\kappa_q, \kappa_w)$ in the central bank's reaction function (26) that maximize welfare (33) under either public or secret interventions. The maximization problem assumes that the central bank is aware of the agents' behavioral biases, as, *e.g.*, in Gabaix (2020). The communication choice effectively consists in choosing the type of public signal $z$ that agents have access to: under public communication, agents access public signal $z^p$ in equation (28); under secret interventions, agents access the public signal $z^s$ in equation (32).

**Planner's problem** *The central bank's problem is formally defined as:*

$$\max_{\kappa_q,\kappa_w,z} \qquad \widetilde{\mathbb{W}} = -\left[\Omega_1 \operatorname{var}(q_0^i - \bar{q}_0) + \Omega_2 \operatorname{var}(E_0^i a_1 - a_1)\right]$$

$$\text{subject to:} \qquad q_0^i - \bar{q}_0 = \frac{\omega_1}{\omega_3}(r_0 - \Delta E_0^i q_1) - \frac{\omega_2}{\omega_3}(E_0^i a_1 - a_1); \tag{34}$$

$$E_0^i a_1 \equiv E[a_1|v^i, z] = \frac{(1+\delta)\beta_v v^i + \beta_z z}{\beta_v + \beta_z + \beta_a}; \qquad \textit{with } z = \{z^p, z^s\}; \tag{35}$$

$$f_1^\star = \kappa_q q_0 + \kappa_w n_1^\star, \tag{36}$$

*where expected excess (home) currency returns are:*

$$r_0 - \Delta E_0^i q_1 = \underbrace{\frac{\Gamma}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\left(\tilde{\theta}\omega_2 \bar{E}_0 a_1 + \omega_3(n_1^\star + f_1^\star)\right)}_{(r_0 - \Delta\bar{E}_0 q_1)} - \frac{1-\gamma}{\theta}(E_0^i - \bar{E}_0)[a_1 + \chi\Gamma(n_1^\star + f_1^\star)]. \tag{37}$$

A key aspect of the planner's problem lies in the dual effects of FX interventions. First, these interventions influence the flows in the foreign exchange market, which financiers must intermediate, thereby affecting both the equilibrium exchange rate and expected excess currency returns. These effects depend on the choice of $(\kappa_q, \kappa_w)$ and operate through standard portfolio balance channels. Second, interventions have informational effects that vary based on whether they are publicly announced or conducted in secret, shaping how the private sector forms expectations.

Online Appendix C.1 shows that welfare can be expressed solely as a function of the choice of policy coefficients $(\kappa_q, \kappa_w)$, communication format—public $(z^p)$ or secret $(z^s)$—and the precision of the ensuing public signal, denoted by $\beta_z$.[28]

[28] The auxiliary welfare function, denoted by $\widetilde{\mathbb{W}}(\kappa_q, \kappa_w, \beta_z)$, is also influenced by the covariance between

**Auxiliary planner's problem**

$$\max_{\kappa_q,\kappa_w,\beta_z} \qquad \widetilde{\mathbb{W}}(\kappa_w,\kappa_q,\beta_z) = -\left[\Omega_1 \operatorname{var}(q_0^i - \bar{q}_0) + \Omega_2 \operatorname{var}(E_0^i a_1 - a_1)\right] \tag{38}$$

$$\text{subject to:} \qquad \begin{cases} \beta_z = \infty, & \text{if } z = z^p \\ \beta_z = \left(\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\frac{\omega_2}{\Gamma\omega_1}\right)^2 (1+\kappa_w)^{-2}\beta_n, & \text{if } z = z^s \end{cases}$$

*where closed-form expressions for* $\operatorname{var}(q_0^i - \bar{q}_0)$ *as function of* $(\kappa_q, \kappa_w, \beta_z)$ *and for* $\operatorname{var}(E_0^i a_1 - a_1)$ *as function of only* $\beta_z$*, are provided in Online Appendix C.1.*

The planner's constraint in the auxiliary problem capture the distinct informational effects of public and secret interventions. Public interventions provide an infinitely precise signal of average productivity expectations, while the precision of the public signal under secret interventions is determined by the sensitivity of the exchange rate to productivity, which in turn depends on $\kappa_q$ and $\kappa_w$, as established in Proposition 3.

In what follows, we show that it is impossible to eliminate both sources of welfare loss in eq. (33), and thus attain the first best (except in the specific case of rational expectations where a "divine coincidence" occurs). We start by characterizing the optimal choice of publicly-announced FX interventions.

**Proposition 4.** *Optimal publicly-announced interventions are characterized by*

$$\kappa_q^p = \tilde{\theta} - \frac{\omega_3\delta}{\Gamma\omega_1}; \qquad \kappa_w^p = -1.$$

*At the optimum:*

$$\operatorname{var}(q_0^i - \bar{q}_0) = 0; \qquad \operatorname{var}(E_0^i a_1 - a_1) = \frac{\delta^2}{\beta_a} \geq 0, \qquad \widetilde{\mathbb{W}}^p = -\Omega_2\frac{\delta^2}{\beta_a} \leq 0.$$

*Proof.* See Online Appendix D. □

When interventions are publicly announced, the policy parameters $\kappa_q$ and $\kappa_w$ do not influence the accuracy of public signals, or expectations more generally, but only the flows in the FX market. For this reason, optimal public interventions focus exclusively on mitigating inefficient fluctuations in the FX market. To do so, the central

noise trading shocks and the noise in the public signal. Under secret and public communication regimes, this covariance term can be expressed as a distinct function of $(\kappa_q, \kappa_w, \beta_z)$. We thus write the welfare function as $\widetilde{\mathbb{W}}(\kappa_q, \kappa_w, \beta_z)$, with a slight abuse of notation.

bank completely offsets noise-trading fluctuations in exchange rates (*i.e.*, $\kappa_w^p = -1$). Moreover, the central bank sets $\kappa_q^p$ to eliminate the remaining inefficient fluctuations in external borrowing in response to productivity shocks. That is, it sets $\kappa_q^p$ such that expected excess currency returns completely offset the effects of misaligned expectations on external borrowing, as per equation (34). At the optimum, however, because average productivity expectations remain misaligned, the allocation of domestic absorption into consumption and investment is distorted and results in a welfare loss.[29]

We now turn to examining the welfare properties of secretly conducted FX interventions. A full analytical characterization is difficult because now the policy parameters also influence the informativeness of the exchange rate, and thus the properties of productivity expectations. Nevertheless, we analytically establish a number of important results. To this end, we break down the problem of optimal secret interventions in two steps, similarly to Angeletos et al. (2020).

**Two-step approach (Secret foreign exchange interventions)**

> **Step 1.** *Choose* $(\kappa_q, \kappa_w)$ *to maximize welfare* $\widetilde{\mathbb{W}}(\kappa_q, \kappa_w, \beta_z)$ *subject to* $\beta_z = \bar{\beta}_z$ *and* $\beta_z = \left(\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\frac{\omega_2}{\Gamma\omega_1}\right)^2 (1+\kappa_w)^{-2}\beta_n$. *Define* $(\kappa_q^s(\bar{\beta}_z), \kappa_w^s(\bar{\beta}_z))$ *as the solution to this step of the problem.*

> **Step 2.** *Choose* $\bar{\beta}_z$ *to maximize welfare* $\widetilde{\mathbb{W}}(\kappa_q^s(\bar{\beta}_z), \kappa_w^s(\bar{\beta}_z), \bar{\beta}_z)$. *Define* $\beta_z^s$ *as the solution to this step of the problem.*

Because $(\kappa_q, \kappa_w)$ do not affect the variance of forecast errors for a given $\beta_z = \bar{\beta}_z$ as noted in the auxiliary planner's problem, in the first step the planner effectively chooses the combination of $(\kappa_q, \kappa_w)$ that minimizes inefficient fluctuations in external borrowing subject to maintaining a given informational content of the exchange rate. In the second step of the problem, the planner then optimizes over the informational content of the exchange rate, $\beta_z$. The second step is what distinguishes the secret intervention from the public intervention. As established in Proposition 3, secret conduct of FX interventions allows the planner to leverage the informational content of the exchange rate and thus influence the variance of productivity forecast errors, effectively altering the agents' information structure. The next proposition analytically establishes that, under certain conditions, secret policy can achieve a superior allocation than public policy precisely by exploiting this lever.

[29] As outlined above, the $\sigma = \theta = 1$ calibration represents the exception since misaligned productivity expectations do not have independent welfare effects.

**Proposition 5.** *Secret policy can always attain the level of welfare of the optimal public policy. In addition, if $\delta > \hat{\delta}$, where $\hat{\delta}$ is defined in Online Appendix D, optimal secret FX intervention delivers higher welfare than optimal public intervention policy.*

*The optimum can be attained with:*

$$\kappa_q^s = \frac{(\Gamma\tilde{\theta}\omega_1 + \omega_3)}{\Gamma\omega_1} - \frac{1}{\Gamma\omega_1}\left[\frac{\omega_3(1+\delta)(\beta_z^s + \beta_v)}{\beta_a + \beta_v + \beta_z^s}\right]\left[1 + \frac{\beta_a}{(1+\delta)^2\beta_z^s}\right] \tag{39}$$

$$\kappa_w^s = -1 + \left(\frac{\beta_v}{\beta_a + \beta_v + \beta_z^s}\frac{\omega_2}{\Gamma\omega_1}\right)\left|\sqrt{\frac{\beta_n}{\beta_z^s}}\right| \tag{40}$$

*and $\beta_z^s$ solves*

$$\frac{\partial\widetilde{\mathbb{W}}(\kappa_q^s(\beta_z^s), \kappa_w^s(\beta_z^s), \beta_z^s)}{\partial\beta_z^s} = 0 \implies -\frac{\Omega_1}{\Omega_2} = \frac{\partial(\mathrm{var}(E_0^i a_1 - a_1))/\partial\beta_z^s}{\partial(\mathrm{var}(q_0^i - \bar{q}_0))/\partial\beta_z^s} \tag{41}$$

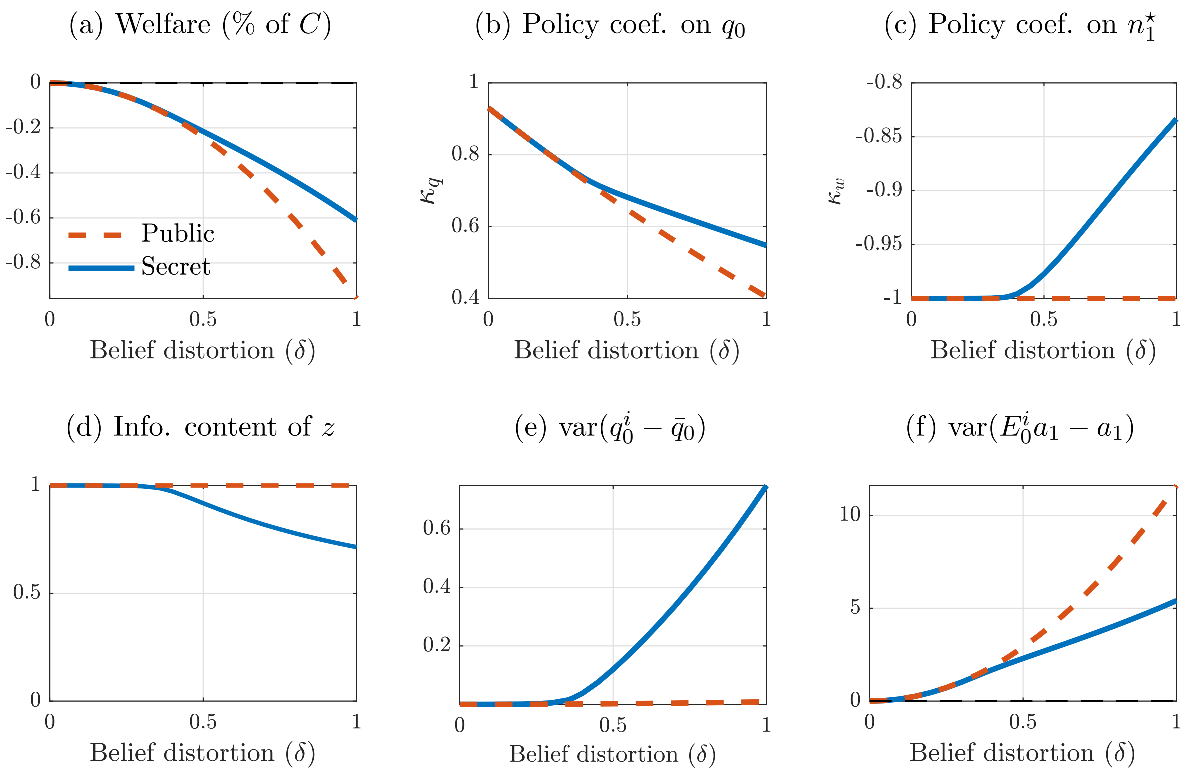
*Proof.* See Online Appendix D. □

Proposition 5 provides several key insights, which we will now explore in more detail, while Figure 5 depicts the characteristics of both the optimal secret and public intervention policies under an illustrative calibration of the model.

A first insight of Proposition 5 is that secret FX intervention policy can always implement the allocation of the *optimal* public FX intervention policy. It can do so by setting $\kappa_q = \kappa_q^p$ and $\kappa_w = \kappa_w^p$, where $(\kappa_q^p, \kappa_w^p)$ are the policy coefficients chosen by the planner under public communication (see Proposition 4). By fully eliminating noise-trading fluctuations ($\kappa_w = \kappa_w^p = -1$), the equilibrium exchange rate reflects only productivity, regardless of whether agents observe the quantity of bonds purchased by the central bank ($\beta_z = \infty$). In turn, by setting $\kappa_q = \kappa_q^p$, the secret policy can also replicate its overall allocation, and thus attain the same level of welfare.

The key insight of Proposition 5 is that secret interventions can yield a *higher* level of welfare compared to optimal public interventions when overreaction is sufficiently strong. To understand this result, recall the two-step auxiliary problem: the planner first chooses $(\kappa_q, \kappa_w)$ for given information structure $\beta_z = \bar{\beta}_z$ and then it optimizes over $\bar{\beta}_z$. For any given information structure encoded in $\bar{\beta}_z$, we learnt that the planner would completely eliminate inefficient fluctuations in external borrowing. However, when the planner can also choose over $\bar{\beta}_z$, it deviates from this benchmark when overreaction is sufficiently strong. In this case, optimal secret interventions trade off efficiency in

Figure 5: Outcomes under optimal policies




*Notes:* The figure reports values of different variables under optimal FX intervention policies for different levels of belief distortions ($\delta$). The rest of parameters are set according to Table G.1 (see Online Appendix).

external borrowing for the sake of limiting the amount of mis-learning, by deliberately limiting the informativeness of the exchange rate. We now elaborate on these points.

The next proposition establishes that a *less* informative exchange rate can indeed reduce inefficient fluctuations in productivity expectations.

**Proposition 6.** *If* $\delta \leq \bar{\delta}$ *then* $\partial \mathrm{var}(E_0^i a_1 - a_1)/\partial \beta_z < 0$. *If instead* $\delta > \bar{\delta}$, $\mathrm{var}(E_0^i a_1 - a_1)$ *is non-monotonic in* $\beta_z$. *Specifically:*

$$\frac{\partial \mathrm{var}(E_0^i a_1 - a_1)}{\partial \beta_z} \begin{cases} < 0 & \text{for } \beta_z < \hat{\beta}_z, \\ = 0 & \text{for } \beta_z = \hat{\beta}_z, \\ > 0 & \text{for } \beta_z > \hat{\beta}_z. \end{cases} \quad \text{where} \quad \hat{\beta}_z \equiv (\beta_a + \beta_v)\frac{1 - 2(1+\delta)}{1 - 2\delta(1+\delta)}. \tag{42}$$

*Proof.* See Online Appendix D. □

To interpret this result, recall that in our model expectations are imperfect due to both noisy private signals and distorted beliefs. These frictions lead to underreaction and overreaction, respectively. When the distortion in beliefs is high, the cost of higher reliance on a biased public signal can be greater that the benefits of moving expectations away from the prior and from noisy private signal.[30] In this case, the forecast error variance is minimized at a finite level of public signal precision, $\hat{\beta}_z$. For $\beta_z > \hat{\beta}_z$, making the public signal *less* precise (*i.e.*, reducing its informational content) actually improves expectations by mitigating overreaction.

Therefore, the level of public information embedded in publicly-announced interventions is suboptimal.[31] In contrast, secret interventions can improve outcomes by limiting the informativeness of the exchange rate, thereby dampening inefficient fluctuations in productivity expectations. To do so, the central bank let the equilibrium exchange rate reflect some degree of noise trading, *i.e.* by setting $\kappa_w^s \neq -1$.[32]

Optimal policy thus chooses the informativeness of the exchange rate such that the welfare benefit of making the exchange rate marginally less informative (thus taming overreaction) exactly equals its marginal welfare cost, *i.e.*, the cost of inefficient external borrowing. We stress that only the policymaker that conducts secret interventions can exploit this trade off, and thus achieve a higher level of welfare than the public intervention policy.

Our paper thus formalizes a new setting in which public information can be socially harmful, namely in the presence of overreaction to endogenous public signals.[33] This mechanism is conceptually distinct from others in the literature in which the decrease in posterior accuracy due to public information is a direct consequence of learning from prices (Amador and Weill, 2010). Indeed, Online Appendix A shows that, because we have abstracted from complementarities, in our model an exogenous increase in public information precision always results in higher posterior precision, and thus smaller

[30] To the contrary, when the distortion in beliefs is low, the marginal benefit of higher reliance on a public signal (even if biased) is always greater than the cost of beliefs being more anchored to the prior or dispersed due to idiosyncratic noise.

[31] This is true not only in the baseline case, when $\beta_z = \infty$ under public interventions, but as long as the central bank has enough information on noise-trading flows, and thus $\beta_z > \hat{\beta}_z$ (see Online Appendix E).

[32] Any two values of $\kappa_w^s$ symmetric around $-1$ yield the same allocation and welfare.

[33] The threshold $\bar{\delta} \approx 0.35$ above which more information increases the variance of forecast errors is below the empirical estimates of $\delta$. Estimates from forecast data are 0.91 for credit spreads (Bordalo et al., 2018), 1.22 for stock prices (Bordalo et al., 2019) and 0.53 on average for several macroeconomic and financial variables (Bordalo et al., 2020). Estimates from macroeconomic models range from around 0.75 (L'Huillier et al., 2024; Na and Yoo, 2024) to around 2 (Bianchi et al., 2024a).

variance of forecast errors, even though the exchange rate becomes less informative.

Finally, we stress that the potential for secret interventions to deliver higher welfare than public ones depends, in part, on the policymaker's ability to select both the reaction function and the communication strategy. When the choice of reaction coefficients is constrained—such as by institutional limits on exchange rate volatility or on sales of foreign reserves—public interventions may result in superior welfare outcomes.[34]

**Discussion** An interesting implication of our analysis is that whether exchange rates reflect the economy's fundamental or noise is an equilibrium outcome that depends on the design and communication of FX interventions. In our model, if public interventions are optimally designed, they should, *ceteris paribus*, result in an exchange rate that is closely tied to future economic conditions. In contrast, optimally secret interventions can lead to an exchange rate that is partly influenced by noise trading. Letting some degree of noise trading is *the* mean to reduce the informational content of the exchange rate. Thus, contrary to models solved under FIRE (see, for example, Itskhoki and Mukhin, 2022), eliminating noise-trading fluctuations is not always optimal.

Importantly, even though optimal secret interventions preserve some noise-trading fluctuations into the exchange rate, the resulting exchange rate volatility need not be higher than under the public communication policy. A less informative exchange rate mitigates the impact of expectations overreaction, which can reduce the overall volatility of macroeconomic aggregates and exchange rates due to productivity shocks.[35]

Finally, both secret and public optimal FX intervention policies involve intervening against inflows in the FX market to some extent, aligning with the empirical observation of "systematic managed floating." This practice sees central banks consistently responding to changes in total market pressure, with part of the adjustment reflected in the exchange rate and the remainder absorbed through changes in foreign exchange reserves (Frankel, 2019).

[34] The relative welfare properties of public communication hinge on whether the full-information economy under- or over-responds to productivity and noise-trading shocks, which depends on the values of $\kappa$'s.

[35] Under the calibration in Table G.1 (see Online Appendix), the optimal secret intervention policy results in lower exchange rate volatility compared to the optimal public intervention policy.

# 5 Conclusions

The debate on foreign exchange interventions presents two seemingly conflicting features. The consensus view among policymakers—as shown in surveys of central bankers—suggests that FX interventions primarily influence markets by shaping expectations through a signaling channel (Patel and Cavallino, 2019). At the same time, most intervention activities in the FX market have historically been, and continue to be, largely secret. This paper demonstrates that these observations can coexist without contradiction. Our formal analysis is based on two intuitive premises: (i) exchange rates serve an informational role by aggregating beliefs about economic fundamentals, and their informational content is influenced by how exchange rate policy is conducted; and (ii) households and financial market participants tend to overreact to news embedded in exchange rate movements. Together, these elements make exchange rate policy a complex task with non-trivial trade-offs. In particular, the tendency for expectations to overreact naturally implies that central banks should exercise caution in using publicly announced interventions—precisely *because* of their signaling effects. In addition, it may be optimal for the central bank to influence the weight that the private sector places on exchange rate information by secretly intervening in the market, and tolerating some noise trading in exchange rate fluctuations.

Although our analysis focuses on foreign exchange interventions, the core mechanism—beliefs overreacting to news embedded in market prices—likely applies to other contexts. For instance, during asset purchase programs in sovereign bond markets, policymakers actively shape public signals, and market reactions depend crucially on how these signals are interpreted. Extending our framework to such settings could inform the design of optimal communication strategies when market systematically misread public information. Moreover, these extensions could help understand how specific market structures and institutional constraints interact with expectation formation, and ultimately influence the transmission of policy interventions.

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# Appendix

## A Derivations of small open economy model

### A.1 Financiers' demand for foreign currency bonds

We follow Fanelli and Straub (2021) and assume that there exists a continuum of risk-neutral financiers, labeled by $j \in [0, \infty)$, in each island $i$. Financiers hold a zero-capital portfolio in home and foreign bonds denoted $(d^i_{j,1}, {d^i_{j,1}}^\star)$. Financier's investment decisions are subject to two restrictions. First, each intermediary is subject to a net open position limit of size $D > 0$. Second, intermediaries face heterogeneous participation costs, as in Alvarez et al. (2009). Each intermediary $j$ active in the foreign bond market at time $t$ is obliged to pay a participation cost of exactly $j$ per unit of foreign currency invested. (Participation costs constitute transfers to households in the home island economy. Thus, no extra cost terms enter the household's budget constraint.) As a result, intermediary $j$ on island $i$ chooses ${d^i_{j,1}}^\star$ that solves:

$$\max_{\frac{{d^i_{j,1}}^\star}{R^\star_0} \in [-D, D]} \frac{{d^i_{j,1}}^\star}{R^\star_0} E^i_0 \left( \tilde{R}^\star_1 \right) - j \left| \frac{{d^i_{j,1}}^\star}{R^\star_0} \right|,$$

where $\tilde{R}^\star_1 \equiv R^\star_0 - R_0 \frac{\mathcal{S}_0}{\mathcal{S}_1}$ is the return on one foreign-currency unit holding expressed in foreign currency and $E^i_0$ is the same expectation operator as the island-$i$ household's.

Intermediary $j$'s expected cash flow conditional on investing is $D \left| E^i_0 \left( \tilde{R}^\star_1 \right) \right|$ while participation costs are $jD$. Investing is optimal for all intermediaries $j \in [0, \bar{j}]$, with the marginal active intermediary $\bar{j}$ given by $\bar{j} = \left| E^i_0 \left( \tilde{R}_1{}^\star \right) \right|$. The aggregate investment volume is then: $\frac{{D^i_1}^\star}{R^\star_0} = \bar{j} D \, \mathrm{sign} \left\{ E^i_0 \left( \tilde{R}_1{}^\star \right) \right\}$. Defining $\hat{\Gamma} \equiv D^{-1}$ and substituting out $\bar{j}$, we obtain the total demand for foreign-currency bonds on island $i$, ${D^i_1}^\star = \int {d^i_{j,1}}^\star \mathrm{d}j$:

$$\frac{{D^i_1}^\star}{R^\star_0} = \frac{1}{\hat{\Gamma}} E^i_0 \left( R^\star_0 - R_0 \frac{\mathcal{S}_0}{\mathcal{S}_1} \right). \tag{A.1}$$

which is equation (6) in the text.

## A.2 Island equilibrium

The log-linearized version of the household's optimality conditions are:

$$\sigma(E_0^i c_1^i - c_0^i) = r_0 - (E_0^i p_1^i - p_0^i), \tag{A.2}$$

$$(1-\alpha)k_1^i = E_0^i(s_1 - p_1^i) + E_0^i a_1 - r_0 + (E_0^i p_1^i - p_0^i), \tag{A.3}$$

$$q_t^i \equiv s_t - p_t^i = -\frac{1-\gamma}{\theta} y_{T,t}^i, \tag{A.4}$$

Island-$i$ budget in (5) can be combined and loglinearized as:

$$\frac{1+\phi}{\beta} y_{T,0}^i = a_1 + \alpha k_1^i - y_{T,1}^i \tag{A.5}$$

where $\phi = \beta\alpha\gamma$. The final good aggregator in (3) yields:

$$\frac{1}{1+\phi} c_0^i + \frac{\phi}{1+\phi} k_1^i = \gamma y_{T,0}^i \qquad c_1^i = \gamma y_{T,1}^i. \tag{A.6}$$

The log-linear optimality condition of financiers (7):

$$\Gamma \int d_1^{i\star} \,\mathrm{d}i = \bar{E}_0 s_1 - s_0 - r_0 \qquad \text{w/:} \quad d_1^{i\star} \equiv {}^{\mathrm{d}D_1^{i\star}}/_{Y_{T,1}^{ss}}, \quad \Gamma \equiv \hat{\Gamma} \cdot Y_{T,1}^{ss} \cdot \beta^2. \tag{A.7}$$

Finally, bond market clearing, (11) implies:

$$\int d_1^{i\star} \,\mathrm{d}i = -\frac{1+\phi}{\beta} \int y_{T,0}^i \,\mathrm{d}i - n_1^\star - f_1^\star \tag{A.8}$$

We normalize the average price of the consumption basket ($\int p_t^i \,\mathrm{d}i = 0$, for $t = [0,1]$). The aggregate real exchange rate thus equals the nominal exchange rate: $q_t = s_t$.

## A.3 Equilibrium exchange rate

Using equations (A.2), (A.3) (A.4), (A.5), and (A.6), one can express island-$i$ price level and tradable demand as:

$$p_0^i = s_0 - \frac{\omega_1}{\omega_3}\big(r_0 - (E_0^i s_1 - s_0)\big) + \frac{\omega_2}{\omega_3} E_0^i a_1 \tag{A.9}$$

$$y_{T,0}^i = -\frac{\theta}{1-\gamma}\frac{\omega_1}{\omega_3}\left(r_0 - (E_0^i s_1 - s_0)\right) + \frac{\theta}{1-\gamma}\frac{\omega_2}{\omega_3}E_0^i a_1 \tag{A.10}$$

where $\omega_1 > 0, \omega_2 > 0, \omega_3 > 0$ are all convolution of parameters:

$$\omega_1 \equiv [\theta\sigma\alpha\gamma(1+\beta) + (1-\alpha)\theta + (1-\gamma)\alpha] \qquad \omega_2 \equiv [(1-\gamma) + \sigma\gamma\theta(1+\alpha\beta)] \tag{A.11}$$

$$\omega_3 \equiv \frac{\theta(1-\alpha)(1+\phi)}{\beta(1-\gamma)}[\sigma\gamma\theta(1+\beta) + (1-\gamma)] + \theta\sigma\alpha\gamma(1+\beta) + (1-\alpha)\theta + (1-\gamma)\alpha$$

Sum (A.9) across islands and use $\int p_t^i \,\mathrm{d}i = 0$, for $t = [0,1]$, to express the small open economy (real and nominal) exchange rate as:

$$q_0 = s_0 = \frac{\omega_1}{\omega_3}\left(r_0 - (\bar{E}_0 s_1 - s_0)\right) - \frac{\omega_2}{\omega_3}\bar{E}_0 a_1 \tag{A.12}$$

Using eqs. (A.2), (A.5), (A.6) and (A.10), one can express the average expectation of aggregate excess home-currency returns as:

$$r_0 - \left(\bar{E}_0 s_1 - s_0\right) = \frac{\Gamma\tilde{\theta}\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\bar{E}_0 a_1 + \frac{\Gamma\omega_3}{\Gamma\tilde{\theta}\omega_1 + \omega_3}(n_1^\star + f_1^\star) \tag{A.13}$$

where $\tilde{\theta} \equiv \frac{(1+\alpha\beta\gamma)\theta}{\beta(1-\gamma)} > 0$. Using (A.13) in (A.12), one obtains eq. (12) in the paper.

**Other equilibrium equations** Using the equilibrium conditions in Appendix A.2, It is useful to express $E_0^i q_1 - \bar{E}_0 q_1$ as a function of shocks and expectations thereof:

$$\begin{aligned} s_1 - p_1^i = \frac{1-\gamma}{\theta}a_1 &+ \left[\frac{1-\gamma}{\theta}\frac{\alpha}{1-\alpha}\left(\frac{\omega_1}{\omega_3} - 1\right) + \frac{1+\phi}{\beta}\frac{\omega_1}{\omega_3}\right]\left(r_0 - (E_0^i s_1 - s_0)\right) + \\ &- \left[\frac{1-\gamma}{\theta}\frac{\alpha}{1-\alpha}\left(\frac{\omega_2}{\omega_3} - 1\right) + \frac{1+\phi}{\beta}\frac{\omega_2}{\omega_3}\right]E_0^i a_1 \end{aligned} \tag{A.14}$$

The aggregate version of eq. (A.14) implies:

$$\begin{aligned} q_1 = s_1 = \frac{1-\gamma}{\theta}a_1 &+ \left[\frac{1-\gamma}{\theta}\frac{\alpha}{1-\alpha}\left(\frac{\omega_1}{\omega_3} - 1\right) + \frac{1+\phi}{\beta}\frac{\omega_1}{\omega_3}\right]\left(r_0 - (\bar{E}_0 s_1 - s_0)\right) + \\ &- \left[\frac{1-\gamma}{\theta}\frac{\alpha}{1-\alpha}\left(\frac{\omega_2}{\omega_3} - 1\right) + \frac{1+\phi}{\beta}\frac{\omega_2}{\omega_3}\right]\bar{E}_0 a_1. \end{aligned} \tag{A.15}$$

Using equations (A.13), (12) and (A.15), one obtains:

$$E_0^i q_1 - \bar{E}_0 q_1 = \frac{1-\gamma}{\theta}(E_0^i a_1 - \bar{E}_0 a_1) + \left[\frac{1-\gamma}{\theta}\frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\right](E_0^i - \bar{E}_0)\bar{E}_0[a_1]. \tag{A.16}$$

# Online Appendix

## A Public Information and Posterior Precision

In this appendix section, we illustrate how the precision of exogenous public information affects the overall posterior precision. To do so, we modify the baseline environment to allow for an exogenous source of public information. While the direct effect of exogenous public information on the posterior precision is generally positive, Amador and Weill (2010) present an environment in which its indirect effect (via learning from prices) may be negative. Below, we show that this is not the case in our framework.

The household starts with a prior that $a_1 \sim \mathcal{N}(0, \bar{\beta}_a^{-1})$. Before market opens, the central bank sends a public signal $g = a_1 + \varepsilon_g$, with $\varepsilon_g \sim \mathcal{N}(0, (\beta_a - \bar{\beta}_a)^{-1})$. Thus, after the signal is observed but before markets open, the precision of the household's information is $\beta_a$, as in the baseline environment. When the precision of this signal is zero, one also recovers the baseline setting. WLOG, we derive the main result under rational expectations ($\delta = 0$). We guess that the exchange rate solution is:

$$q_0 = \lambda_a a_1 + \lambda_n n_1^\star + \lambda_g g \tag{AA.1}$$

Given this process, individuals form posterior beliefs using the signal:

$$z^p | a_1 = \frac{q_0 - \lambda_g g}{\lambda_a} \bigg| a_1 \sim \mathcal{N}(a_1, \left(\frac{\lambda_n}{\lambda_a}\right)^2 \beta_n^{-1}) \tag{AA.2}$$

Next, define $D = \beta_a + \beta_v + \beta_z$ and use the perceived market clearing condition (19):

$$q_0 \overset{p}{=} \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3} n_1^\star - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \left( \frac{\beta_v a_1 + \beta_z \frac{q_0 - \lambda_g g}{\lambda_a} + (\beta_a - \bar{\beta}_a) g}{D} \right)$$

The undetermined coefficients are thus: $\lambda_a = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3} \frac{\beta_v + \beta_z}{D}$ and $\lambda_n = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3} \frac{\beta_v+\beta_z}{\beta_v}$. Define $\Lambda \equiv \lambda_a / \lambda_n$, and note that:

$$\Lambda = -\frac{\omega_2}{\Gamma\omega_1} \frac{\beta_v}{\beta_v + \beta_a + \Lambda^2 \beta_n}; \qquad \frac{\lambda_g}{\lambda_a} = \frac{\beta_a - \bar{\beta}_a}{\beta_v + \beta_z}. \tag{AA.3}$$

As a result, the solution for the exchange rate process in eq. (AA.1) is:

$$q_0 = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \frac{\beta_v + \beta_z + \beta_a - \bar{\beta}_a}{D} a_1 + \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \frac{\beta_v + \beta_z}{\beta_v} n_1^\star - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \frac{\beta_a - \bar{\beta}_a}{D} \varepsilon_g$$

Let us now explore how an increase in the precision of public signal affects the posterior precision. First, take the following derivative:

$$\frac{\partial(\beta_v + \beta_a + \Lambda^2\beta_n)}{\partial\beta_a} = 1 + 2\Lambda\beta_n\frac{\partial\Lambda}{\partial\beta_a} \tag{AA.4}$$

Rewrite the fixed point equation (AA.3) as $\Lambda(\beta_v + \beta_a + \Lambda^2\beta_n) = -\frac{\omega_2}{\Gamma\omega_1}$, and differentiate to obtain $\frac{\partial\Lambda}{\partial\beta_a} = -\frac{\Lambda}{(\beta_v+\beta_a+3\Lambda^2\beta_n)}$, which implies:

$$\frac{\partial(\beta_v + \beta_a + \Lambda^2\beta_n)}{\partial\beta_a} = 1 - \frac{2\Lambda^2\beta_n}{\beta_v + \beta_a + 3\Lambda^2\beta_n} > 0 \tag{AA.5}$$

Thus, if one were to increase the precision of exogenous public signals, the precision of private information is unaffected (by assumption) and the precision of overall public information increases. This is *despite* the fact that the precision of the endogenous public signal decreases (because agents place less weight on private signals), *i.e.* despite $\frac{d\Lambda}{d\beta_a} < 0$. In other words, the direct effect dominates. This rules out negative welfare effects of public information of the types discussed in Amador and Weill (2010).

# B Alternative belief formation models

Our baseline assumption E2 implies that agents underestimate others' responses, generating endogenous overreaction to public signals. In this appendix, we show that our results also hold under two alternative formulations of beliefs: (i) no higher-order bias, but diagnostic beliefs about both private and public signals; (ii) Partial Equilibrium Thinking as in Bastianello and Fontanier (2024), where agents process private signals optimally but neglect others' learning from prices.

## B.1 Diagnostic expectation without higher order belief bias

First, we assume that the representativeness heuristic applies to both exogenous private signals and endogenous public signal ($q_0$ under laissez faire). Therefore, we have:

$$E^i[a_1|v^i, q_0] = (1+\delta)\mathbb{E}^i[a_1|v^i, q_0]. \tag{B.1}$$

In this alternative framework, we assume that each agent knows that others are behaviorally biased (while they incorrectly think themselves as rational). As a result, agents correctly perceive the exchange rate as following (12) (with $f_1^\star = 0$, under laissez faire). We thus conjecture that the equilibrium exchange rate process follows:

$$q_0 = \lambda_a a_1 + \lambda_n n_1^\star, \tag{B.2}$$

The relationship between the exchange rate and the two shocks, governed by $(\lambda_a,\ \lambda_n)$, is determined as the solution of a fixed point problem. Rewrite eq. (B.2) as ${}^{q_0}/_{\lambda_a} = a_1 + {}^{\lambda_n}/_{\lambda_a} n_1^\star$, where ${}^{q_0}/_{\lambda_a}$ represents an unbiased signal centered around future productivity $a_1$ with a error variance of $\beta_z^{-1} \equiv ({}^{\lambda_n^2}/_{\lambda_a^2})\beta_n^{-1}$. Thus, agents' posterior is:

$$E^i[a_1|v^i, q_0] = (1+\delta)\frac{\beta_v v^i + \beta_z \frac{q_0}{\lambda_a}}{\beta_v + \beta_z + \beta_a}, \tag{B.3}$$

We can average posterior beliefs $\bar{E}[a_1|v^i, q_0] \equiv \int^i E^i[a_1|v^i, q_0]\ \mathrm{d}i$ using $\int^i v^i\ \mathrm{d}i = a_1$ and substitute back in the exchange rate process, eq. (19), to verify the conjecture, eq. (B.2). The next proposition characterizes the equilibrium of this model:

**Proposition B.1.** *The symmetric linear market equilibrium is unique and the exchange rate process is described by eq.* (B.2) *with coefficients*

$$\lambda_a = -(1+\delta)\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\frac{\beta_v + \Lambda^2\beta_n}{\beta_a + \beta_v + \Lambda^2\beta_n} \qquad \lambda_n = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\frac{\beta_v + \Lambda^2\beta_n}{\beta_v} \tag{B.4}$$

*where* $\Lambda \equiv {}^{\lambda_a}/_{\lambda_n}$, *and* $\Lambda^2$ *is unique and implicitly defined by*

$$\Lambda^2 = \left((1+\delta)\frac{\omega_2}{\Gamma\omega_1}\right)^2 \frac{\beta_v^2}{(\beta_a + \beta_v + \Lambda^2\beta_n)^2} \tag{B.5}$$

*Proof.* It follows from the Proof of Proposition 1 in Appendix D. □

**Forecast error variance** Agents overreact to both private and public signal according to parameter $\delta \geq 0$. The forecast error variance in this model is:

$$\mathrm{var}(E_0^i a_1 - a_1) = \frac{[\delta(\beta_v + \beta_z) - \beta_a]^2}{(\beta_v + \beta_z + \beta_a)^2}\frac{1}{\beta_a} + \frac{(1+\delta)^2(\beta_z + \beta_v)}{(\beta_v + \beta_z + \beta_a)^2} \tag{B.6}$$

If the exchange rate is perfectly accurate, $\beta_z \to \infty$, the consensus forecast error variance remains positive, $\mathrm{var}(\bar{E}_0 a_1 - a_1) = \delta^2 \frac{1}{\beta_a}$.

More generally, for large enough diagnosticity $\delta$, higher public signal precision has two opposing effects on the forecast error variance. First, like in the rational expectation case, higher precision provides more information, lowering the forecast error variance. Second, higher precision increases the relative weight on the public signal in posterior belief, and therefore it amplifies the degree of overreaction, increasing the forecast error variance. We show the first effect prevails for low values of signal accuracy, while the second effect prevails for larger values of accuracy.

**Proposition B.2** (Forecast error variance with diagnostic expectations)**.** *If $\delta < \bar{\delta}$, then the unconditional variance of the individual forecast error on productivity is minimized with perfectly informative public signal $\beta_z \to \infty$. If $\delta > \bar{\delta}$, then the unconditional variance of the individual forecast error on productivity is minimized with perfectly uninformative public signal $\beta_z \to 0$, where $\bar{\delta} \equiv 1$.*

*Proof.* It follows from eq. (B.6). □

The qualitative implications of this alternative framework, in terms of properties of forecast errors, are thus analogous to the baseline model, *i.e.* Proposition 6.

## B.2 Partial equilibrium thinking

We now consider a belief updating model, where the only bias is in the formation of higher order beliefs. Similarly to the Partial Equilibrium Thinking model of Bastianello and Fontanier (2024), suppose that agents update their belief rationally with respect to the exogenous signal, but fail to realize that the other agents are also learning from the exchange rate. As a result, they make mistakes in interpreting the endogenous signal. In particular, they perceive the exchange rate process to be:

$$q_0 \overset{p}{=} \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3} n_1^\star - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \bar{\mathbb{E}}_0[a_1|v^i] \tag{B.7}$$

where $\bar{\mathbb{E}}_0[a_1|v^i]$ is the average Bayesian posterior conditional *only* on the exogenous private signal: $\bar{\mathbb{E}}_0[a_1|v^i] = \frac{\beta_v}{\beta_a+\beta_v} a_1$. The *perceived* exchange rate thus follows $q_0 \overset{p}{=} \lambda_a a_1 + \lambda_n n_1^\star$, from which they extract the signal

$$\frac{q_0}{\lambda_a} \overset{p}{=} a_1 + \frac{\lambda_n}{\lambda_a} n_1^\star \quad \text{with} \quad \lambda_a = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \frac{\beta_v}{\beta_a + \beta_v} \qquad \lambda_n = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \tag{B.8}$$

where the *perceived* accuracy of the exchange rate is $\tilde{\beta}_z \equiv (\lambda_a^2/\lambda_n^2)\beta_n = \left(\omega_2/(\Gamma\omega_1)\beta_v/(\beta_a+\beta_v)\right)^2 \beta_n$. Note that, unlike in the baseline model, the perceived accuracy of the public signal, $\tilde{\beta}_z$ can differ from its actual accuracy, $\tilde{\beta}_z$. Actual expectations are:

$$\bar{E}_0[a_1] = \frac{\beta_v a_1 + \tilde{\beta}_z \frac{q_0}{\lambda_a}}{\beta_a + \beta_v + \tilde{\beta}_z} \tag{B.9}$$

The solution for the *actual* exchange rate process, therefore, is:

$$q_0 = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\frac{\beta_v^2}{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}a_1 + \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\frac{\beta_v(\beta_a+\beta_v+\tilde{\beta}_z)}{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}n_1^\star, \tag{B.10}$$

Higher-order belief bias leads to misperception of the exchange rate process, and therefore mistakes in information extraction. The signal agents actually use is:

$$\frac{q_0}{\lambda_a} = \frac{\beta_v(\beta_v+\beta_a)}{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}a_1 - \frac{\Gamma\omega_1}{\omega_2}\frac{(\beta_v+\beta_a+\tilde{\beta}_z)(\beta_v+\beta_a)}{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}n_1^\star, \tag{B.11}$$

Equation (B.11) reveals that the actual accuracy of the signal they use is

$$\beta_z = \left(\frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}{(\beta_v+\beta_a+\tilde{\beta}_z)(\beta_v+\beta_a)}\right)^2\beta_n,$$

which is distinct from the perceived accuracy $\tilde{\beta}_z$:

$$\tilde{\beta}_z = \beta_z\left(\frac{\beta_v(\beta_v+\beta_a+\tilde{\beta}_z)}{\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z}\right)^2 \tag{B.12}$$

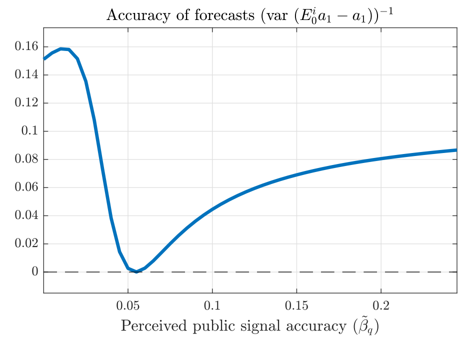
There are two differences between the perceived signal (B.8) and the actual signal (B.11). First, the signal is not centered around the realization $a_1$, but it is multiplied by the term $\Psi_1 \equiv \frac{\beta_v(\beta_v+\beta_a)}{\beta_v(\beta_v+\beta_a)-\beta_a\tilde{\beta}_z}$. Agents thus misperceive the mean of the endogenous signal, similarly to our baseline model of beliefs. Second, in this case agents also misperceive the accuracy of the endogenous signal by a factor: $\Psi_2 \equiv \left(\frac{\beta_v(\beta_v+\beta_a+\tilde{\beta}_z)}{\beta_v(\beta_v+\beta_a)-\beta_a\tilde{\beta}_z}\right)^2$.

The forecast error variance is:

$$\text{var}(E_0^i a_1 - a_1) = \frac{\left[\tilde{\beta}_z\left(\frac{\beta_a\tilde{\beta}_z}{\beta_v(\beta_v+\beta_a)-\beta_a\tilde{\beta}_z}\right) - \beta_a\right]^2\beta_a^{(-1)} + \tilde{\beta}_z\left(\frac{\beta_v(\beta_v+\beta_a+\tilde{\beta}_z)}{\beta_v(\beta_v+\beta_a)-\beta_a\tilde{\beta}_z}\right)^2 + \beta_v}{(\beta_v+\beta_a+\tilde{\beta}_z)^2}$$

Consider what happens when the perceived accuracy varies. If $\tilde{\beta}_z = 0$, the perceived signal equal the actual signal: $\text{var}(E_0^i a_1 - a_1) = \frac{1}{\beta_v+\beta_a}$. Indeed, the bias comes from agents thinking other agents do not use the public signal to update, and in this case they are correct. If $\beta_v(\beta_v+\beta_a) - \beta_a\tilde{\beta}_z > 0$, agents misinterpret the endogenous signal for two reasons. First, as $\Psi_1 > 0$, agents think the public signal loads less on the shock than it actually does. Similarly to the baseline model, this leads to overreaction to the endogenous signal. Second, the term $\Psi_2 > 0$ leads agents to overweight the public signal and therefore also overreacting to it. As the perceived precision of the public signal increases (that is, $\beta_v(\beta_v+\beta_a)-\beta_a\tilde{\beta}_z \to 0$, and both

Figure B.1: Forecast error accuracy with Partial Equilibrium Thinking




*Notes:* This figure reports the precision of forecasts, $(\text{var}(E_0^i a_1 - a_1))^{-1}$ as a function of the perceived public signal accuracy $\tilde{\beta}_z$. The rest of parameters are set according to Table G.1.

$\Psi_1$ and $\Psi_2$ go to infinity), this implies infinite overreaction and thus: $\text{var}(E_0^i a_1 - a_1) \to \infty$. As the perceived precision of the public signal increases further (that is $\beta_v(\beta_v + \beta_a) - \beta_a \tilde{\beta}_z < 0$ and therefore $\Psi_1 > 0$) agents incorrectly think the signal positively correlates with productivity, and thus they update in the opposite direction. If $\tilde{\beta}_z \to \infty$, the signal does not load on productivity $a_1$, as the only equilibrium is one in which neither expectations nor the exchange rate respond to changes in the productivity, and: $\text{var}(E_0^i a_1 - a_1) = \frac{1}{\beta_a}$.

Thus, for certain values of $\tilde{\beta}_z$, this model of beliefs also generates overreaction—due to misperceptions of both signal mean and precision. While this result echoes Proposition 6, its characterization is not available in closed form solution. Therefore, we provide a numerical illustration in Figure B.1.

# C Welfare criterion

We evaluate welfare using an utilitarian criterion. Since all islands are ex-ante identical, we can simply maximize the welfare of an individual island $i$ given in equation (1). We takr second-order approximation of the welfare function around the steady state::

$$\mathbb{W} \equiv C^{1-\sigma}\mathbb{E}\left\{[\hat{c}_0^i + \frac{1}{2}(1-\sigma)(\hat{c}_0^i)^2] + \beta[\hat{c}_1^i + \frac{1}{2}(1-\sigma)(\hat{c}_1^i)^2]\right\} + t.i.p. + \mathcal{O}(||\xi||^3), \tag{C.1}$$

where hatted variables are expressed in log-deviations from steady state, $t.i.p.$ stands for terms independent of policy, and $\mathcal{O}(||\xi||^3)$ denotes terms that are of third or higher order. Utility is maximized when consumption takes on its efficient values

$$\overline{\mathbb{W}} \approx \mathbb{E}\left[[\bar{c}_0 + \frac{1}{2}(1-\sigma)\bar{c}_0^2] + \beta[\bar{c}_1 + \frac{1}{2}(1-\sigma)\bar{c}_1^2]\right] \tag{C.2}$$

where barred variables are log-deviations from the steady state in the efficient allocation (which is the same for every island). In general, this maximum may not be attainable.

To lighten notation we drop the $i$ subscript in the following derivations. Take the second-order approximation of the resource constraint at period 1 (eq. (3)):

$$\hat{c}_1 + \frac{1}{2}\hat{c}_1^2 = \gamma \hat{y}_{T,1} + \frac{1}{2}\frac{\gamma}{\theta}(\gamma + \theta - 1)\hat{y}_{T,1}^2 \tag{C.3}$$

Take the square of equation (C.3), and keep only terms that are at most of order 2:

$$c_1^2 = \gamma^2 y_{T,1}^2 \tag{C.4}$$

Take the second-order approximation of the resource constraint at period 0 (eq. (3)):

$$\hat{c}_0 + \frac{1}{2}\hat{c}_0^2 + \phi(\hat{k}_1 + \frac{1}{2}\hat{k}_1^2) = (1+\phi)\gamma \hat{y}_{T,0} + \frac{1}{2}(1+\phi)\frac{\gamma}{\theta}(\gamma + \theta - 1)\hat{y}_{T,0}^2 \tag{C.5}$$

Take the square of equation (C.5), and keep only terms that are at most of order 2:

$$\hat{c}_0^2 = \phi^2 \hat{k}_1^2 + (1+\phi)^2\gamma^2 \hat{y}_{T,0}^2 - 2\gamma(1+\phi)\phi \hat{k}_1 \hat{y}_{T,0} \tag{C.6}$$

Take the second-order approximation of equations (2) and (5):

$$\hat{y}_{T,1} + \frac{1}{2}\hat{y}_{T,1}^2 = -\frac{1+\phi}{\beta}\left(\hat{y}_{T,0} + \frac{1}{2}\hat{y}_{T,0}^2\right) + \left(\hat{a}_1 + \alpha \hat{k}_1 + \frac{1}{2}\left(\alpha^2 \hat{k}_1^2 + 2\alpha \hat{a}_1 \hat{k}_1 + \hat{a}_1^2\right)\right) \tag{C.7}$$

Take the square of equation (C.7), and keep only terms that are at most of order 2:

$$\hat{y}_{T,1}^2 = \frac{1}{\beta^2}(1+\phi)^2 \hat{y}_{T,0}^2 + \hat{a}_1^2 + \alpha^2 \hat{k}_1^2 - 2\frac{1}{\beta}(1+\phi)\hat{y}_{T,0}\hat{a}_1 - 2\frac{1}{\beta}(1+\phi)\alpha \hat{y}_{T,0}\hat{k}_1 + 2\alpha \hat{a}_1 \hat{k}_1 \tag{C.8}$$

Combine (C.3) with (C.4):

$$\hat{c}_1 = \gamma \hat{y}_{T,1} + \frac{1}{2}\gamma(1-\gamma)\frac{\theta - 1}{\theta}\hat{y}_{T,1}^2 \tag{C.9}$$

Combine (C.5) with (C.6):

$$\hat{c}_0 = \gamma(1+\phi)\hat{y}_{T,0} - \phi k_1 - \tfrac{1}{2}\phi(1+\phi)\hat{k}_1^2 + \tfrac{1}{2}(1+\phi)\gamma\left[(1-\gamma)\tfrac{\theta-1}{\theta} - \gamma\phi\right]\hat{y}_{T,0}^2 + \gamma(1+\phi)\phi \hat{k}_1 \hat{y}_{T,0}. \tag{C.10}$$

Multiply eq. (C.9) by $\beta$ and substitute $y_{T,1}$ from eq. (C.7) and $\hat{y}^2_{T,1}$ from eq. (C.8):

$$\beta\hat{c}_1 = -\gamma(1+\phi)\hat{y}_{T,0} + \gamma\beta\hat{a}_1 + \gamma\alpha\beta\hat{k}_1 + \frac{\beta\gamma}{2}\left\{\alpha^2\hat{k}_1^2 + 2\alpha\hat{a}_1\hat{k}_1 + \hat{a}_1^2 - \frac{(1+\phi)}{\beta}\hat{y}^2_{T,0}\right\}$$
$$+\frac{1}{2}\beta\gamma\left[(1-\gamma)\frac{\theta-1}{\theta}-1\right]\left(\frac{1}{\beta^2}(1+\phi)^2\hat{y}^2_{T,0} + \hat{a}_1^2 + \alpha^2\hat{k}_1^2 - 2\frac{1}{\beta}(1+\phi)\hat{y}_{T,0}\hat{a}_1 - 2\frac{1}{\beta}(1+\phi)\alpha\hat{y}_{T,0}\hat{k}_1 + 2\alpha\hat{a}_1\hat{k}_1\right) \tag{C.11}$$

Summing eqs. (C.10) and (C.11) one obtains:

$$\hat{c}_0 + \beta\hat{c}_1 = \gamma\beta\hat{a}_1 + \frac{\beta\gamma}{2}(1+\chi)\hat{a}_1^2 - \frac{1}{2}\phi(1+\phi-\alpha(1+\chi))\hat{k}_1^2 + \frac{1}{2}\Omega\hat{y}^2_{T,0}$$
$$+\phi(1+\chi)\hat{a}_1\hat{k}_1 - \gamma\chi(1+\phi)\hat{y}_{T,0}\hat{a}_1 + \gamma(1+\phi)(\phi-\chi\alpha)\hat{k}_1\hat{y}_{T,0} \tag{C.12}$$

where $\Omega = \gamma(1+\phi)\left[\chi\left(1+{}^{(1+\phi)}/_{\beta}\right)-\gamma\phi\right]$ and $\chi = \left[(1-\gamma)^{(\theta-1)}/_{\theta}-1\right]$.

Now recognize that for any variable $\hat{x} = \tilde{x} + \bar{x}$ we have $\hat{x}^2 = \tilde{x}^2 + 2\tilde{x}\bar{x} + \bar{x}^2$, where $\bar{x}^2$ can be ignored because it is independent of policy. Use these relationships in (C.12), and drop terms independent of policy:

$$\hat{c}_0 + \beta\hat{c}_1 = -\frac{1}{2}\phi(1+\phi-\alpha(1+\chi))(\tilde{k}_1^2 + 2\bar{k}_1\tilde{k}_1) + \frac{1}{2}\Omega(\tilde{y}^2_{T,0} + 2\bar{y}_{T,0}\tilde{y}_{T,0}) \tag{C.13}$$
$$+\phi(1+\chi)\hat{a}_1(\tilde{k}_1) - \gamma\chi(1+\phi)\tilde{y}_{T,0}\hat{a}_1 + \gamma(1+\phi)(\phi-\chi\alpha)(\tilde{k}_1+\bar{k}_1)(\tilde{y}_{T,0}+\bar{y}_{T,0}).$$

Now expand and use efficient allocation relationships:

$$\bar{y}_{T,0} = \frac{\theta}{1-\gamma}\frac{\omega_2}{\omega_3}\hat{a}_1; \qquad \bar{k}_1 = \frac{1}{1-\alpha}\left(\frac{\omega_3-\omega_2}{\omega_3}\right)\hat{a}_1, \tag{C.14}$$

which imply:

$$\bar{k}_1 = \frac{1-\gamma}{\theta(1-\alpha)}\left(\frac{\omega_3-\omega_2}{\omega_2}\right)\bar{y}_{T,0}. \tag{C.15}$$

One thus obtains:

$$\hat{c}_0 + \beta\hat{c}_1 = -\frac{1}{2}\phi(1+\phi-\alpha(1+\chi))(\tilde{k}_1^2 + 2\bar{k}_1\tilde{k}_1) + \frac{1}{2}\Omega(\tilde{y}^2_{T,0} + 2\bar{y}_{T,0}\tilde{y}_{T,0})$$
$$+\phi(1+\chi)(1-\alpha)\left(\frac{\omega_3}{\omega_3-\omega_2}\right)\bar{k}_1\tilde{k}_1 - \gamma\chi(1+\phi)\frac{(1-\gamma)}{\theta}\frac{\omega_3}{\omega_2}\tilde{y}_{T,0}\bar{y}_{T,0}$$
$$+\gamma(1+\phi)(\phi-\chi\alpha)\tilde{k}_1\tilde{y}_{T,0} + \gamma(1+\phi)(\phi-\chi\alpha)\frac{\theta(1-\alpha)}{1-\gamma}\left(\frac{\omega_2}{\omega_3-\omega_2}\right)\tilde{k}_1\bar{k}_1 +$$
$$\gamma(1+\phi)(\phi-\chi\alpha)\frac{1-\gamma}{\theta(1-\alpha)}\left(\frac{\omega_3-\omega_2}{\omega_2}\right)\bar{y}_{T,0}\tilde{y}_{T,0} \tag{C.16}$$

Using eq. (C.6), as well as eqs. (C.4) and (C.8), one obtains:

$$\frac{1}{2}(1-\sigma)(\hat{c}_0)^2+\beta\frac{1}{2}(1-\sigma)(\hat{c}_1)^2=$$
$$\frac{1-\sigma}{2}\left[\phi(\phi+\gamma\alpha)\hat{k}_1^2+(1+\frac{1}{\beta})(1+\phi)^2\gamma^2\hat{y}_{T,0}^2-2\gamma(1+\phi)(\phi+\alpha\gamma)\hat{k}_1\hat{y}_{T,0}-2(1+\phi)\gamma^2\hat{y}_{T,0}\hat{a}_1+2\phi\gamma\hat{a}_1\hat{k}_1\right] \tag{C.17}$$

Recognize that for any variable $\hat{x}=\tilde{x}+\bar{x}$ we have $\hat{x}^2=\tilde{x}^2+2\tilde{x}\bar{x}+\bar{x}^2$, where $\bar{x}^2$ can be ignored because it is independent of policy. One can rewrite (C.17) as:

$$\frac{1}{2}(1-\sigma)(\hat{c}_0)^2+\beta\frac{1}{2}(1-\sigma)(\hat{c}_1)^2=\frac{1-\sigma}{2}\left[\phi(\phi+\gamma\alpha)(\tilde{k}_1^2+2\bar{k}_1\tilde{k}_1)+(1+\frac{1}{\beta})(1+\phi)^2\gamma^2(\tilde{y}_{T,0}^2+2\bar{y}_{T,0}\tilde{y}_{T,0})\right]$$
$$+\frac{1-\sigma}{2}\left[-2\gamma(1+\phi)(\phi+\alpha\gamma)(\tilde{k}_1+\bar{k}_1)(\tilde{y}_{T,0}+\bar{y}_{T,0})-2(1+\phi)\gamma^2(\tilde{y}_{T,0}+\bar{y}_{T,0})\hat{a}_1+2\phi\gamma\hat{a}_1(\tilde{k}_1+\bar{k}_1)\right]$$

Applying the substitutions from the efficient allocation in eqs. (C.14) and (C.15), and adding eq. (C.16):

$$\hat{c}_0+\beta\hat{c}_1+\frac{1}{2}(1-\sigma)(\hat{c}_0)^2+\beta\frac{1}{2}(1-\sigma)(\hat{c}_1)^2=-\frac{1}{2}\phi\left[1+\phi-\alpha(1+\chi)-(1-\sigma)(\phi+\gamma\alpha)\right](\tilde{k}_1^2+2\bar{k}_1\tilde{k}_1)$$
$$+\frac{1}{2}\left[\Omega+(1-\sigma)(1+\frac{1}{\beta})(1+\phi)^2\gamma^2\right](\tilde{y}_{T,0}^2+2\bar{y}_{T,0}\tilde{y}_{T,0})+\phi[(1+\chi)+(1-\sigma)\gamma](1-\alpha)\left(\frac{\omega_3}{\omega_3-\omega_2}\right)\bar{k}_1\tilde{k}_1$$
$$-\gamma(1+\phi)[\chi+(1-\sigma)\gamma]\frac{(1-\gamma)}{\theta}\frac{\omega_3}{\omega_2}\bar{y}_{T,0}\tilde{y}_{T,0}+\gamma(1+\phi)[(\phi-\chi\alpha)-(1-\sigma)(\phi+\alpha\gamma)]\tilde{k}_1\tilde{y}_{T,0}+$$
$$+\gamma(1+\phi)[(\phi-\chi\alpha)-(1-\sigma)(\phi+\alpha\gamma)]\left[\frac{\theta(1-\alpha)}{1-\gamma}\left(\frac{\omega_2}{\omega_3-\omega_2}\right)\tilde{k}_1\bar{k}_1+\frac{1-\gamma}{\theta(1-\alpha)}\left(\frac{\omega_3-\omega_2}{\omega_2}\right)\bar{y}_{T,0}\tilde{y}_{T,0}\right] \tag{C.18}$$

Note that the following two set of relationships among parameters hold:

$$\phi\left[1+\phi-\alpha(1+\chi)-(1-\sigma)(\phi+\gamma\alpha)\right]=\phi[(1+\chi)+(1-\sigma)\gamma](1-\alpha)\left(\frac{\omega_3}{\omega_3-\omega_2}\right)$$
$$+\gamma(1+\phi)[(\phi-\chi\alpha)-(1-\sigma)(\phi+\alpha\gamma)]\frac{\theta(1-\alpha)}{1-\gamma}\left(\frac{\omega_2}{\omega_3-\omega_2}\right)$$
$$-\left[\Omega+(1-\sigma)(1+\frac{1}{\beta})(1+\phi)^2\gamma^2\right]=-\gamma(1+\phi)[\chi+(1-\sigma)\gamma]\frac{(1-\gamma)}{\theta}\frac{\omega_3}{\omega_2}$$
$$+\gamma(1+\phi)[(\phi-\chi\alpha)-(1-\sigma)(\phi+\alpha\gamma)]\frac{1-\gamma}{\theta(1-\alpha)}\left(\frac{\omega_3-\omega_2}{\omega_2}\right)$$

Use these two conditions in eq. (C.18), and plug it in (C.1):

$$
\begin{aligned}
(C^{\sigma-1})\mathbb{W} = &-\frac{1}{2}\phi\left[1+\phi-\alpha(1+\chi)-\phi(1-\sigma)(1+1/\beta)\right]\mathbb{E}(\tilde{k}_1^i)^2 \\
&-\frac{1}{2}\gamma(1+\phi)\left[\gamma\phi-\chi\left(1+\frac{1+\phi}{\beta}\right)-(1-\sigma)(1+1/\beta)(1+\phi)\gamma\right]\mathbb{E}(\tilde{y}_{T,0}^i)^2 \\
&+\gamma(1+\phi)[(\phi-\chi\alpha)-\phi(1-\sigma)(1+1/\beta)]\mathbb{E}\tilde{k}_1^i\tilde{y}_{T,0}^i + t.i.p. + \mathcal{O}(||\xi||^3),
\end{aligned}
\tag{C.19}
$$

where $\chi = \left[(1-\gamma)\frac{\theta-1}{\theta}-1\right]$ and $C^{\sigma-1} = \left(\frac{1}{\gamma}\right)^{\sigma-1}(\alpha\beta)^{\frac{\alpha(\sigma-1)}{1-\alpha}}$. Rewrite (C.19) as:

$$
\begin{aligned}
(C^{\sigma-1})\mathbb{W} = &-\frac{1}{2}\frac{\phi}{\theta}\omega_1\mathbb{E}(\tilde{k}_1^i)^2-\frac{1}{2}(1+\phi)\frac{\gamma}{(1-\alpha)(1-\gamma)}\left[(\omega_3-\omega_1)+(1-\alpha)\theta\right]\mathbb{E}(\tilde{q}_0^i)^2 \\
&-\frac{\phi}{(1-\alpha)\theta}(\omega_3-\omega_1)\mathbb{E}\tilde{k}_1^i\tilde{q}_0^i + t.i.p. + \mathcal{O}(||\xi||^3),
\end{aligned}
$$

The island-level real exchange rate and capital relative to their frictionless values are:

$$
\tilde{q}_0^i = \frac{\omega_1}{\omega_3}\left(r_0-(E_0^iq_1-q_0)\right)-\frac{\omega_2}{\omega_3}(E_0^ia_1-a_1); \qquad \tilde{k}_1^i = \frac{1}{1-\alpha}\left[\left(\frac{\omega_1}{\omega_3}-1\right)\left(r_0-(E_0^iq_1-q_0)\right)-\left(\frac{\omega_2}{\omega_3}-1\right)(E_0^ia_1-a_1)\right],
$$

implying

$$
\tilde{k}_1^i = \frac{1}{1-\alpha}\left[-\left(\frac{\omega_3-\omega_1}{\omega_1}\right)\tilde{q}_0^i+\left(\frac{\omega_1-\omega_2}{\omega_1}\right)(E_0^ia_1-a_1)\right] \tag{C.20}
$$

Use (C.20) into (C.20), and note that (C.20) is zero when evaluated at the first best, since, by definition, all gaps are zero. Thus we obtain eq. (33) in Section 4.

$$
\widetilde{\mathbb{W}} \equiv \mathbb{W}-\overline{\mathbb{W}} = -\Omega_1\mathbb{E}(\tilde{q}_0^i)^2-\Omega_2\mathbb{E}(E_0^ia_1-a_1)^2 + t.i.p. + \mathcal{O}(||\xi||^3), \tag{C.21}
$$

$$
\Omega_1 \equiv \frac{1}{2C^{\sigma-1}}\frac{\gamma}{(1-\gamma)}(1+\phi)\theta\frac{\omega_3}{\omega_1}; \qquad \Omega_2 \equiv \frac{1}{2C^{\sigma-1}}\frac{\phi}{\theta}\left(\frac{1}{1-\alpha}\right)^2\frac{(\omega_1-\omega_2)^2}{\omega_1}. \tag{C.22}
$$

## C.1 Auxiliary welfare representation

This appendix derives the auxiliary welfare representation, presented in Section 4.2. In doing so, it uses the fact that the public signal $z$ satisfies the following representation (regardless of whether interventions are conducted publicly or secretly):

$$
z|a_1 \sim \mathcal{N}\left((1+\delta)a_1, \beta_z^{-1}\right) \tag{C.23}
$$

**Auxiliary expression for** $\text{var}(E_0^i a_1 - a_1)$ Equations (C.23) and (16) imply:

$$\text{var}(E_0^i a_1 - a_1) = \left(\frac{1}{\beta_v + \beta_a + \beta_z}\right)^2 \left[(1+\delta)^2\beta_v + \beta_z + (\delta\beta_z + \delta\beta_v - \beta_a)^2 \frac{1}{\beta_a}\right] \tag{C.24}$$

**Auxiliary expression for** $\text{var}(q_0^i - \bar{q}_0)$ First, find an auxiliary expression for island-expected excess currency returns. To do so, use eqs. (A.16) and

$$(E_0^i - \bar{E}_0)\bar{E}_0[a_1] = \frac{\beta_v}{\beta_v + \beta_a + \beta_z}(E_0^i a_1 - \bar{E}_0 a_1); \qquad E_0^i a_1 - \bar{E}_0 a_1 = \frac{(1+\delta)\beta_v}{\beta_v + \beta_a + \beta_z}\epsilon^i$$

into eq. (37) to obtain:

$$\begin{aligned} r_0 - \left(E_0^i s_1 - s_0\right) = &-\frac{1-\gamma}{\theta}\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a + \beta_v + \beta_z}\right]\frac{\beta_v}{\beta_v + \beta_a + \beta_z}(1+\delta)\epsilon^i \\ &+ \frac{\Gamma\tilde{\theta}\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\bar{E}_0 a_1 + \frac{\Gamma\omega_3}{\Gamma\tilde{\theta}\omega_1 + \omega_3}(n_1^\star + f_1^\star). \end{aligned} \tag{C.25}$$

Use eq. (26) in eq. (C.25), and plug the resulting equation in eq. (34):

$$\begin{aligned} q_0^i - \bar{q}_0 = &-\frac{1-\gamma}{\theta}\frac{\omega_1}{\omega_3}\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a + \beta_v + \beta_z}\right]\frac{\beta_v}{\beta_v + \beta_a + \beta_z}(1+\delta)\epsilon^i + \\ &+ \frac{\Gamma\tilde{\theta}\omega_2 - \Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)}\frac{\omega_1}{\omega_3}\bar{E}_0 a_1 + \frac{\Gamma\omega_3}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\frac{\omega_1}{\omega_3}[(1+\kappa_w)n_1^\star] - \frac{\omega_2}{\omega_3}(E_0^i a_1 - a_1); \end{aligned}$$

Using equations (C.23) and (16) into the last equation, one obtains:

$$\begin{aligned} q_0^i - \bar{q}_0 = &-\frac{1}{\omega_3}\left\{\frac{1-\gamma}{\theta}\omega_1\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a + \beta_v + \beta_z}\right] + \omega_2\right\}\frac{\beta_v}{\beta_v + \beta_a + \beta_z}(1+\delta)\epsilon^i + \\ &\left(\frac{1}{\omega_3}\right)\left[\frac{\Gamma\tilde{\theta}\omega_2 - \Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)}\omega_1(1+\delta)(\beta_v + \beta_z) - \omega_2(\delta\beta_z + \delta\beta_v - \beta_a)\right]\frac{1}{(\beta_v + \beta_z + \beta_a)}a_1 + \\ &- \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\frac{\beta_z}{\beta_v + \beta_z + \beta_a}\epsilon^z + \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}(1+\kappa_w)n_1^\star \end{aligned} \tag{C.26}$$

and therefore:

$$
\operatorname{var}(q_0^i - \bar{q}_0) = \begin{bmatrix} \left(\frac{1}{\omega_3}\right)^2 \left\{\frac{1-\gamma}{\theta}\omega_1\left[1+\frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\right]+\omega_2\right\}^2 \frac{1}{(\beta_v+\beta_a+\beta_z)^2}(1+\delta)^2\beta_v + \\ \left(\frac{1}{\omega_3}\right)^2 \left[\frac{\Gamma\tilde{\theta}\omega_2-\Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)}\omega_1(1+\delta)(\beta_v+\beta_z)-\omega_2(\delta\beta_z+\delta\beta_v-\beta_a)\right]^2 \frac{1}{(\beta_v+\beta_z+\beta_a)^2}\frac{1}{\beta_a}+ \\ +\frac{\omega_2^2}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}\frac{1}{(\beta_v+\beta_z+\beta_a)^2}\beta_z \\ +\frac{(\Gamma\omega_1)^2}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}(1+\kappa_w)^2\frac{1}{\beta_n}-2\frac{\omega_2\Gamma\omega_1}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}\frac{\beta_z}{\beta_v+\beta_z+\beta_a}(1+\kappa_w)\operatorname{cov}(n_1^\star,\epsilon^z) \end{bmatrix} \tag{C.27}
$$

where $\epsilon^z$ denotes the noise in the public signal $z$.

**Auxiliary welfare expression** Having derived the auxiliary expressions for the welfare components (eqs. (C.24) and (C.27)), the auxiliary expression for welfare is:

$$
\widetilde{\mathbb{W}}(\kappa_q,\kappa_w,\beta_z) = -\begin{bmatrix} \Omega_1 \underbrace{\begin{bmatrix} \left(\frac{1}{\omega_3}\right)^2 \left\{\frac{1-\gamma}{\theta}\omega_1\left[1+\frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\right]+\omega_2\right\}^2 \frac{1}{(\beta_v+\beta_a+\beta_z)^2}(1+\delta)^2\beta_v + \\ \left(\frac{1}{\omega_3}\right)^2 \left[\frac{\Gamma\tilde{\theta}\omega_2-\Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)}\omega_1(1+\delta)(\beta_v+\beta_z)-\omega_2(\delta\beta_z+\delta\beta_v-\beta_a)\right]^2 \frac{1}{(\beta_v+\beta_z+\beta_a)^2}\frac{1}{\beta_a}+ \\ +\frac{\omega_2^2}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}\frac{1}{(\beta_v+\beta_z+\beta_a)^2}\beta_z \\ +\frac{(\Gamma\omega_1)^2}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}(1+\kappa_w)^2\frac{1}{\beta_n}-2\frac{\omega_2\Gamma\omega_1}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}\frac{\beta_z}{\beta_v+\beta_z+\beta_a}(1+\kappa_w)\operatorname{cov}(n_1^\star,\epsilon^z) \end{bmatrix}}_{\operatorname{var}(q_0^i-\bar{q}_0)} + \\ \Omega_2 \underbrace{\left(\frac{1}{\beta_v+\beta_a+\beta_z}\right)^2\left[(1+\delta)^2\beta_v+\beta_z+(\delta\beta_z+\delta\beta_v-\beta_a)^2\frac{1}{\beta_a}\right]}_{\operatorname{var}(E_0^i a_1 - a_1)} \end{bmatrix} \tag{C.28}
$$

The auxiliary welfare function, $\widetilde{\mathbb{W}}(\kappa_q,\kappa_w,\beta_z)$, is influenced by the covariance between noise trading shocks and the noise in the public signal, $\operatorname{cov}(n_1^\star,\epsilon^z)$. Under both communication regimes, this covariance is a (distinct) function of $(\kappa_q,\kappa_w,\beta_z)$. We thus write the auxiliary welfare function as in eq. (C.28), with a slight abuse of notation.

**Auxiliary welfare expression under public communication** Using public signal (28) in (C.28) the auxiliary welfare representation under public communication is:

$$
\widetilde{\mathbb{W}}(\kappa_q,\kappa_w;\beta_z) = -\Omega_1 \underbrace{\left[\left[\frac{\Gamma\tilde{\theta}\omega_2-\Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)}\frac{\omega_1}{\omega_3}(1+\delta)-\frac{\omega_2}{\omega_3}\delta\right]^2\frac{1}{\beta_a}+\left[\frac{\Gamma\omega_1}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)}\right]^2(1+\kappa_w)^2\frac{1}{\beta_n}\right]}_{\operatorname{var}(q_0^i-\bar{q}_0)} - \Omega_2 \underbrace{\frac{\delta^2}{\beta_a}}_{\operatorname{var}(E_0^i a_1 - a_1)} \tag{C.29}
$$

**Auxiliary welfare expression under secret conduct of FX interventions** Using public signal (32), along with the results of Proposition 3, in (C.28) one can express the auxiliary welfare representation under secret communication as:

$$\widetilde{\mathbb{W}}(\kappa_q,\kappa_w;\beta_z) = -\left[\begin{array}{l} \underbrace{\Omega_1 \left[\begin{array}{l} \left(\frac{1}{\omega_3}\right)^2 \left\{\frac{1-\gamma}{\theta}\omega_1\left[1+\frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\right]+\omega_2\right\}^2 \frac{1}{(\beta_v+\beta_a+\beta_z)^2}(1+\delta)^2\beta_v + \\ \left(\frac{1}{\omega_3}\right)^2\left[\frac{\Gamma\tilde{\theta}\omega_2-\Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)}\omega_1(1+\delta)(\beta_v+\beta_z)-\omega_2(\delta\beta_z+\delta\beta_v-\beta_a)\right]^2\frac{1}{(\beta_v+\beta_z+\beta_a)^2}\frac{1}{\beta_a}+ \\ +\frac{\omega_2^2}{(\Gamma\tilde{\theta}\omega_1+\omega_3-\Gamma\omega_1\kappa_q)^2}\frac{1}{(\beta_a+\beta_v+\beta_z)^2}\frac{(\beta_v+\beta_z)^2}{\beta_z} \end{array}\right]}_{\mathrm{var}(q_0^i-\bar{q}_0)} + \\ \underbrace{\Omega_2\left(\frac{1}{\beta_v+\beta_a+\beta_z}\right)^2\left[(1+\delta)^2\beta_v+\beta_z+(\delta\beta_z+\delta\beta_v-\beta_a)^2\frac{1}{\beta_a}\right]}_{\mathrm{var}(E_0^i a_1-a_1)} \end{array}\right] \tag{C.30}$$

Note that eq. (C.30) already internalizes the relevant constraint in the auxiliary planner's problem (38), thereby expressing welfare solely as a function of $\kappa_q$ and $\beta_z$.

# D Proofs

**Proof of Proposition 1.** Define $D \equiv \beta_a + \beta_z + \beta_v$. Plug the signal under laissez faire in the solution for the perceived exchange rate process (19):

$$q_0 \overset{p}{=} \left[1+\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_z}{D\lambda_a^l}\right]^{-1}\left\{\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}n_1^\star - \left[\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v}{D}\right]a_1\right\} \tag{D.1}$$

To find the undetermined coefficients, set (D.1) equal to the guess (17):

$$\lambda_a^l = -\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v+\beta_z}{D}; \qquad \lambda_n^l = \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v+\beta_z}{\beta_v}, \tag{D.2}$$

and therefore:

$$\frac{\lambda_a^l}{\lambda_n^l} = -\frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{D} \tag{D.3}$$

Define $\Lambda_l \equiv \lambda_a^l/\lambda_n^l$. Then:

$$\Lambda_l = -\frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{\beta_v+\beta_a+\Lambda_l^2\beta_n}, \quad \text{or} \quad \Lambda_l^3 + \left(\frac{\beta_v+\beta_a}{\beta_n}\right)\Lambda_l + \frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{\beta_n} = 0 \tag{D.4}$$

Define $\rho_1 \equiv \frac{\beta_v+\beta_a}{\beta_n}$ and $\rho_2 \equiv \frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{\beta_n}$. Thus, rewrite (D.4) as:

$$\Lambda_l^3 + \rho_1\Lambda_l + \rho_2 = 0 \tag{D.5}$$

Cubics of this form are said to be "depressed." Cardano's formula states that, if: (a) the cubic equation is of the form in (D.5); (b) $\rho_1$ and $\rho_2$ are real numbers; (c) $\frac{\rho_2^2}{4} + \frac{\rho_1^3}{27} > 0$ (which is satisfied in our context for any real value of $\omega_3/\Gamma\omega_2$); then then, equation (D.5) has (two complex conjugate roots and) one real root given by: $\sqrt[3]{-\frac{\rho_2}{2}+\sqrt{\frac{\rho_2^2}{4}+\frac{\rho_1^3}{27}}}+\sqrt[3]{-\frac{\rho_2}{2}-\sqrt{\frac{\rho_2^2}{4}+\frac{\rho_1^3}{27}}}$. To find the solution for the *actual* exchange rate process (20) substitute the actual average posterior beliefs $\int_0^1 E^i[a_1|v^i,z^l]di = \frac{(1+\delta)\beta_v a_1+\beta_z z^l}{\beta_v+\beta_z+\beta_a}$ and the coefficients $(\lambda_a^l, \lambda_n^l)$ in the actual market clearing condition (12). □

**Proof of Proposition 2.** Under public communication, agents observe both the volume of interventions, eq. (26), and the equilibrium exchange rate, eq. (12). These two public signals can be combined into a single composite signal, $z^p$, which fully reveals average expectations. Specifically, as long as $\kappa_w \neq 0$ in eq. (26), the following holds:

$$z^p \equiv \frac{1}{\omega_2\kappa_w}\left(\Gamma\omega_1(1+\kappa_w)f_1^\star - \left[(\Gamma\tilde{\theta}\omega_1+\omega_3)\kappa_w + \Gamma\omega_1\kappa_q\right]q_0\right) = \bar{E}_0 a_1 \tag{D.6}$$

When average expectations $\bar{E}_0 a_1$ are perfectly observable, agents in all islands solely rely on this value, as they perceive it to reveal actual $a_1$. As a result, individual expectations coincide with the average, and average expectations are:

$$\bar{E}_0 a_1 = (1+\delta)a_1 \tag{D.7}$$

Substituting equations (D.7) and (26) into equation (12), we obtain the solution for the equilibrium exchange rate, as reported in Proposition 2. □

**Proof of Proposition 3.** Consider the case in which the central bank does *not* reveal the size of FX interventions. Substituting the central bank's reaction function (26) in the exchange rate equation (12), one obtains

$$q_0 = \left(1-\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}\kappa_q\right)^{-1}\left[\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}(1+\kappa_w)n_1^\star - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\bar{E}_0 a_1\right] \tag{D.8}$$

Accordingly, we begin with a guess for the *perceived* exchange rate process:

$$q_0 \overset{p}{=} \lambda_a^s a_1 + \lambda_n^s n_1^\star \tag{D.9}$$

Combining the information of the private signal $v^i$ with the prior and and the public signal under secret interventions, $z^s$, we obtain individual agents' posterior beliefs:

$$E_0^i a_1 \equiv E[a_1|v^i, z^s] = \frac{(1+\delta)\beta_v v^i + \beta_z z^s}{\beta_v+\beta_z+\beta_a} \tag{D.10}$$

where $z^s|a_1 \equiv \frac{q_0}{\lambda_a^s}\Big|a_1 \sim \mathcal{N}((1+\delta)a_1, \beta_z^{-1})$. Because agents are unaware of others' belief distortion (E2), they perceive the market clearing condition (D.8) to be:

$$q_0 \overset{p}{=} \left(1 - \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\kappa_q\right)^{-1} \left[\frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1 + \omega_3}(n_1^\star + \kappa_w n_1^\star) - \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\bar{\mathbb{E}}[a_1|v^i, z^s]\right] \tag{D.11}$$

where

$$\bar{\mathbb{E}}[a_1|v^i, z^s] \equiv \int^i \mathbb{E}^i[a_1|v^i, z^s]\,\mathrm{d}i = \frac{\beta_v a_1 + \beta_z z^s}{\beta_v + \beta_z + \beta_a} \tag{D.12}$$

are average rational (instead of diagnostic) expectations.

Define $D \equiv \beta_a + \beta_v + \beta_z$. Plug (D.12) and (E.7) in the solution for perceived exchange rate process (D.11):

$$q_0 \overset{p}{=} \left(1 - \tfrac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}\kappa_q\right)^{-1} \left\{ - \tfrac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\tfrac{\beta_v+\beta_z}{\beta_v+\beta_z+\beta_a}a_1 + \tfrac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}((1+\kappa_w)n_1^\star) - \tfrac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\tfrac{\beta_z\frac{\lambda_n^s}{\lambda_a^s}n_1^\star}{\beta_v+\beta_z+\beta_a} \right\} \tag{D.13}$$

To find the undetermined coefficients in eq. (30), set (D.13) equal to the guess (D.9). By Cardano's formula (see Proof of Proposition 1), eq. (31) has a unique real solution. Last, to find the solution for the *actual* exchange rate process, eq. (30) substitute the actual average posterior beliefs obtained by summing over (D.10) and the coefficients $(\lambda_a^s, \lambda_n^s)$ in the actual market clearing condition (12). □

**Proof of Proposition 4.** We let the planner solve the auxiliary problem as defined in Section 4, where the auxiliary representation of the welfare function under public communication is given by (C.29), which internalizes the relevant constraint in the auxiliary planner's problem (38). The coefficients $\kappa_q^p$ and $\kappa_w^p$ in the Proposition result from the first-order conditions of the resulting problem with respect to $\kappa_q$ and $\kappa_w$, respectively. To find the welfare level at this optimum, plug $(\kappa_q^p, \kappa_w^p)$ in the auxiliary welfare function (C.29). □

**Proof of Proposition 5.** We let the planner solve the *two-step* auxiliary problem outlined in Section 4, where the auxiliary representation of the welfare function under secret interventions is given by (C.30), which already internalizes the relevant constraint in the auxiliary planner's problem (38), thereby expressing the welfare solely as a function of $\kappa_q$ and $\beta_z$. Because of this, Step 1 of the auxiliary problem boils down to finding the welfare-maximizing value of $\kappa_q$ for any given $\beta_z = \bar{\beta}_z$, that is $\kappa_q^s(\bar{\beta}_z)$. The corresponding maximizing value of $\kappa_w^s(\bar{\beta}_z)$ can be then determined by the relevant constraint in the auxiliary planner's problem (38). The first-order condition of (C.30) with respect to $\kappa_q$, lead one to find the optimal $\kappa_q^s(\bar{\beta}_z)$ in eq. (39). By replacing $\kappa_q$ with its optimal value $\kappa_q^s(\bar{\beta}_z)$ into welfare function (C.30) we obtain:

$$\widetilde{\mathbb{W}}(\kappa_q^s(\bar{\beta}_z), \kappa_w^s(\bar{\beta}_z), \bar{\beta}_z) = -\,\Omega_1 \underbrace{\left[\begin{matrix}\frac{\omega_2^2}{\omega_3^2}\left\{\frac{1-\gamma}{\theta}\left[\frac{\omega_1}{\omega_2} + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_2}\right)\frac{\beta_v}{\beta_a+\beta_v+\bar{\beta}_z}\right]+1\right\}^2 \frac{(1+\delta)^2\beta_v}{(\beta_v+\beta_a+\bar{\beta}_z)^2} \\ +\frac{\omega_2^2}{\omega_3^2}\frac{1}{\beta_a+(1+\delta)^2\bar{\beta}_z}\end{matrix}\right]}_{\mathrm{var}(\tilde{q}_0^i)}$$

$$-\,\Omega_2 \underbrace{\left(\frac{\delta^2}{\beta_a} + \frac{(1-\delta^2)\beta_a + (1-\delta^2)\beta_v + (1-2\delta-2\delta^2)\bar{\beta}_z}{(\beta_v+\beta_a+\bar{\beta}_z)^2}\right)}_{\mathrm{var}(E_0^i a_1 - a_1)}, \tag{D.14}$$

where $\Omega_1 > 0$ and $\Omega_2 \geq 0$ are defined in equation (C.22).

Eq. (D.14) represents the level of welfare for any value of $\bar{\beta}_z$, where the values of $(\kappa_q, \kappa_w)$ are set to their welfare-maximizing values $(\kappa_q^s(\bar{\beta}_z), \kappa_w^s(\bar{\beta}_z))$.

Eq. (D.14) yields two important results. First, secret policy can always attain the level of welfare of the optimal public policy. To see this, consider the limit of eq. (D.14):

$$\lim_{\bar{\beta}_z\to\infty} \widetilde{\mathbb{W}}(\kappa_q^s(\bar{\beta}_z), \kappa_w^s(\bar{\beta}_z), \bar{\beta}_z) = -\frac{1}{2}(C^{1-\sigma})\left[\frac{\alpha\beta\gamma(\omega_2-\omega_1)^2}{\theta(1-\alpha)^2\omega_1}\delta^2\frac{1}{\beta_a}\right].$$

Note that this level of welfare is equivalent to the one attained by the optimal public FX intervention policy, $\widetilde{\mathbb{W}}^p$, reported in Proposition 4, which can be obtained by setting $(\kappa_q, \kappa_w)$ according to $(\kappa_q^p, \kappa_w^p)$ in Proposition 4.

Second, secret interventions can attain a higher welfare than public ones if $\delta$ is sufficiently large. To see this, examine the asymptotic behavior of eq. (D.14) as $\bar{\beta}_z \to \infty$. To begin with, identify and collect the leading terms in eq. (D.14):

$$-\frac{\gamma\beta}{\omega_1}\frac{\alpha(\omega_2-\omega_1)^2}{\theta(1-\alpha)^2}\frac{1}{(\beta_v+\beta_a+\bar{\beta}_z)^2}(1-2\delta-2\delta^2)\bar{\beta}_z - \frac{\gamma\beta}{\omega_1}\tilde{\theta}\frac{\omega_2^2}{\omega_3}\frac{1}{\beta_a+(1+\delta)^2\bar{\beta}_z}$$

The sign of the dominant terms is positive if:

$$2\delta^4 + 6\delta^3 + 5\delta^2 > 1 + \tilde{\theta}\frac{\omega_2^2}{\omega_3}\frac{\theta(1-\alpha)^2}{\alpha(\omega_2-\omega_1)^2} \tag{D.15}$$

The left-hand side of eq. (D.15) is monotonically increasing in $\delta$ for $\delta > 0$, while the right-hand side of eq. (D.15) is a positive constant. Define $\hat{\delta} \geq 0$ as the smallest positive root of the quartic eq. (D.15). If $\delta > \hat{\delta}$, then the welfare function (D.14) reaches the optimal-public level from above as $\beta_z \to \infty$. This means that $\delta > \hat{\delta}$ is a sufficient condition for secret interventions to dominate optimal public interventions. □

**Proof of Proposition 6.** Take the limit of eq. (C.24) for $\beta_z \to \infty$ and $\beta_z \to 0$:

$$\lim_{\beta_z \to \infty} \mathrm{var}(E_0^i a_1 - a_1) = \delta^2 \tfrac{1}{\beta_a}; \qquad \lim_{\beta_z \to 0} \mathrm{var}(E_0^i a_1 - a_1) = \tfrac{(\delta\beta_v - \beta_a)^2 \frac{1}{\beta_a} + (1+\delta)^2 \beta_v}{(\beta_v + \beta_a)^2} \tag{D.16}$$

Thus, $\lim_{\beta_z \to \infty} \mathrm{var}(E_0^i a_1 - a_1) < \lim_{\beta_z \to 0} \mathrm{var}(E_0^i a_1 - a_1)$ if $\delta < 1$. Moreover,

$$\tfrac{\partial\, \mathrm{var}(E_0^i a_1 - a_1)}{\partial \beta_z} = \tfrac{1}{(\beta_v + \beta_z + \beta_a)^3} \left\{ (\beta_v + \beta_a)[1 - 2(1+\delta)] - \beta_z [1 - 2\delta(1+\delta)] \right\}. \tag{D.17}$$

We can distinguish two regions. If $\Leftrightarrow$ $\delta \leq \bar{\delta} \equiv \frac{-1+\sqrt{3}}{2}$, *i.e.* $[1 - 2\delta(1+\delta)] > 0$, then ${}^{\partial\, \mathrm{var}(E_0^i a_1 - a_1)}/_{\partial \beta_z} < 0$, and the forecast error variance declines in public signal accuracy. Instead, if $\delta > \bar{\delta}$, then ${}^{\partial\, \mathrm{var}(E_0^i a_1 - a_1)}/_{\partial \beta_z} < 0$ as long as $\beta_z \leq (\beta_a + \beta_v) \frac{1-2(1+\delta)}{1-2\delta(1+\delta)}$ and ${}^{\partial\, \mathrm{var}(E_0^i a_1 - a_1)}/_{\partial \beta_z} > 0$ as long as $\beta_z > (\beta_a + \beta_v) \frac{1-2(1+\delta)}{1-2\delta(1+\delta)}$. As a result, if $\delta \leq {}^{(-1+\sqrt{3})}/_2$ the global minimum forecast error variance obtains at $\beta_z = \infty$. Instead, if $\delta > {}^{(-1+\sqrt{3})}/_2$, the global minimum obtains at $\beta_z = (\beta_a + \beta_v) \frac{1-2(1+\delta)}{1-2\delta(1+\delta)}$. $\square$

# E Central bank observes noise trading imperfectly

We revisit Sections 3 and 4 under the relaxed assumption that the central bank observes noise trading flows imperfectly. Specifically, the FX intervention rule (26) becomes:

$$f_1^\star = \kappa_q q_0 + \kappa_w w, \tag{E.1}$$

where $w$ denotes the central bank's signal of the noise-trading flows $n_1^\star$:

$$w = n_1^\star + \eta, \qquad \text{with} \quad \eta \sim \mathcal{N}(0, \beta_\eta^{-1}). \tag{E.2}$$

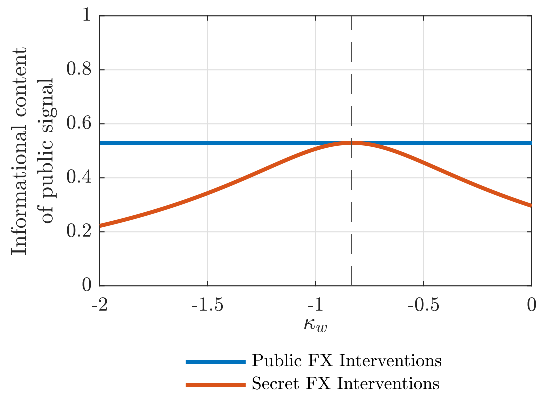
The intervention rule thus reads:

$$f_1^\star = \kappa_q q_0 + \kappa_w (n_1^\star + \eta). \tag{E.3}$$

Sections E.1 and E.2 characterize this extended setting, while Section E.3 presents the results (in the form of propositions analogous to those obtained in the baseline model).

The key insights from this extension are as follows. First, under public interventions, the resulting public signal is no longer perfectly informative. Nevertheless, the accuracy of this signal remains independent of the policy coefficients $(\kappa_q, \kappa_w)$ (Proposition E.1). Second, under secret interventions, the central bank can still influence the informational content of the exchange rate through its choice of policy coefficients $(\kappa_q, \kappa_w)$. This result is established

Figure E.1: Informational context of public signal in extended model of central bank




*Notes:* The figure reports values of the informational content of the public signal ($\beta_z/(\beta_a+\beta_v+\beta_z)$) for different values of the central bank's reaction function coefficient $\kappa_w$ under both public and secret FX intervention policy under rational expectations (that is, $\delta = 0$), when the central bank observes noise trading imperfectly (i.e. $\beta_\eta^{-1} > 0$). The rest of the parameters are set according to Table G.1.

in Proposition E.2 and illustrated numerically in Figure E.1. Furthermore, as in the baseline model, a secret FX intervention policy can replicate the informational content of a public intervention policy. This is achieved by selecting $\kappa_w$ to appropriately weight the two independent sources of noise—noise trading and central bank noise—such that the resulting exchange rate signal matches the informativeness of the public signal $z^p$, which optimally combines the noise sources in a Bayesian manner (Corollary 1).

From a normative perspective, we confirm that secret intervention can yield higher welfare outcomes when expectations exhibit sufficient overreaction. This result is formally characterized in Propositions E.3 and E.4, and numerically illustrated in Figure E.2. The economic intuitions derived from the baseline framework apply, *mutatis mutandis*, to this extended framework with central bank noise.

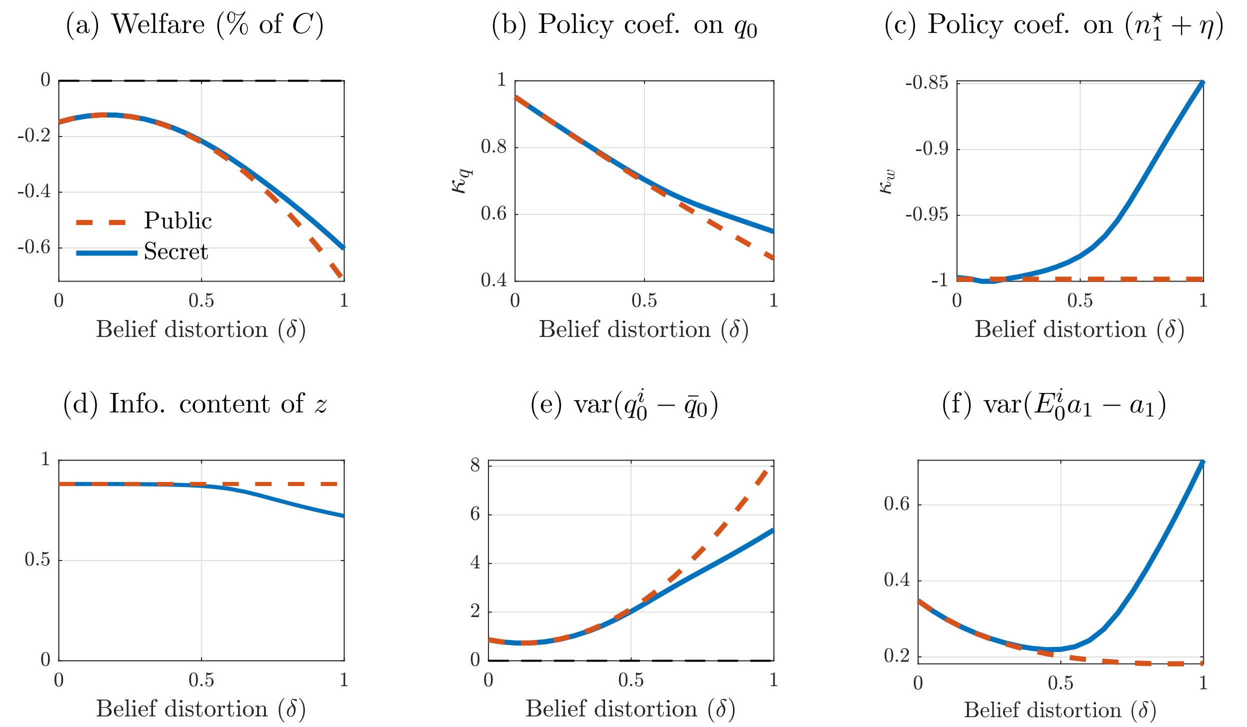
## E.1 Public signals in extended model

In this section, we characterize the properties of the public signal under public FX interventions, $z^p$, and secret FX interventions, $z^s$.

**Public interventions** Under public interventions, agents access two public signals, the exchange rate and the observed intervention volume (eqs. (12) and (E.3)). Consider the following guesses for the *perceived* exchange rate and intervention processes, respectively:

$$q_0 \overset{p}{=} \lambda_a^p a_1 + \lambda_n^p n_1^\star + \lambda_\eta^p \eta; \qquad f_1^\star \overset{p}{=} \gamma_a^p a_1 + \gamma_n^p n_1^\star + \gamma_\eta^p \eta. \tag{E.4}$$

Figure E.2: Outcomes under optimal policies in extended model of central bank




*Notes:* The figure reports values of different variables under optimal FX intervention policies for different levels of belief distortions ($\delta$). The rest of parameters are set according to Table G.1.

Using these equations one can construct the following signals:

$$u^p \equiv \frac{q_0 - \lambda_\eta^p \left(\frac{f_1^\star - \kappa_q q_0}{\kappa_w}\right)}{\lambda_a^p} \overset{p}{=} a_1 + \frac{(\lambda_n^p - \lambda_\eta^p)}{\lambda_a^p} n_1^\star; \quad g^p \equiv \frac{q_0 - \lambda_n^p \left(\frac{f_1^\star - \kappa_q q_0}{\kappa_w}\right)}{\lambda_a^p} \overset{p}{=} a_1 + \frac{(\lambda_\eta^p - \lambda_n^p)}{\lambda_a^p} \eta.$$

The conditional precisions of these two *independent* signals are:

$$\beta_u \equiv \left(\frac{\lambda_a^p}{(\lambda_n^p - \lambda_\eta^p)}\right)^2 \beta_n \qquad \beta_g \equiv \left(\frac{\lambda_a^p}{(\lambda_n^p - \lambda_\eta^p)}\right)^2 \beta_\eta$$

Consider unique public signal $z^p$:

$$z^p \equiv \left(\frac{\beta_u}{\beta_u + \beta_g} u^p + \frac{\beta_g}{\beta_u + \beta_g} g^p\right) = \left(\frac{\beta_n}{\beta_n + \beta_\eta} u^p + \frac{\beta_\eta}{\beta_n + \beta_\eta} g^p\right)$$

$$\Rightarrow \qquad z^p \overset{p}{=} a_1 + \frac{(\lambda_n^p - \lambda_\eta^p)}{\lambda_a^p} \frac{\beta_n}{\beta_n + \beta_\eta} n_1^\star + \frac{(\lambda_\eta^p - \lambda_n^p)}{\lambda_a^p} \frac{\beta_\eta}{\beta_n + \beta_\eta} \eta. \tag{E.5}$$

The variance and precision of $z^p$ are:

$$\beta_z^{-1} = \left(\frac{(\lambda_n^p - \lambda_\eta^p)}{\lambda_a^p}\right)^2 \frac{1}{(\beta_n + \beta_\eta)} \qquad \text{or} \quad \beta_z \equiv \left(\frac{\lambda_a^p}{(\lambda_n^p - \lambda_\eta^p)}\right)^2 (\beta_n + \beta_\eta) \tag{E.6}$$

**Secret interventions** In the extended model, there is an analogous representation of signal $z^s$ in eq. (16):

$$z^s \equiv \frac{q_0}{\lambda_a^s} \overset{p}{=} a_1 + \frac{\lambda_n^s}{\lambda_a^s} n_1^\star + \frac{\lambda_\eta^s}{\lambda_a^s}\eta, \quad \text{w/:} \quad \beta_z^{-1} = \left(\frac{\lambda_n^s}{\lambda_a^s}\right)^2 \beta_n^{-1} + \left(\frac{\lambda_\eta^s}{\lambda_a^s}\right)^2 \beta_\eta^{-1} \tag{E.7}$$

## E.2 Auxiliary planner's problem in extended model

Under Propositions E.1 and E.2, the auxiliary planner problem is:

**Auxiliary planner's problem (under central bank noise)**

$$\max_{\kappa_q, \kappa_w, \beta_z} \qquad \widetilde{\mathbb{W}}(\kappa_w, \kappa_q, \beta_z) = -\left[\Omega_1 \operatorname{var}(q_0^i - \bar{q}_0) + \Omega_2 \operatorname{var}(E_0^i a_1 - a_1)\right] \tag{E.8}$$

$$\text{subject to:} \qquad \begin{cases} \beta_z = \left(\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\frac{\omega_2}{\Gamma\omega_1}\right)^2 (\beta_n + \beta_\eta), & \text{if } z = z^p \\ \beta_z = \left(\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\frac{\omega_2}{\Gamma\omega_1}\right)^2 \left[(1+\kappa_w)^2\frac{1}{\beta_n} + \kappa_w^2\frac{1}{\beta_\eta}\right]^{-1}, & \text{if } z = z^s \end{cases}$$

*where closed-form expressions for* $\operatorname{var}(q_0^i - \bar{q}_0)$ *and* $\operatorname{var}(E_0^i a_1 - a_1)$ *are provided below.*

Following the derivations in Appendix C.1 one can show that, in this extended model, the external borrowing gap is:

$$\begin{aligned} q_0^i - \bar{q}_0 = &-\frac{1-\gamma}{\theta}\frac{\omega_1}{\omega_3}\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\right]\frac{\beta_v}{\beta_v+\beta_a+\beta_z}(1+\delta)\epsilon^i \\ &+ \frac{\Gamma\tilde{\theta}\omega_2 - \Gamma\omega_2\kappa_q}{(\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)}\frac{\omega_1}{\omega_3}\bar{E}_0 a_1 + \frac{\Gamma\omega_3}{\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q}\frac{\omega_1}{\omega_3}[(1+\kappa_w)n_1^\star + \kappa_w\eta] - \frac{\omega_2}{\omega_3}(E_0^i a_1 - a_1); \end{aligned} \tag{E.9}$$

which we further characterize in Propositions E.3 and E.4 depending on the type of communication format. Whereas, $\operatorname{var}(E_0^i a_1 - a_1)$ remains the one in eq. (C.24).

## E.3 Propositions

**Proposition E.1.** *Suppose the central bank intervenes according to* (E.1) *and* (E.2)*, where* $f_1^\star$ *is directly observed. Then:*

$$q_0 = \lambda_a^p(1+\delta)a_1 + \lambda_n^p n_1^\star + \lambda_\eta^p \eta, \tag{E.10}$$

$$\textit{with:} \qquad \lambda_a^p = -\left(1 - \frac{\Gamma\omega_1\kappa_q}{\Gamma\tilde{\theta}\omega_1 + \omega_3}\right)^{-1} \frac{\omega_2}{\Gamma\tilde{\theta}\omega_1 + \omega_3} \frac{\beta_v + \beta_z}{\beta_v + \beta_a + \beta_z}$$

$$\lambda_n^p = -\lambda_a^p \left( \frac{\Gamma\omega_1}{\omega_2} \frac{\beta_v + \beta_a + \beta_z}{\beta_v + \beta_z} (1 + \kappa_w) - \frac{\beta_z}{\beta_v + \beta_z} \frac{(\lambda_n^p - \lambda_\eta^p)}{\lambda_a^p} \frac{\beta_n}{\beta_n + \beta_\eta} \right)$$

$$\lambda_\eta^p = -\lambda_a^p \left( \frac{\Gamma\omega_1}{\omega_2} \frac{\beta_v + \beta_a + \beta_z}{\beta_v + \beta_z} (\kappa_w) - \frac{\beta_z}{\beta_v + \beta_z} \frac{(\lambda_\eta^p - \lambda_n^p)}{\lambda_a^p} \frac{\beta_\eta}{\beta_n + \beta_\eta} \right)$$

*with* $\beta_z \equiv \Lambda_p^2(\beta_n + \beta_\eta)$ *and a unique* $\Lambda_p \equiv \lambda_a^p / (\lambda_n^p - \lambda_\eta^p)$ *implicitly defined by:*

$$\Lambda_p^3 + \frac{(\beta_v + \beta_a)}{(\beta_n + \beta_\eta)} \Lambda_p + \frac{\omega_2}{\Gamma\omega_1} \frac{\beta_v}{(\beta_n + \beta_\eta)} = 0 \tag{E.11}$$

*The informational content of public signals (the Bayesian weight on public signals) is independent of the central banks' choice of* $(\kappa_q,\, \kappa_w)$*, as long as* $\kappa_w \neq 0$*.*

$$\mathcal{I}_R = \frac{\beta_z}{\beta_v + \beta_a + \beta_z} = \frac{\Lambda_p^2(\beta_n + \beta_\eta)}{\beta_v + \beta_a + \Lambda_p^2(\beta_n + \beta_\eta)}, \tag{E.12}$$

*where* $\Lambda_p$ *is independent of the central banks' choice of* $(\kappa_q,\, \kappa_w)$*, as evident from* (E.11).

*Proof.* Combining the information of the private signal $v^i$ with the prior and and the public signal, $z^p$, now given by eq. (E.5), find individual agents' posterior beliefs:

$$E_0^i a_1 \equiv E[a_1 | v^i, z^p] = \frac{(1+\delta)\beta_v v^i + \beta_z z^p}{\beta_v + \beta_a + \beta_z} \tag{E.13}$$

The average rational (instead of diagnostic) expectations are:

$$\bar{\mathbb{E}}[a_1 | v^i, z^p] \equiv \int^i \mathbb{E}^i[a_1 | v^i, z^p, g^p]\, \mathrm{d}i = \frac{\beta_v v^i + \beta_z z^p}{\beta_v + \beta_a + \beta_z}, \tag{E.14}$$

where $z^p$ is provided in eq. (E.5).

Use average rational posterior (E.14) in the system (12)-(E.3), which agents perceive to hold under rational expectations. After algebraic manipulations, one can find the fixed point of $(\lambda_a^p, \lambda_n^p, \lambda_\eta^p)$, reported in the Proposition. By Cardano's formula, eq. (E.11) has a unique real solution. Last, to find the solution for the actual exchange rate process (E.10) substitute the actual average posterior beliefs obtained by summing over (E.13) and the coefficients $(\lambda_a^p, \lambda_n^p, \lambda_\eta^p)$ in the actual market clearing condition (12). □

**Proposition E.2.** *Suppose the central bank intervenes according to* (E.1) *and* (E.2) *and* $f_1^\star$ *is not directly observed. Then:*

$$q_0 = \lambda_a^s(1+\delta)a_1 + (1+\kappa_w)\lambda_j^s n_1^\star + \kappa_w \lambda_j^s \eta, \tag{E.15}$$

$$\textit{with:}\quad \lambda_a^s = -\left(1 - \frac{\Gamma\omega_1}{\Gamma\tilde{\theta}\omega_1+\omega_3}\kappa_q\right)^{-1}\left(\frac{\omega_2}{\Gamma\tilde{\theta}\omega_1+\omega_3}\frac{\beta_v+\beta_z}{\beta_v+\beta_z+\beta_a}\right);$$

$$\lambda_n^s = \underbrace{-\lambda_a^s\frac{\Gamma\omega_1}{\omega_2}\frac{\beta_v+\beta_z+\beta_a}{\beta_v}}_{\lambda_j^s}(1+\kappa_w); \qquad \lambda_\eta^s = \underbrace{-\lambda_a^s\frac{\Gamma\omega_1}{\omega_2}\frac{\beta_v+\beta_z+\beta_a}{\beta_v}}_{\lambda_j^s}(\kappa_w);$$

*where* $\beta_z \equiv \Lambda_s^2\left((1+\kappa_w)^2\beta_n^{-1} + (\kappa_w)^2\beta_\eta^{-1}\right)^{-1}$, *and a unique* $\Lambda_s \equiv \left(\frac{\lambda_a^s}{\lambda_j^s}\right)$ *such that*

$$\Lambda_s^3 + \frac{\beta_v+\beta_a}{\left((1+\kappa_w)^2\beta_n^{-1} + (\kappa_w)^2\beta_\eta^{-1}\right)^{-1}}\Lambda_s + \frac{\omega_2}{\Gamma\omega_1}\frac{\beta_v}{\left((1+\kappa_w)^2\beta_n^{-1} + (\kappa_w)^2\beta_\eta^{-1}\right)^{-1}} = 0 \tag{E.16}$$

*Under secret FX interventions, the relative informational content of the exchange rate and the overall posterior accuracy depend on the intervention rule via eq.* (E.16).

*Proof.* The proof follow the steps in Proof of Proposition 3, where the sufficient statistic for the public signal under secret interventions, $z^s$, is now given by eq. (E.7). □

**Corollary 1.** *The precisions of public signal* $\beta_z$ *under public interventions (Proposition E.1) and secret interventions (Proposition E.2) are the same if and only if the secret intervention policy sets:* $\kappa_w = -\frac{\beta_\eta}{\beta_n+\beta_\eta}$.

**Proposition E.3.** *When the central bank observes noise trading imperfectly, the optimal public FX intervention policy is characterized by:*

$$\kappa_w^p = -\frac{\beta_\eta}{\beta_n+\beta_\eta}; \qquad \kappa_q^p = \frac{(\Gamma\tilde{\theta}\omega_1+\omega_3)}{\Gamma\omega_1} - \frac{1}{\Gamma\omega_1}\left[\frac{\omega_3(1+\delta)(\beta_z+\beta_v)}{\beta_a+\beta_v+\beta_z}\right]\left[1+\frac{\beta_a}{(1+\delta)^2\beta_z}\right], \tag{E.17}$$

*where* $\beta_z$ *is a parameter independent of policy defined in Proposition E.1.*

*At the optimum, the welfare level is:*

$$\begin{aligned}\widetilde{\mathbb{W}}^p = &-\Omega_1\underbrace{\left[\begin{array}{l}\frac{\omega_2^2}{\omega_3^2}\left\{\frac{1-\gamma}{\theta}\left[\frac{\omega_1}{\omega_2}+\frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_2}\right)\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\right]+1\right\}^2\frac{(1+\delta)^2\beta_v}{(\beta_v+\beta_a+\beta_z)^2}\\ +\left(\frac{\omega_2}{\omega_3}\right)^2\frac{1}{(1+\delta)^2\beta_z+\beta_a}\end{array}\right]}_{\mathrm{var}(\tilde{q}_0^i)}\\ &-\Omega_2\underbrace{\left(\frac{\delta^2}{\beta_a}+\frac{(1-\delta^2)\beta_a+(1-\delta^2)\beta_v+(1-2\delta-2\delta^2)\beta_z}{(\beta_v+\beta_a+\beta_z)^2}\right)}_{\mathrm{var}(E_0^i a_1 - a_1)},\end{aligned} \tag{E.18}$$

*where* $\beta_z$ *is a parameter independent of policy defined in Proposition E.1.*

*Proof.* The proof follows the steps in Proof of Proposition 4 and Proof of Proposition 5, where the sufficient statistic for the public signal under public interventions, $z^p$, is now (E.5). □

**Proposition E.4.** *Secret policy can always attain the level of welfare of the optimal public policy, including when the central bank observed noise trading imperfectly. In addition, if the central bank information about noise trading is sufficiently precise, optimal secret intervention policy attains a higher level of welfare than optimal public intervention policy for $\delta > \hat{\delta}$, where $\hat{\delta}$ is defined in the Proof.*

*Under the optimal secret intervention policy, $\kappa_q^s$ and $\kappa_w^s$ satisfy:*

$$\begin{aligned} \kappa_q^s &= \frac{(\Gamma\tilde{\theta}\omega_1 + \omega_3)}{\Gamma\omega_1} - \frac{1}{\Gamma\omega_1}\left[\frac{\omega_3(1+\delta)(\beta_z^s + \beta_v)}{\beta_a + \beta_v + \beta_z^s}\right]\left[1 + \frac{\beta_a}{(1+\delta)^2\beta_z^s}\right] \\ \left[(1+\kappa_w^s)^2\frac{1}{\beta_n} + (\kappa_w^s)^2\frac{1}{\beta_\eta}\right] &= \left(\frac{\beta_v}{\beta_a + \beta_v + \beta_z^s}\frac{\omega_2}{\Gamma\omega_1}\right)^2 \frac{1}{\beta_z^s} \end{aligned} \tag{E.19}$$

*and $\beta_z^s$ solves* $\partial\widetilde{\mathbb{W}}(\kappa_q^s(\beta_z^s), \kappa_w^s(\beta_z^s), \beta_z^s)/\partial\beta_z^s = 0 \implies -\Omega_1/\Omega_2 = \frac{\partial(\text{var}(E_0^i a_1 - a_1))/\partial\beta_z^s}{\partial(\text{var}(q_0^i - \bar{q}_0))/\partial\beta_z^s}$

*Proof.* Define the two-step approach in this extended model:

**Two-step approach (Secret interventions with imperfect $n_1^\star$ observation)**

**Step 1.** *Choose $(\kappa_w, \kappa_q)$ to maximize welfare $\widetilde{\mathbb{W}}(\kappa_w, \kappa_q, \beta_z)$ subject to $\beta_z = \bar{\beta}_z$ and $\beta_z = \left(\frac{\beta_v}{\beta_a+\beta_v+\beta_z}\frac{\omega_2}{\Gamma\omega_1}\right)^2\left[(1+\kappa_w)^2\frac{1}{\beta_n} + \kappa_w^2\frac{1}{\beta_\eta}\right]^{-1}$. Define $(\kappa_w^s(\bar{\beta}_z), \kappa_q^s(\bar{\beta}_z))$ as the solution to this step of the problem.*

**Step 2.** *Choose $\bar{\beta}_z$ to maximize welfare $\widetilde{\mathbb{W}}(\kappa_w^s(\bar{\beta}_z), \kappa_q^s(\bar{\beta}_z), \bar{\beta}_z)$. Define $\beta_z^s$ as the solution to this step of the problem.*

Use the relevant signal process (eqs. (16) and (E.7)), the constraint under secret interventions in Step 1, into eq. (E.9) to obtain:

$$\begin{aligned} \text{var}(q_0^i - \bar{q}_0) = &\left\{\left\{\frac{1-\gamma}{\theta}\frac{\omega_1}{\omega_3}\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a + \beta_v + \bar{\beta}_z}\right] + \frac{\omega_2}{\omega_3}\right\}\frac{\beta_v}{\beta_v + \beta_a + \bar{\beta}_z}(1+\delta)\right\}^2\frac{1}{\beta_v} \\ &+ \left[\frac{1}{(\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)}\right]^2\left\{\frac{\omega_2}{\omega_3}\left[\frac{(1+\delta)\omega_3(\beta_v + \bar{\beta}_z)}{\beta_v + \bar{\beta}_z + \beta_a} - (\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)\right]\right\}^2\frac{1}{\beta_a} + \\ &+ \left[\frac{1}{(\Gamma\tilde{\theta}\omega_1 + \omega_3 - \Gamma\omega_1\kappa_q)}\right]^2\left[\frac{\omega_2(\bar{\beta}_z + \beta_v)}{\beta_a + \beta_v + \bar{\beta}_z}\right]^2\frac{1}{\bar{\beta}_z} \end{aligned} \tag{E.20}$$

Take the FOC of eq. (E.20) w.r.t. $\kappa_q$ to obtain the optimal $\kappa_q^s(\bar{\beta}_z)$ in eq. (E.19), and use it to rewrite eq. (E.20) as:

$$\text{var}(q_0^i - \bar{q}_0) = \left\{\left\{\frac{1-\gamma}{\theta}\frac{\omega_1}{\omega_3}\left[1 + \frac{\alpha}{1-\alpha}\left(\frac{\omega_1 - \omega_2}{\omega_1}\right)\frac{\beta_v}{\beta_a + \beta_v + \beta_z}\right] + \frac{\omega_2}{\omega_3}\right\}\frac{\beta_v}{\beta_v + \beta_a + \beta_z}(1+\delta)\right\}^2\frac{1}{\beta_v} + \left(\frac{\omega_2}{\omega_3}\right)^2\frac{1}{(1+\delta)^2\beta_z + \beta_a} \tag{E.21}$$

Use eq. (C.24) and eq. (E.21), to rewrite eq. (33) under secret interventions as:

$$
\begin{aligned}
\widetilde{\mathbb{W}}(\kappa_q^s(\bar\beta_z),\kappa_w^s(\bar\beta_z),\bar\beta_z) = &-\Omega_1 \underbrace{\begin{bmatrix} \frac{\omega_2^2}{\omega_3^2}\left\{\frac{1-\gamma}{\theta}\left[\frac{\omega_1}{\omega_2}+\frac{\alpha}{1-\alpha}\left(\frac{\omega_1-\omega_2}{\omega_2}\right)\frac{\beta_v}{\beta_a+\beta_v+\bar\beta_z}\right]+1\right\}^2\frac{(1+\delta)^2\beta_v}{(\beta_v+\beta_a+\bar\beta_z)^2} \\ +\left(\frac{\omega_2}{\omega_3}\right)^2\frac{1}{(1+\delta)^2\bar\beta_z-\beta_a} \end{bmatrix}}_{\mathrm{var}(\tilde q_0^i)} \\
&-\Omega_2\underbrace{\left(\frac{\delta^2}{\beta_a}+\frac{(1-\delta^2)\beta_a+(1-\delta^2)\beta_v+(1-2\delta-2\delta^2)\bar\beta_z}{(\beta_v+\beta_a+\bar\beta_z)^2}\right)}_{\mathrm{var}(E_0^i a_1-a_1)},
\end{aligned}
\tag{E.22}
$$

Equation (E.22) represents the level of welfare for any value of $\bar\beta_z$, where the values of $(\kappa_q,\kappa_w)$ are set to their welfare-maximizing values $(\kappa_q^s(\bar\beta_z),\kappa_w^s(\bar\beta_z))$.

First, secret interventions can always achieve the same level of welfare as the optimal public policy. To see this, observe that if secret interventions deliver the same signal precision $\beta_z$ as public interventions, then they replicates the welfare level attained by the optimal public intervention policy, as reported in Proposition E.3 (compare equations (E.22) and (E.18)). This equivalence is achieved through an appropriate choice of the policy coefficient $\kappa_w$ (see Corollary 1). Comparing Corollary 1 and Proposition E.3, such central bank's policy coefficient value is $\kappa_w=\kappa_w^p$, which, in addition, minimizes the influence of non-fundamental financial flows on exchange rate fluctuations.

Second, secret policy can attain a higher welfare than public if $\delta$ is sufficiently large, under $\beta_\eta\to\infty$ in eq. (E.2). To assess whether secret FX policy achieves strictly higher welfare than optimal public policy, consider the leading-order behavior of equation (E.22) as $\bar\beta_z\to\infty$. The two dominant terms are:

$$
-\Omega_2\frac{(1-2\delta-2\delta^2)\bar\beta_z}{(\beta_a+\beta_v+\bar\beta_z)^2}\quad\text{and}\quad -\Omega_1\frac{\omega_2^2}{\omega_3^2}\frac{1}{\beta_a+(1+\delta)^2\bar\beta_z}. \tag{E.23}
$$

Both scale as $\bar\beta_z^{-1}$ and determine the direction from which $\widetilde{\mathbb{W}}(\bar\beta_z)$ converges to $\widetilde{\mathbb{W}}^p$. If the net coefficient on $\bar\beta_z^{-1}$ is negative, convergence occurs from above, and welfare under secret interventions strictly exceeds $\widetilde{\mathbb{W}}^p$ for large, but finite, $\bar\beta_z$. This occurs if:

$$
2\delta^4+6\delta^3+5\delta^2>1+\tilde\theta\cdot\frac{\omega_2^2}{\omega_3}\cdot\frac{\theta(1-\alpha)^2}{\alpha(\omega_2-\omega_1)^2}. \tag{E.24}
$$

Let $\hat\delta$ be the smallest positive solution to this inequality. Then for all $\delta>\hat\delta$, the function $\widetilde{\mathbb{W}}(\bar\beta_z)$ converges to $\widetilde{\mathbb{W}}^p$ from above.

Now fix a finite, large $\bar\beta_z^*$. By continuity of eq. (E.22) in $\bar\beta_z$, and since the higher-order remainder terms decay faster than $\bar{\beta}_z^{-1}$, there exists a shifted threshold $\hat{\delta}(\bar{\beta}_z^*) > \hat{\delta}$ such that:

$$\delta > \hat{\delta}(\bar{\beta}_z^*) \quad \Rightarrow \quad \widetilde{\mathbb{W}}(\bar{\beta}_z^*) > \widetilde{\mathbb{W}}^p. \tag{E.25}$$

Thus, the result extends to all sufficiently large—but finite—values of $\bar{\beta}_z$, with the sufficient condition on $\delta$ adjusting upward as the signal becomes less informative. □

# F Data sources and robustness

**Exchange rates** Exchange rate series are retrieved from *Bloomberg*, aggregate from daily to monthly frequency (period average). We consider the 12-month growth rate, in line with the literature (Broer and Kohlhas, 2022). We restrict the sample to country whose exchange rate regime is free floating or managed floating, according to the classification in Ilzetzki et al. (2019): we include the classification "Moving band that is narrower than or equal to +/-2%", "De facto moving band +/-5%/ Managed floating" and "Freely floating" (fine classification codes 11, 12 and 13). For robustness, we also include the whole coarse classification code 3, which includes "Pre announced crawling band that is wider than or equal to +/-2%" and "De facto crawling band that is narrower than or equal to +/-5%" (fine classification codes 9 and 10). We stop in 2019 to exclude the COVID-19 pandemic from the sample period.

**GDP** We measure realized outcomes with first release data. For each survey year, we use the realized outcome for real GDP growth from the *International Monetary Fund*'s World Economic Outlook (IMF WEO) published in April of the subsequent year. When the April issue does not contain the actual data but only an "estimate" by the IMF, we use the first available issue that contains the actual data. For robustness, we also consider the latest release data on GDP growth from the IMF International Financial Statistics database.

GDP Forecast data are from *Consensus Economics*, which provides monthly forecasts from a panel of individual economic forecasters. We consider observations for a set of 20 small open economies, from around 1998 to 2019 depending on data availability (see Table F.1). The forecast horizon differs across variables. In case of real GDP growth, in each month the forecasters provide a forecast about the annual growth of GDP in the current and following calendar year. We consider the forecast about the following year. We compute forecast errors as actual realization minus individual forecast. For each country, we discard forecasts at the bottom and top percentile, to avoid outliers. The final sample includes around 840 unique forecasters-country pair with an average of around 80 observations for each forecasters-country.

**Interest rates** We retrieve short-term interest rates series from *Refinitiv*, aggregated from

Table F.1: Summary statistics (real GDP forecasts)

| Country | Obs. | Mean | SD tot | SD within | SD between | Min Date | Max Date |
|---|---|---|---|---|---|---|---|
| Australia | 4544 | -.57 | 1.21 | 1.15 | .99 | 1998m1 | 2019m12 |
| Brazil | 4269 | -1.59 | 2.52 | 2.4 | 1.57 | 2000m10 | 2019m12 |
| Canada | 3199 | -.93 | 1.92 | 1.88 | 1.32 | 2003m6 | 2019m12 |
| Chile | 3904 | -1.17 | 2.48 | 2.33 | 1.86 | 2000m10 | 2019m12 |
| Colombia | 3017 | -.66 | 3.06 | 2.91 | 2.2 | 2000m10 | 2019m12 |
| India | 1239 | -2.7 | 5 | 4.57 | 3.45 | 2013m12 | 2019m12 |
| Indonesia | 1151 | .22 | .87 | .77 | .6 | 2000m5 | 2007m6 |
| Israel | 1605 | -.38 | 1.91 | 1.87 | .87 | 2005m2 | 2019m12 |
| Japan | 4971 | -.66 | 2.07 | 2.01 | 1.35 | 1998m1 | 2019m12 |
| Malaysia | 2541 | -.82 | 3.32 | 3.14 | 3.21 | 1998m8 | 2019m12 |
| Mexico | 4669 | -1.47 | 2.94 | 2.86 | 1.48 | 1998m2 | 2019m12 |
| New Zealand | 3462 | -.37 | 1.64 | 1.61 | .99 | 1998m1 | 2019m12 |
| Norway | 2424 | -1.11 | 2.65 | 2.42 | 2.2 | 1998m6 | 2019m12 |
| Peru | 1257 | 1.51 | 2.85 | 2.8 | .66 | 2004m1 | 2012m6 |
| Singapore | 3101 | -.5 | 4.1 | 3.94 | 2.45 | 1998m1 | 2016m3 |
| South Africa | 1666 | -1.93 | 2.56 | 2.47 | 1.56 | 2005m2 | 2019m12 |
| South Korea | 3905 | -.81 | 1.77 | 1.73 | 1.34 | 1999m7 | 2019m12 |
| Sweden | 1815 | -.18 | 1.67 | 1.51 | 1.69 | 1998m1 | 2019m12 |
| Switzerland | 2676 | -.41 | 1.57 | 1.54 | 1.06 | 1998m6 | 2019m12 |
| Thailand | 3783 | -1.25 | 3.11 | 3 | 2.1 | 1999m2 | 2019m12 |
| UK | 6745 | -.89 | 2.65 | 2.51 | 2.28 | 1998m1 | 2019m12 |

Table F.2: Summary statistics (short term interest rate differential)

| Country | Obs. | Mean | SD tot | SD within | SD between | Min Date | Max Date |
|---|---|---|---|---|---|---|---|
| Australia | 5162 | .1 | .45 | .43 | .28 | 1991m1 | 2019m12 |
| Brazil | 2353 | -1.46 | 1.25 | 1.18 | .79 | 2007m7 | 2019m12 |
| Canada | 2849 | .35 | .43 | .43 | .1 | 2003m6 | 2019m12 |
| Chile | 2408 | .08 | .89 | .88 | .28 | 2006m4 | 2019m12 |
| Colombia | 517 | -.05 | .62 | .61 | .26 | 2016m3 | 2019m12 |
| India | 309 | .34 | .63 | .46 | .69 | 2013m12 | 2019m12 |
| Indonesia | 519 | 1.6 | 2.42 | 2.25 | 1.28 | 2000m5 | 2007m6 |
| Japan | 5021 | .06 | .41 | .4 | .19 | 1991m1 | 2019m12 |
| Malaysia | 1358 | .16 | .48 | .44 | .45 | 1998m8 | 2019m12 |
| New Zealand | 3478 | .17 | .53 | .53 | .19 | 1994m12 | 2019m12 |
| Norway | 1843 | .13 | .63 | .61 | .28 | 1998m8 | 2019m12 |
| Singapore | 2560 | .14 | .65 | .64 | .31 | 1994m12 | 2016m3 |
| South Korea | 1997 | -.55 | .79 | .77 | .44 | 2001m6 | 2019m12 |
| Sweden | 1562 | -.01 | .42 | .4 | .21 | 1995m1 | 2019m12 |
| Switzerland | 2190 | .16 | .41 | .41 | .11 | 1998m6 | 2019m12 |
| Thailand | 1758 | .15 | .97 | .9 | .69 | 1999m2 | 2019m12 |
| UK | 6820 | .14 | .45 | .44 | .18 | 1993m9 | 2019m12 |

*Notes:* This table reports summary statistics and sample considered of (i) next-year real GDP growth forecast error in the upper panel, and (ii) 12-month ahead interest rate differential forecast error in the lower panel.

daily to monthly frequency (period average). We construct the interest differentials between each country and the U.S.. See Table F.2 for summary statistics.

Interest rate forecast data are from *Consensus Economics*. In each month forecasters provide forecasts about average short-term rate in 3 and 12 months. Consistent with the analysis of real GDP growth, we consider the 12-month forecast horizon. To construct the forecast about the interest rate differential with the U.S., we need forecast about U.S. interest rates. Unfortunately, very few forecasters provide forecasts about both small-open economies and the U.S.. Therefore, we proxy individual U.S. interest rate forecast using the consensus (average) forecast, under the assumption that forecasters use similar information source to forecast a large economy such as the U.S.. However, this forecast is strictly speaking not in the information set of individual forecasters in month $t$. For robustness, we also consider the consensus forecast from the previous month, which is instead available to forecasters in current month. For each country, we discard forecasts at the bottom and top percentile.

## F.1 Analysis using interest rates forecasts

The empirical strategy is similar the one utilized for the GDP growth forecasts in regressions (24) and (25). First, we investigate the relationship between exchange rate changes and 3-month interest rate differentials forecasts. To do so, for each country, we consider the regression

$$(F^i_m[i_{j,m+h}] - \bar{F}_m[i_{US,m+h}]) = \alpha_{i,j} + \beta_j(s_{j,m} - s_{j,m-12}) + \varepsilon_{i,j,m}, \tag{F.1}$$

where $i_{j,m+h}$ is the short term interest rate in country $j$ in month $m+h$, $F^i_m[i_{j,m+h}]$ is the forecast provided in month $m$ by forecaster $i$, $\bar{F}_m[i_{US,m+h}] = \sum_{i=1}^{N_{US,m}} F^i_m[i_{US,m+h}]/N_{US,m}$ is the average forecasts about U.S. interest rate, and $\Delta s_{j,m,m-12}$ is the 12-month percentage change of country $j$ exchange rate in unit of dollars. For each country, we include the forecaster fixed effect.

Second, we test whether interest rate expectations are excessively elevated—i.e. they overreact—when the exchange rate has recently appreciated. To do so, for each country, we implement the following regression of 3-month interest rate differential forecast errors on exchange rate changes:

$$(i_{j,m+h} - i_{US,t+h}) - (F^i_m[i_{j,m+h}] - \bar{F}_m[i_{US,m+h}]) = \alpha_{i,j} + \gamma_j(s_{j,m} - s_{j,m-12}) + \varepsilon_{i,j,m}, \tag{F.2}$$

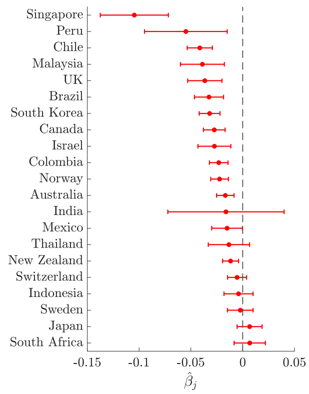
Figure F.2 reports the estimated $\gamma_j$, normalized by coefficient $\beta_j$ in regression (24) such that $\gamma_j < 0$ implies expectations' overreaction. A significant portion of the countries display overreaction to exchange rate appreciations.

In the top panels of Figure F.2, we use $\bar{F}_m[i_{US,m+h}]$ to proxy forecasters' expectations about U.S. interest rate, which is not provided by the large majority of forecasters in the sample, under the assumption that information about the U.S. are public and common. However, a possible concern with this regression is that $\bar{F}_m[i_{US,m+h}]$ is not in the information set of the forecaster in month $m$. To address this concern, in the bottom panels of Figure F.2, we use the lag consensus, $\bar{F}_{m-1}[i_{US,m+h}]$, to proxy forecasters' expectations about the U.S. interest rate. Results are virtually identical.

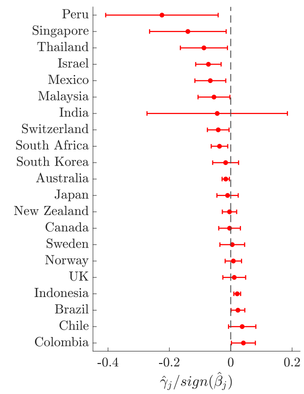
Figure F.1: Real GDP growth forecasts and exchange rate changes: Robustness

**All months**

(a) GDP growth forecasts ($\hat{\beta}_j$)

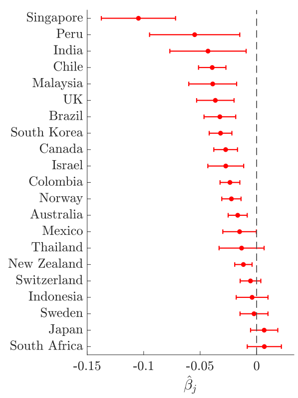



(b) GDP growth forecast errors ($\hat{\gamma}_j$)

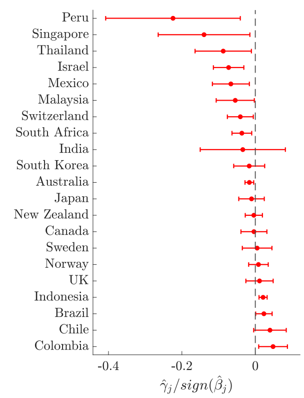



**Including crawling pegs**

(c) GDP growth forecasts ($\hat{\beta}_j$)

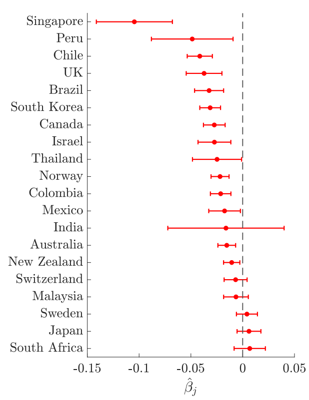



(d) GDP growth forecast errors ($\hat{\gamma}_j$)

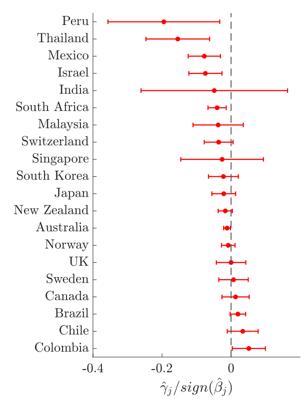



**Data from *IMF International Financial Statistics***

(e) GDP growth forecasts ($\hat{\beta}_j$)

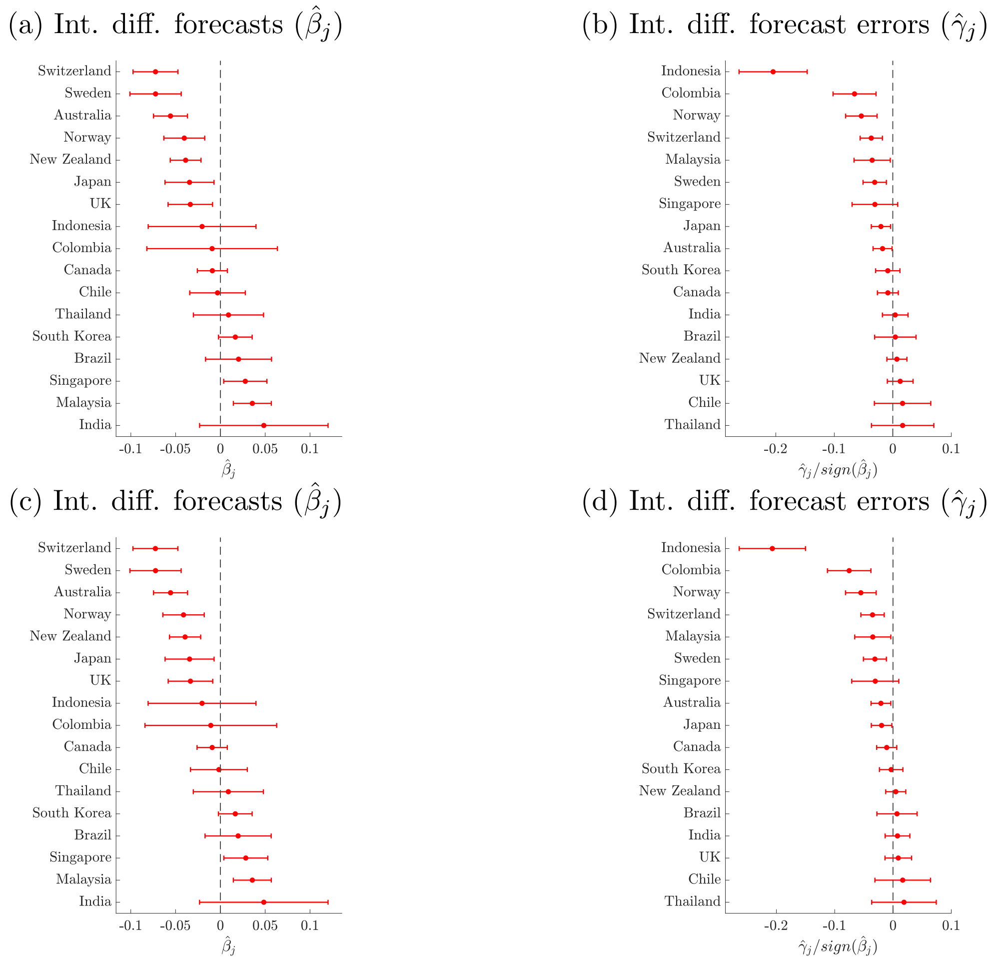



(f) GDP growth forecast errors ($\hat{\gamma}_j$)




*Notes:* These panels reproduce the analysis of Table 3. The top panels use forecasts for all months. The middle panels use an extended sample of countries that includes also "crawling band" exchange rate regimes. The bottom panels, use GDP growth data from the *IMF International Financial Statistics*, instead of first-release data from the annual *IMF World Economic Outlook*.

Figure F.2: Interest rate differential forecasts and exchange rate changes




*Notes:* Panel (a) and (c) report the estimated $\hat{\beta}_j$ from regression (F.1) for each country $j$ (the expected 12-month ahead interest rate differential associated with a 1% 12-month appreciation in the local currency/USD exchange rate). Panel (b) and (d) report the estimated $\hat{\gamma}_j$ from regression (F.2) for each country $j$ normalized by the sign of the estimated $\hat{\beta}_j$ from regression (F.1), *i.e.* $\hat{\gamma}_j/\,\text{sign}(\hat{\beta}_j)$. Panels (a) and (b) use use consensus, $\bar{F}_m[i_{US,m+h}]$, as proxy for expectations of U.S. interest rate, while panels (c) and (d) use lag consensus, $\bar{F}_{m-1}[i_{US,m+h}]$, as proxy for expectations of U.S. interest rate. Bars display the 90% confidence interval with Driscoll and Kraay (1998) standard errors. Data is monthly from 1998-2019.

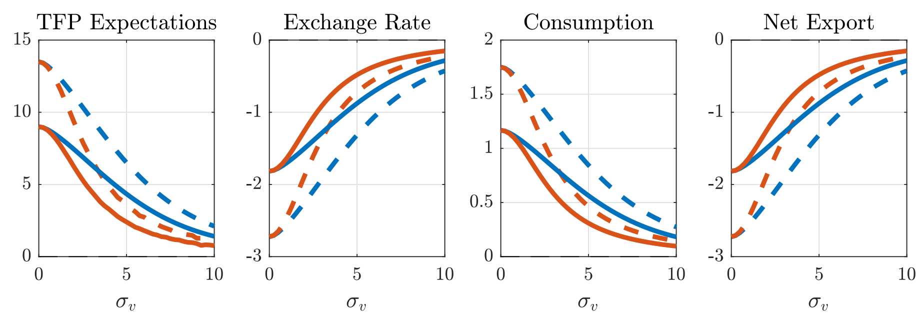
# G Additional Tables and Figures

| Parameter | Interpretation | Value |
|---|---|---|
| $\beta$ | Discount Factor | 0.99 |
| $\alpha$ | Share of Capital in Tradable | 0.33 |
| $\gamma$ | Trade Opennes | 0.33 |
| $\theta$ | Trade Elasticity | 0.67 |
| $\sigma$ | CRRA parameter | 10.00 |
| $\Gamma$ | Intermediation Friction | 5.00 |
| $\sigma_a$ | Std. dev. of productivity shocks | 3.00 |
| $\sigma_v$ | Std. dev. of private signal | 5.00 |
| $\sigma_n$ | Std. dev. of noise-trading shock | 0.25 |

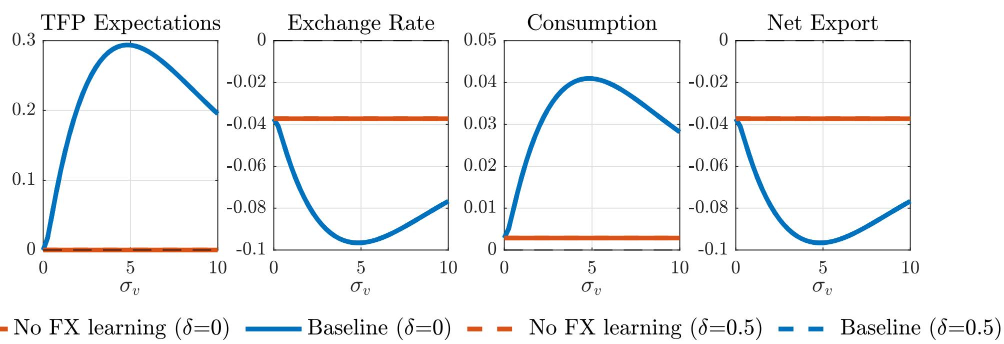
Table G.1: Baseline parameterization for numerical illustrations

Figure G.1: Equilibrium responses to shocks under dispersed information

(a) Response to one-standard deviation productivity shock ($a_1 \uparrow$)




(b) Response to one-standard deviation noise-trading shock ($n_1^\star \downarrow$)




*Notes:* The figure reports the equilibrium response of model variables to productivity and noise-trading shocks for different levels of the noise in private signal, $\sigma_v$, under laissez faire. The parameter governing belief distortions is $\delta = [0; 0.5]$. The rest of the parameters are set according to Table G.1 in the Online Appendix.